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UNIVERSITAT ROVIRA i VIRGILI

UIB Universitat de les Illes Balears

Successive Approximation ADCs: From compact low-power data conversion techniques to identifying system level bottlenecks

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IEEE SSCS Electron Devices Society CAS

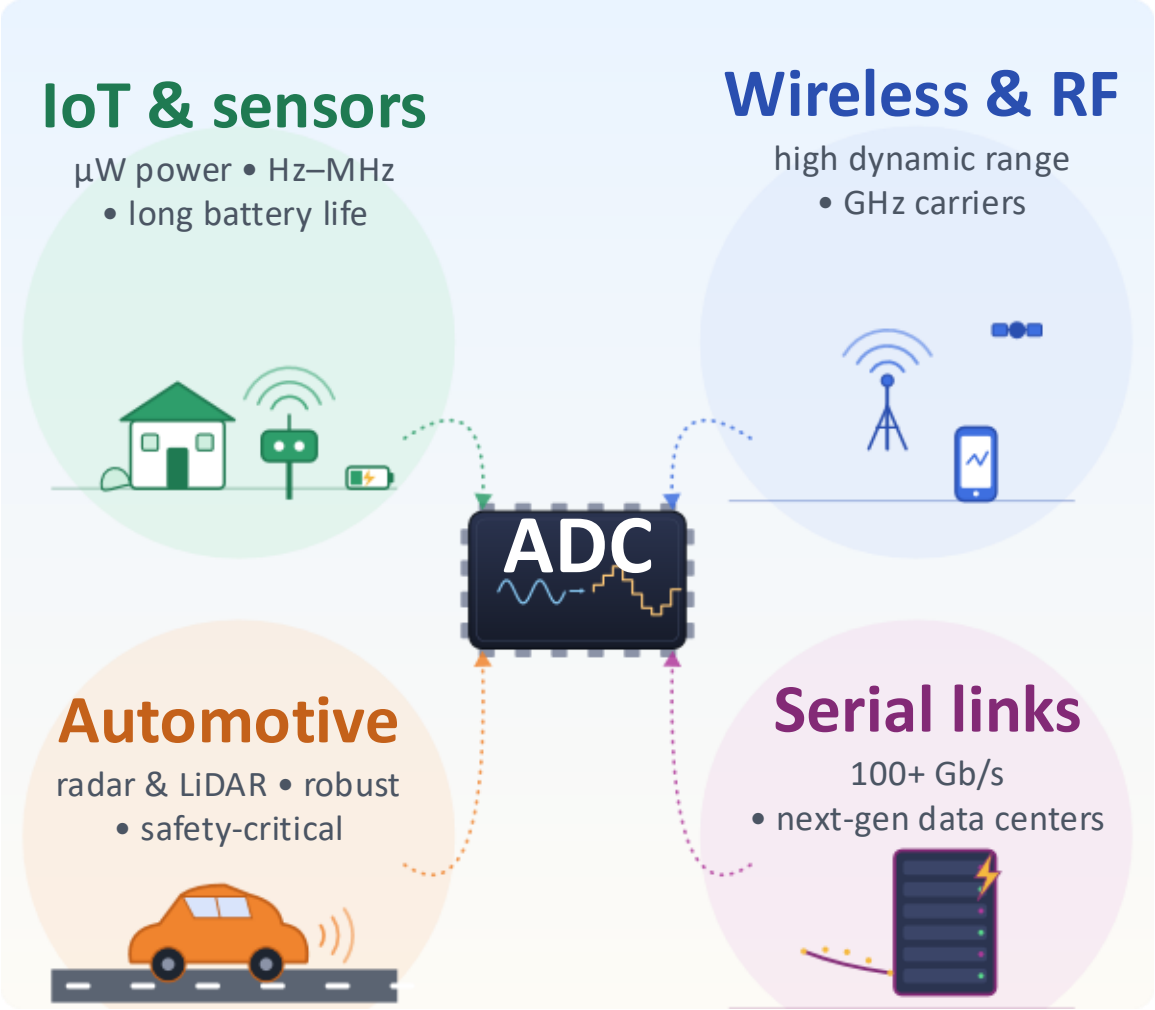
Outline

- Introduction
- Ultra-low-power SAR ADCs overview: Building blocks perspective
- Wide-bandwidth (GS/s) SAR ADCs
- Auxiliary blocks and its requirements
- Conclusion

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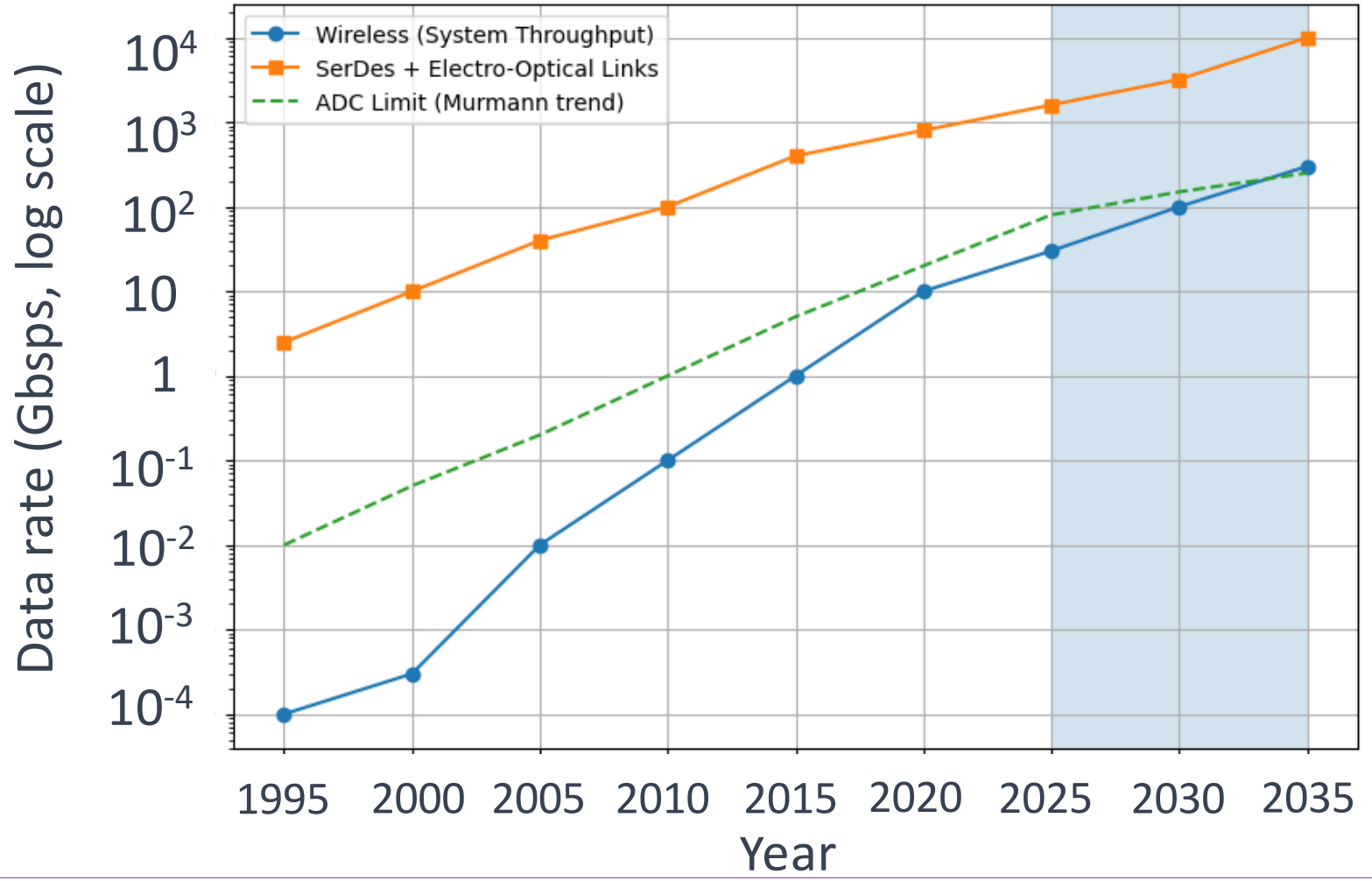
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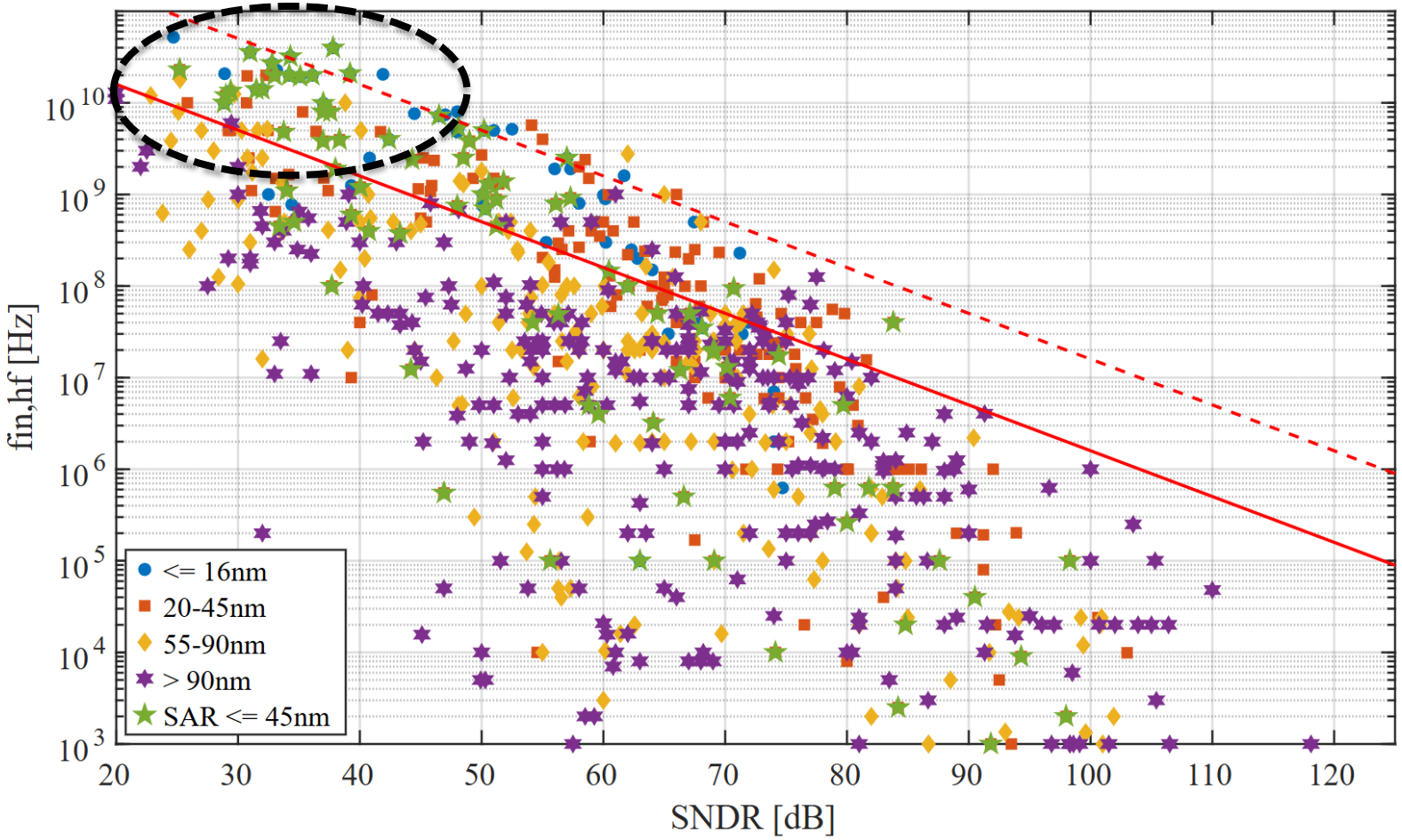


from microwatt sensor nodes to 100+ Gb/s serial links

ADC speed limits vis-à-vis data rate

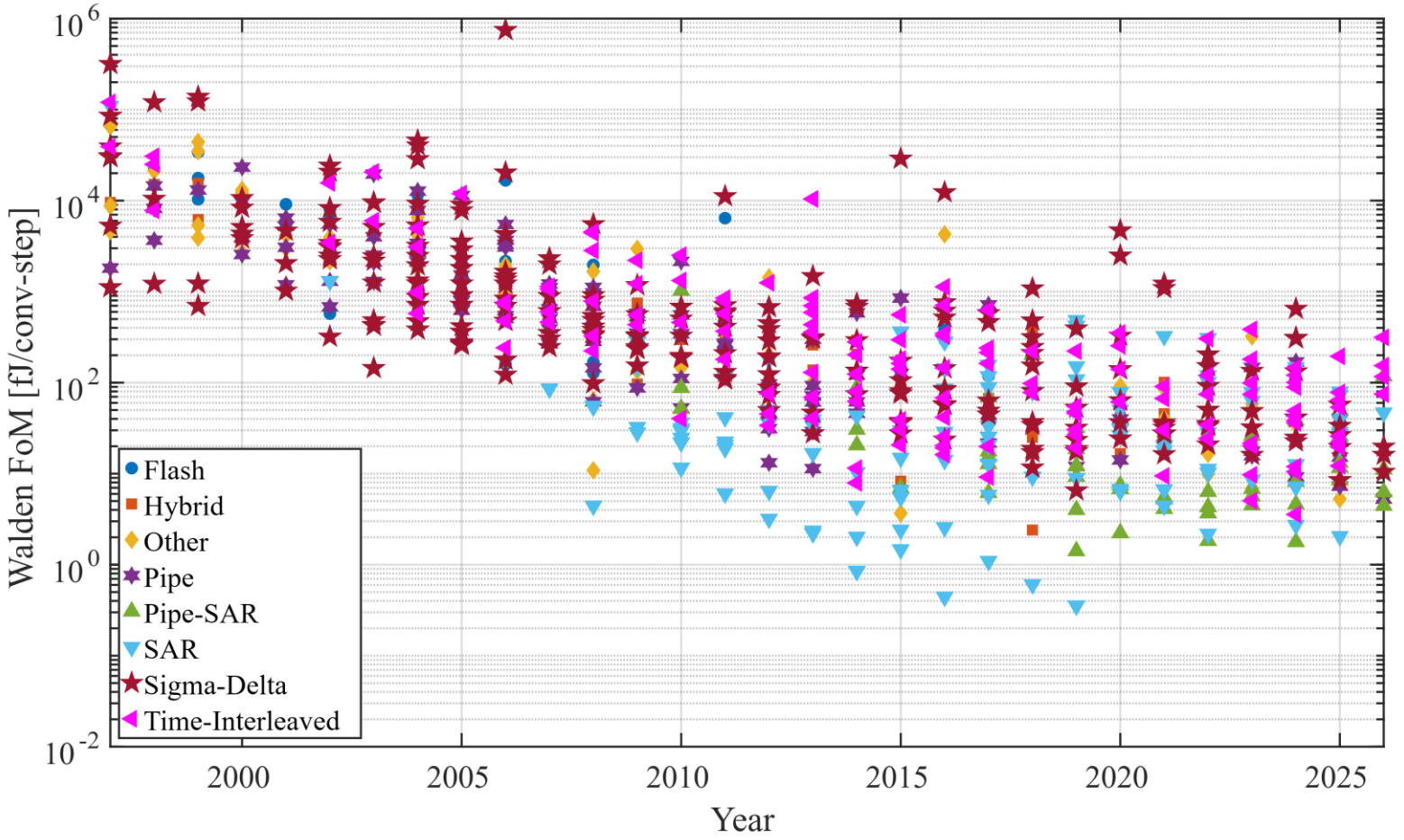


ADC performance limits



SAR ADCs benefit from technology scaling

ADC performance limits

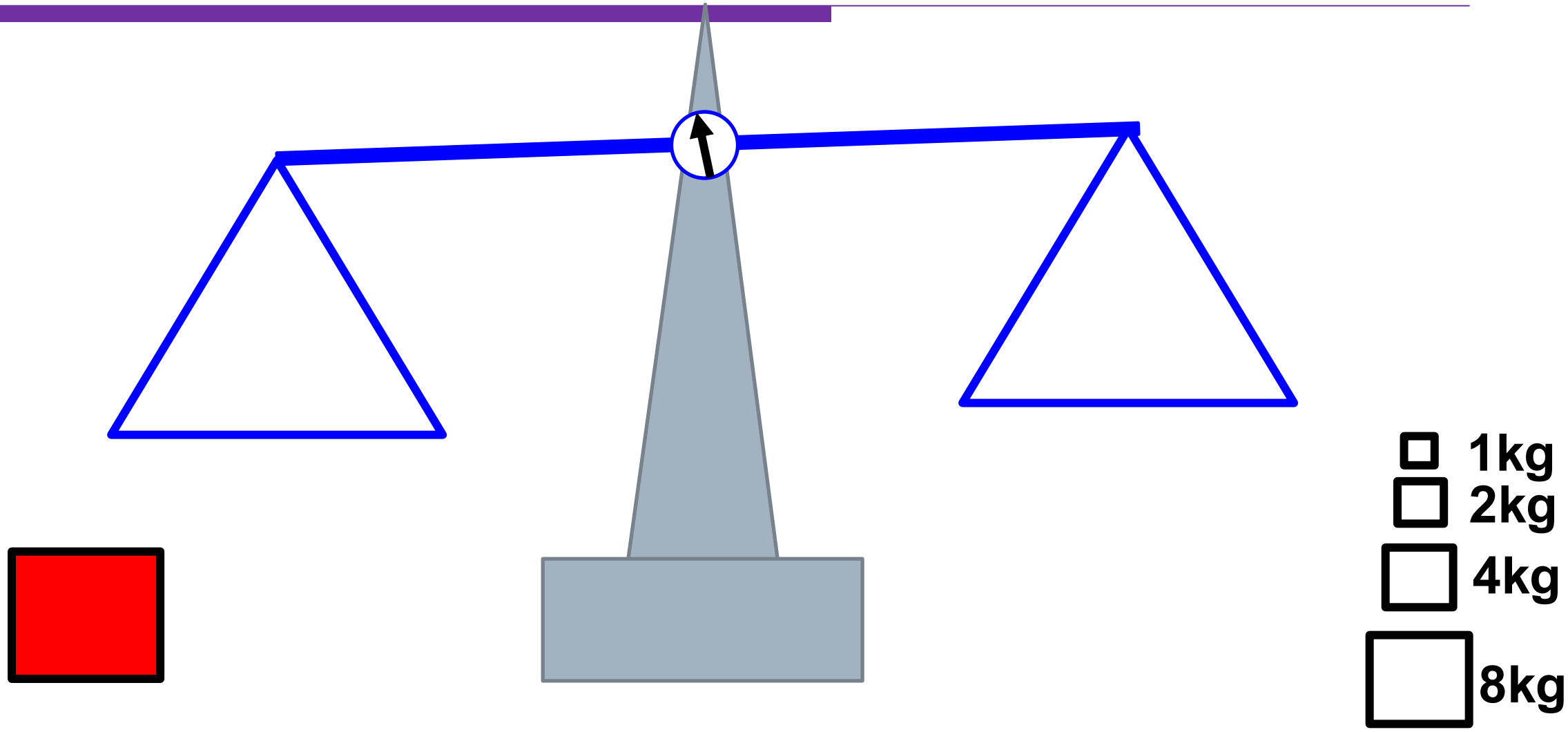


SAR ADCs have excellent energy-efficiency -> almost digital architecture

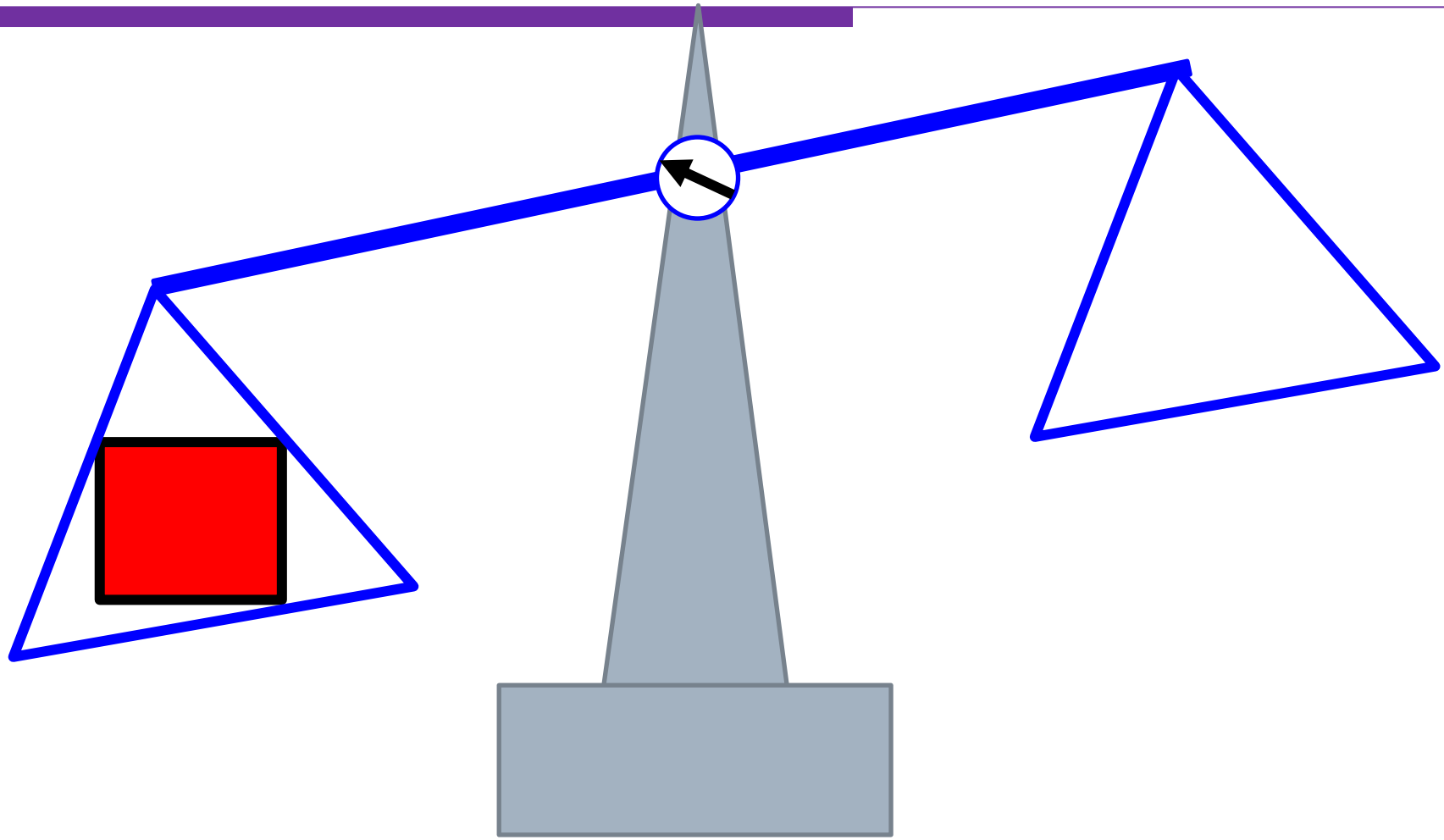
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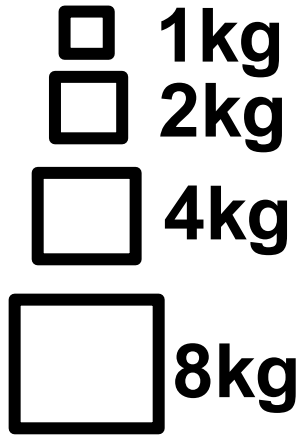
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Successive Approximation - Analogy

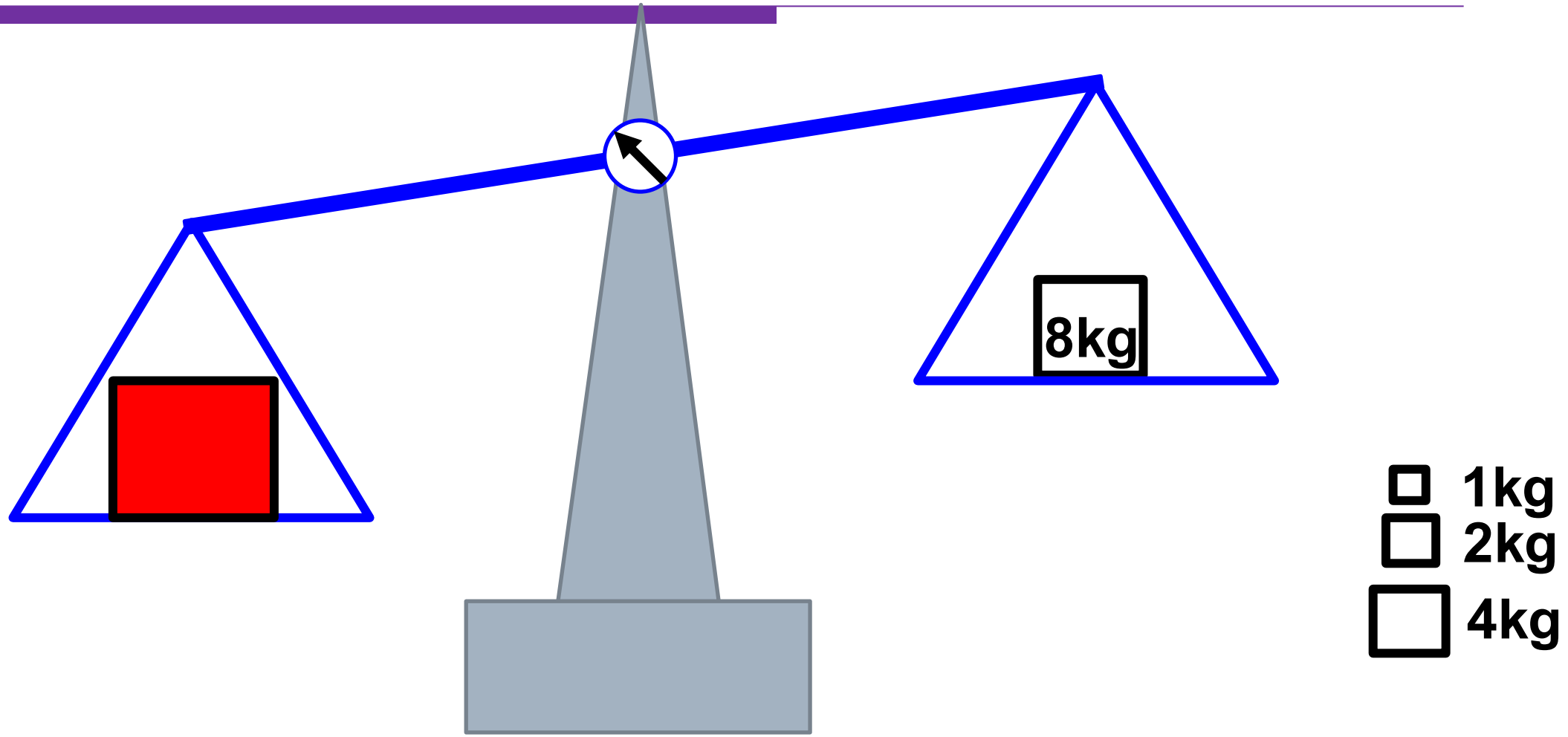


Successive Approximation - Analogy

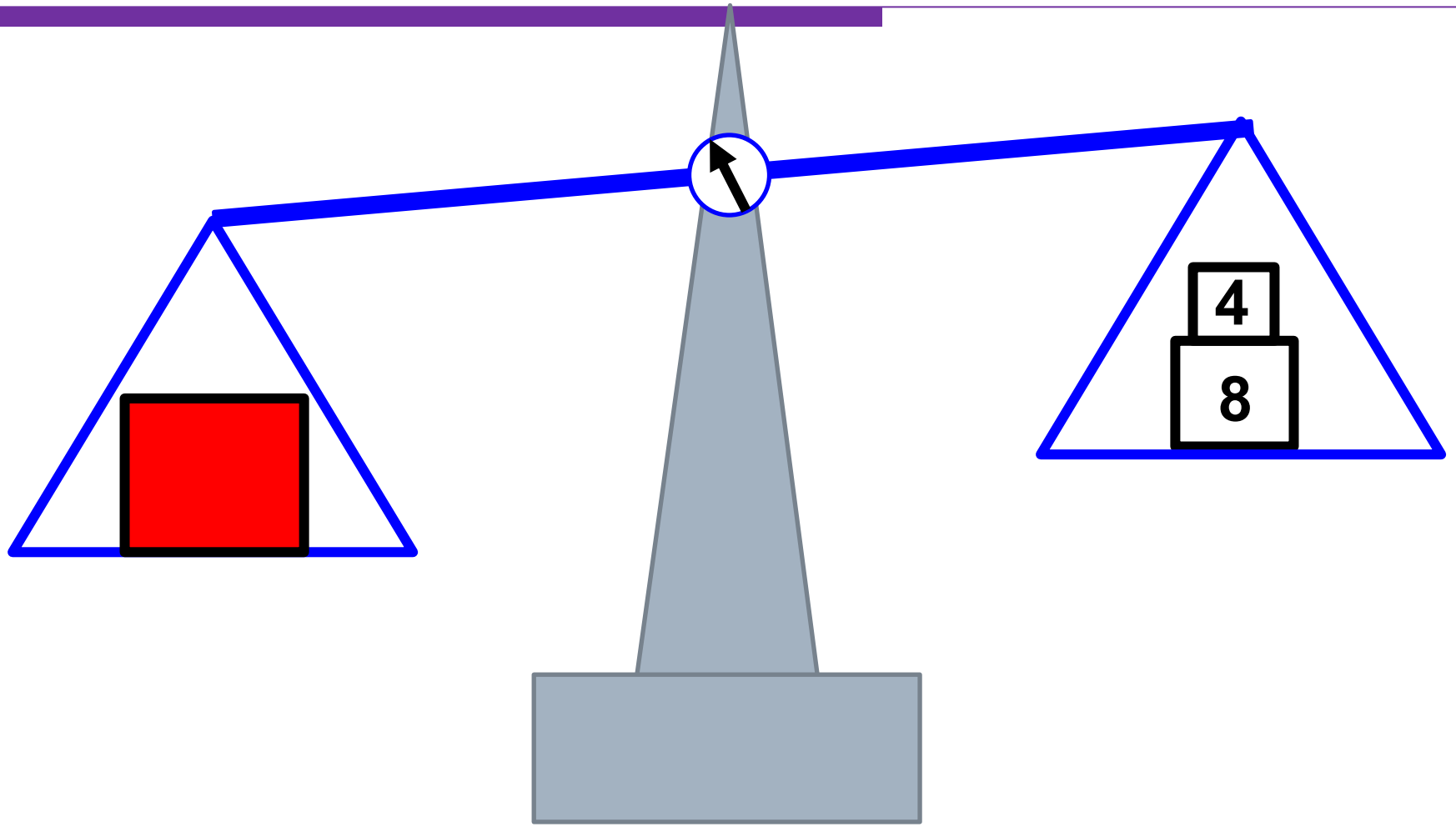




Successive Approximation - Analogy

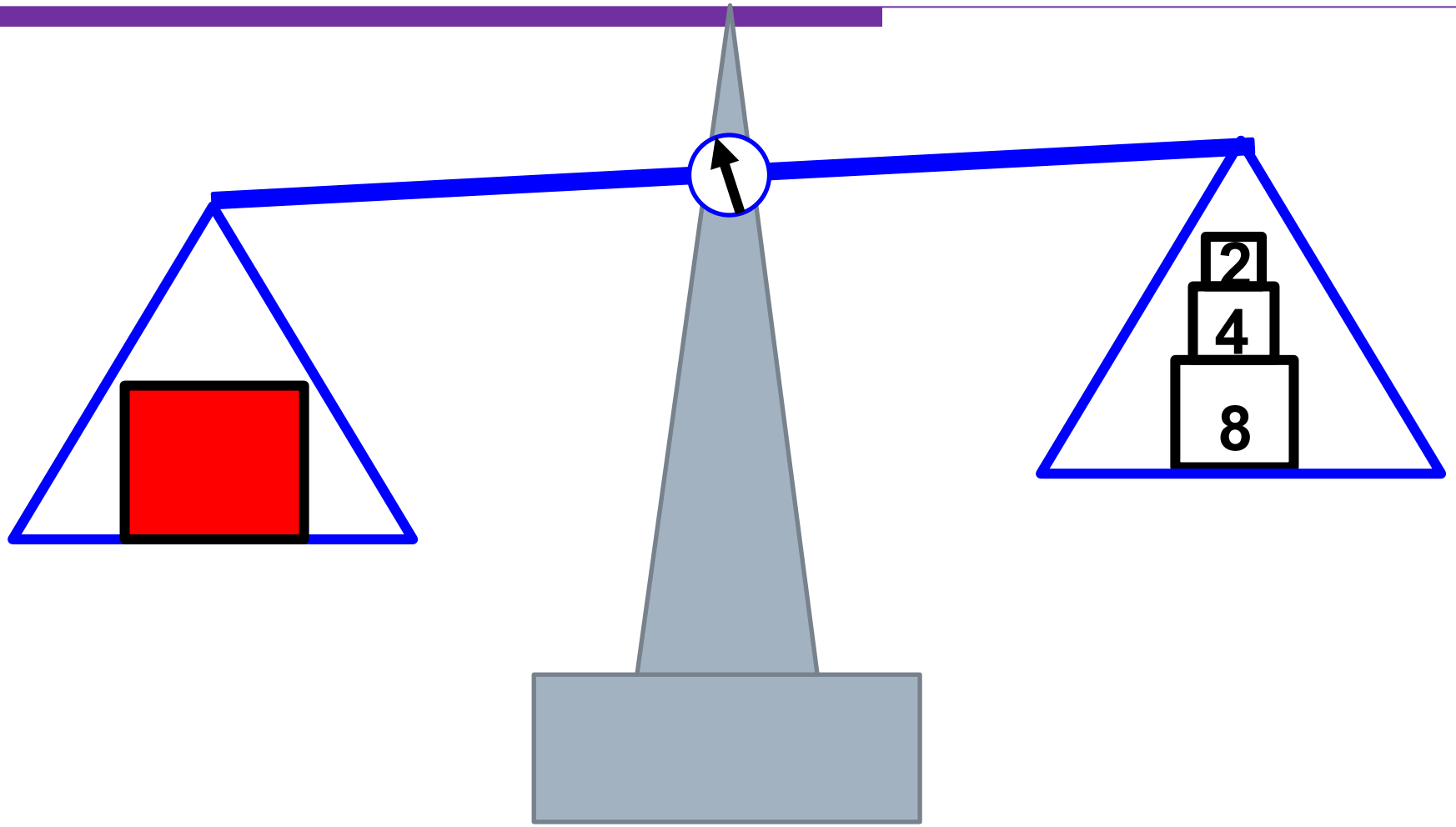


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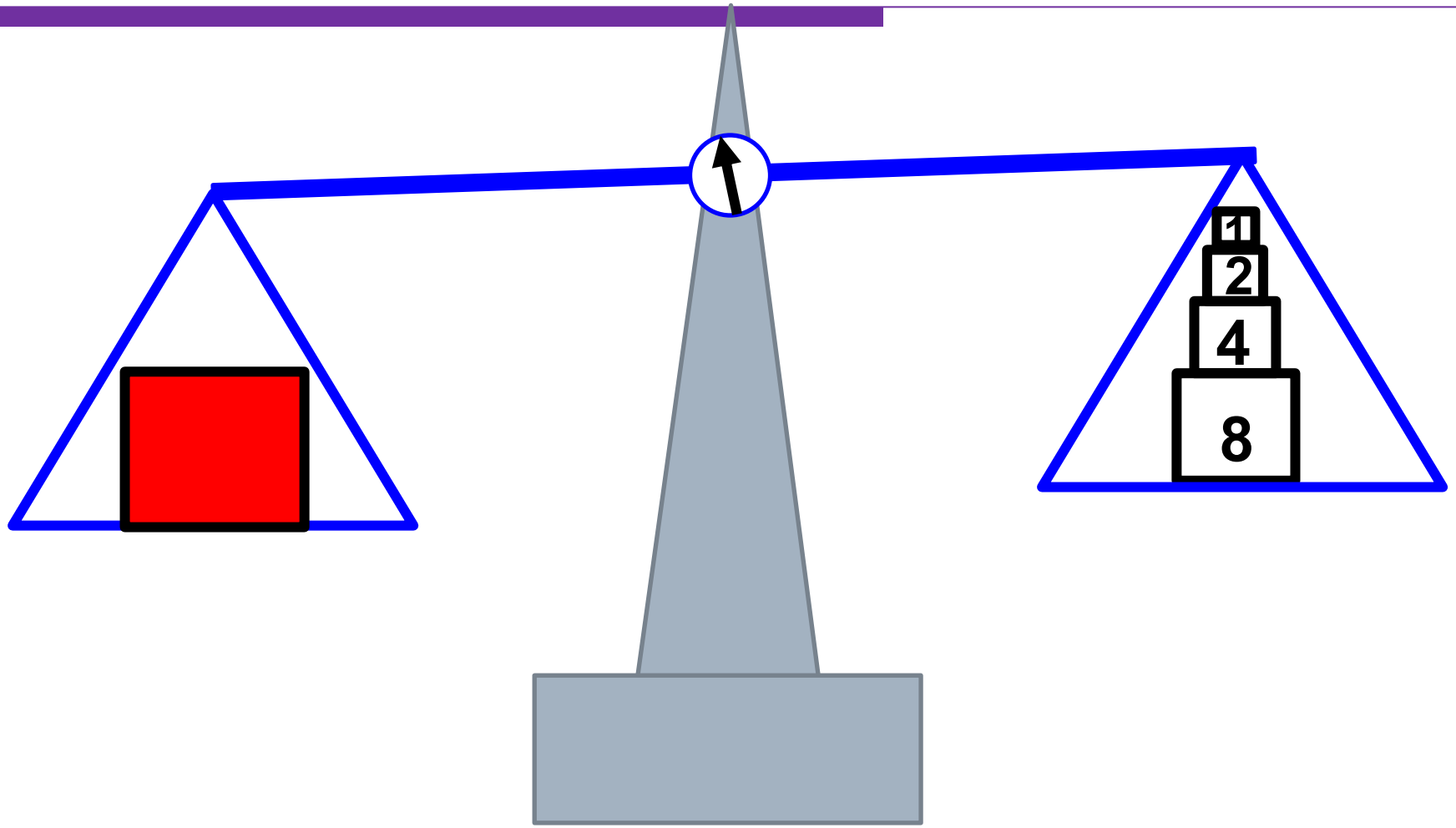


1kg
2kg

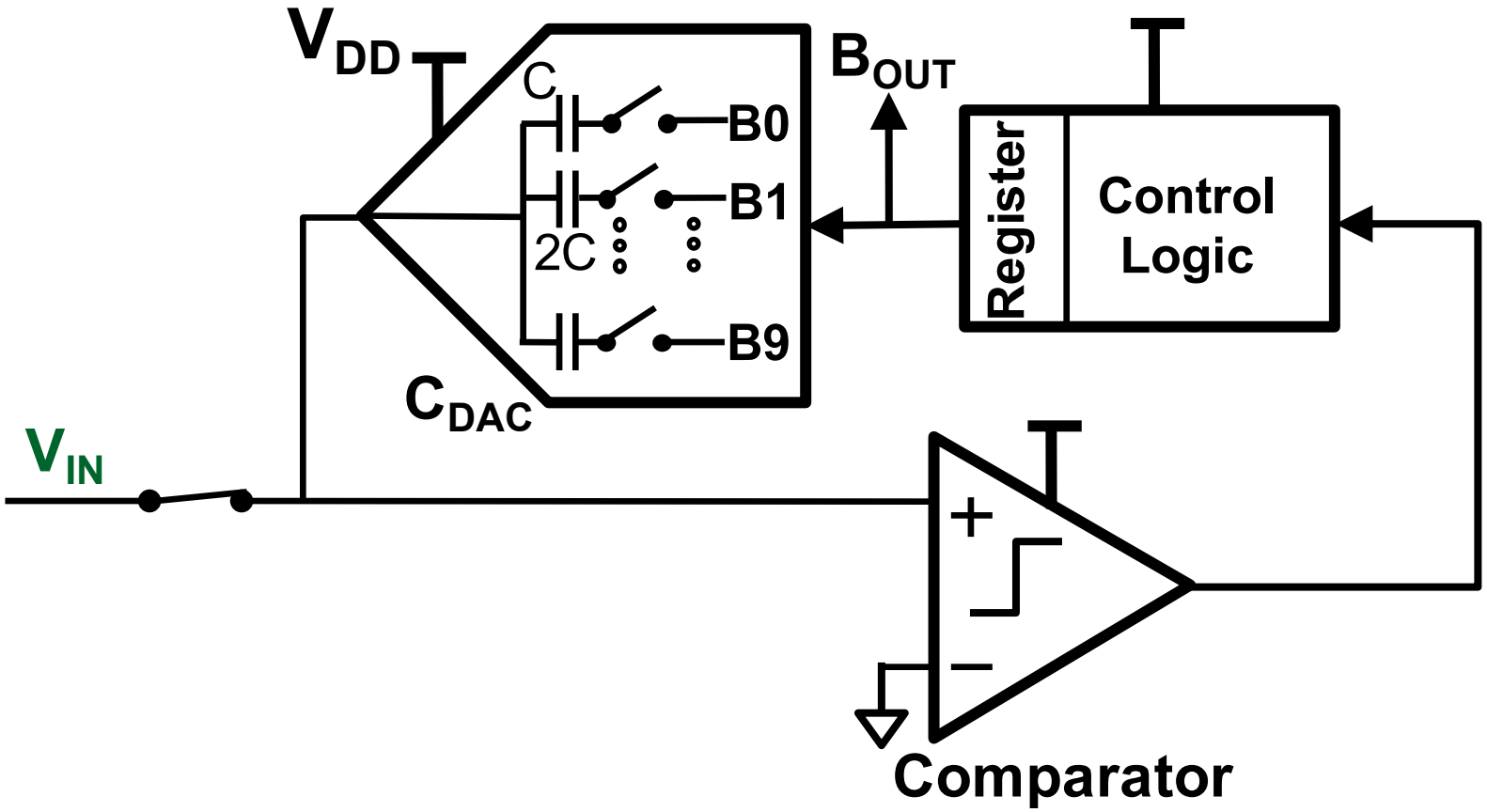
Successive Approximation - Analogy

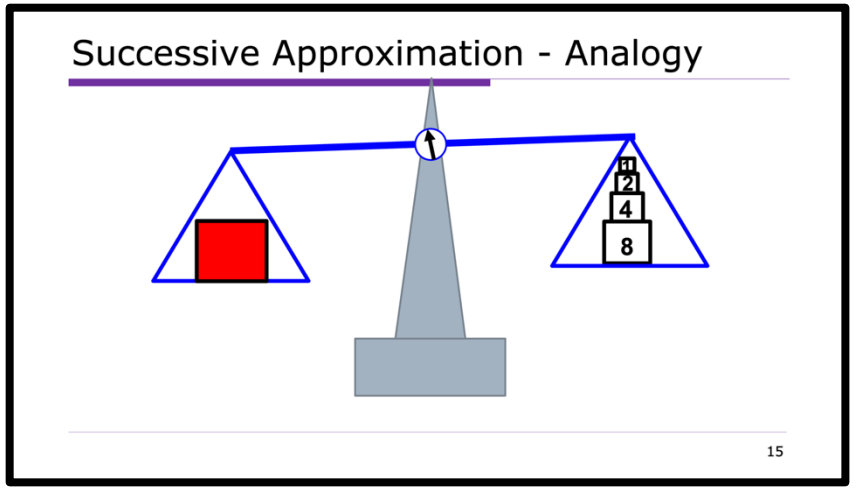


Successive Approximation - Analogy



SAR ADC – Block diagram

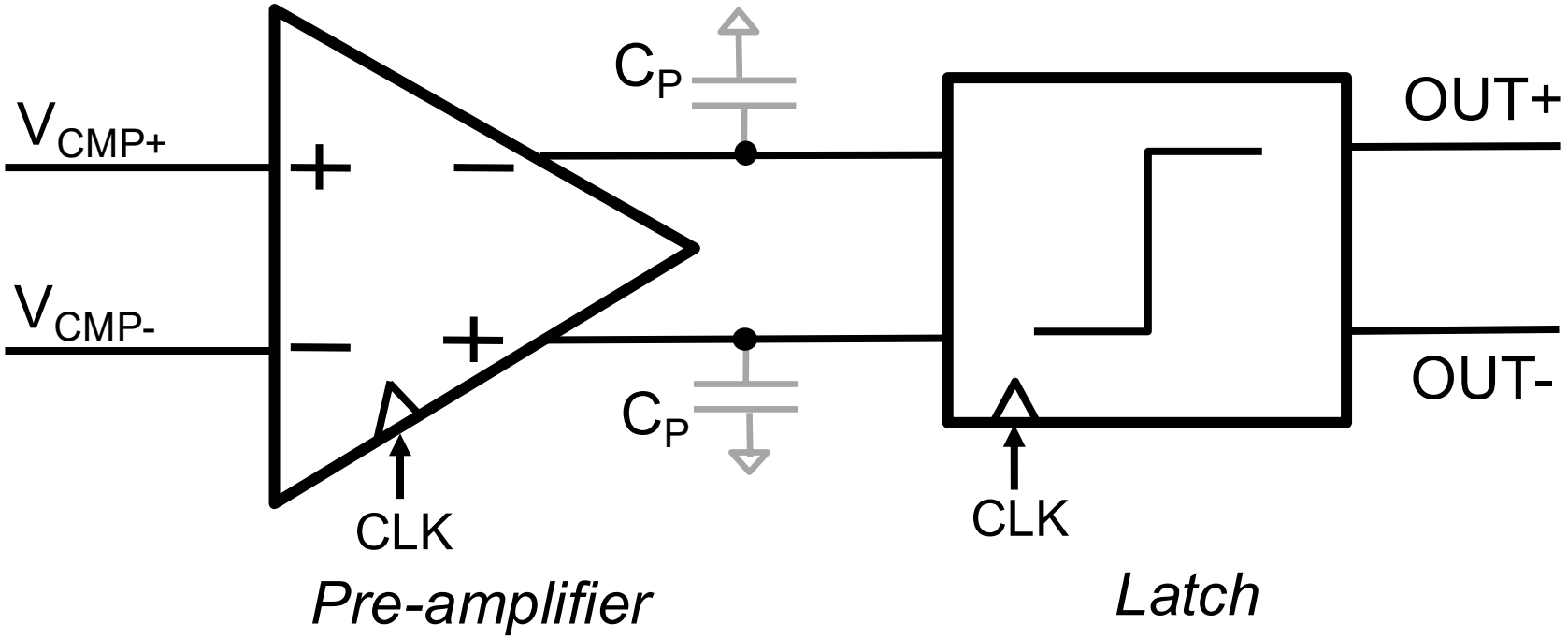




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  - Comparator
  - Design Example
- Wide-bandwidth (GS/s) SAR ADCs
- Auxiliary blocks and its requirements
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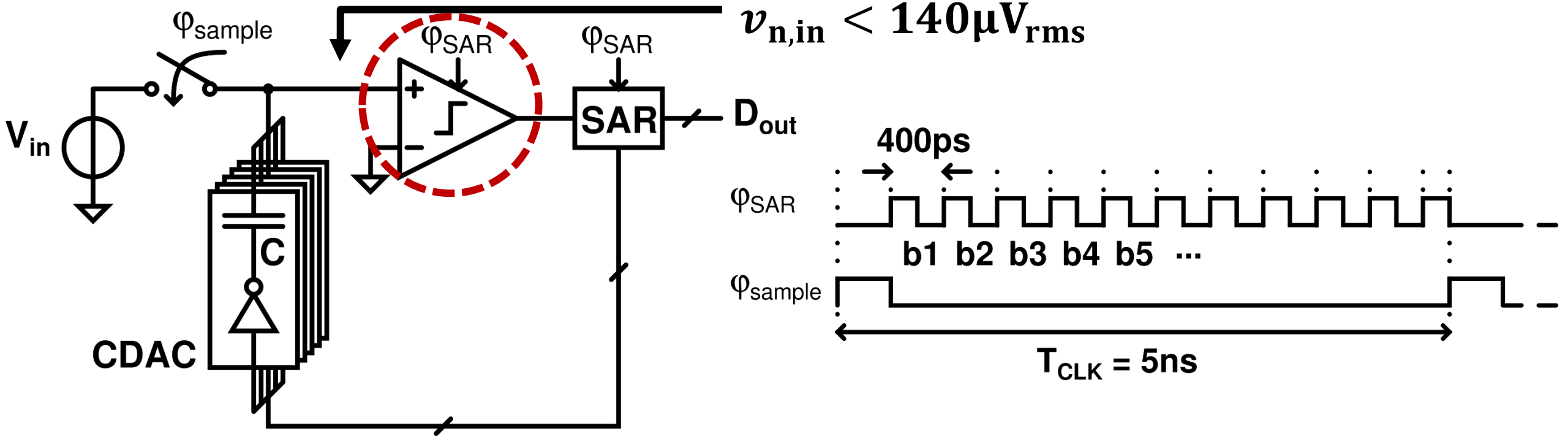
Comparator



- Resolves (small) analog input signal into (large) digital output
- Pre-amplifier is an integrator (C_P sized for noise).
  - Consumes 70-80% of comparator power
- Power consumption scales weakly with technology for a given SNR

17

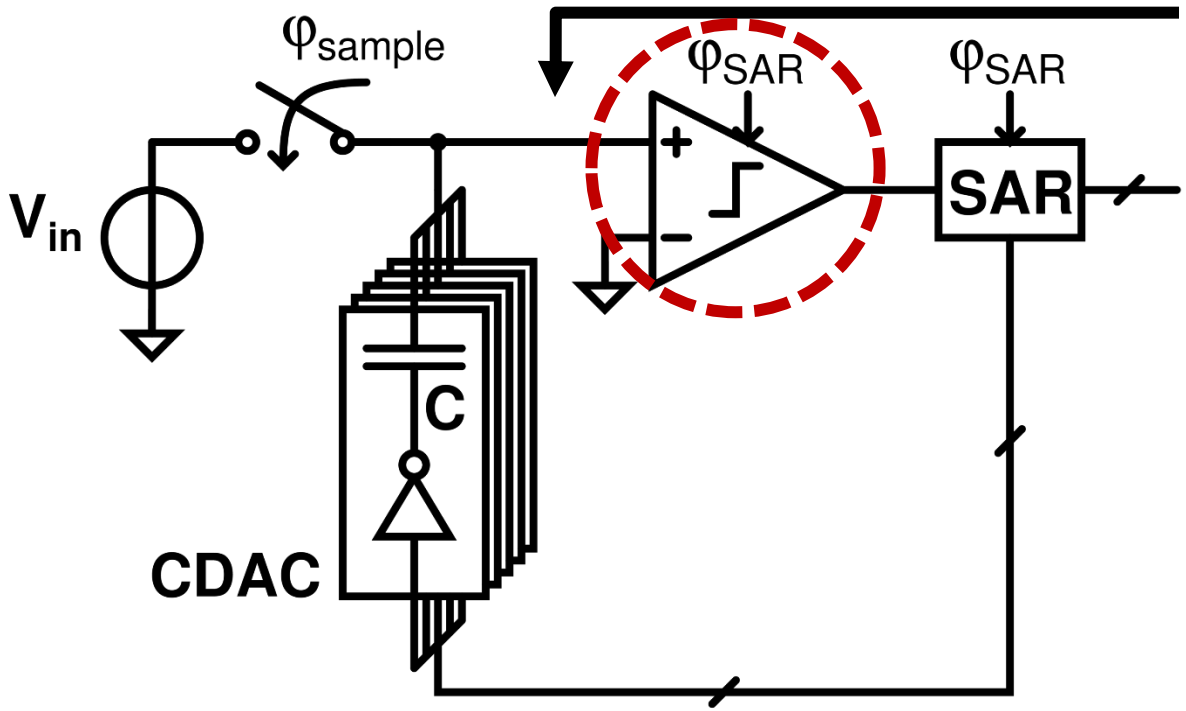
Comparator



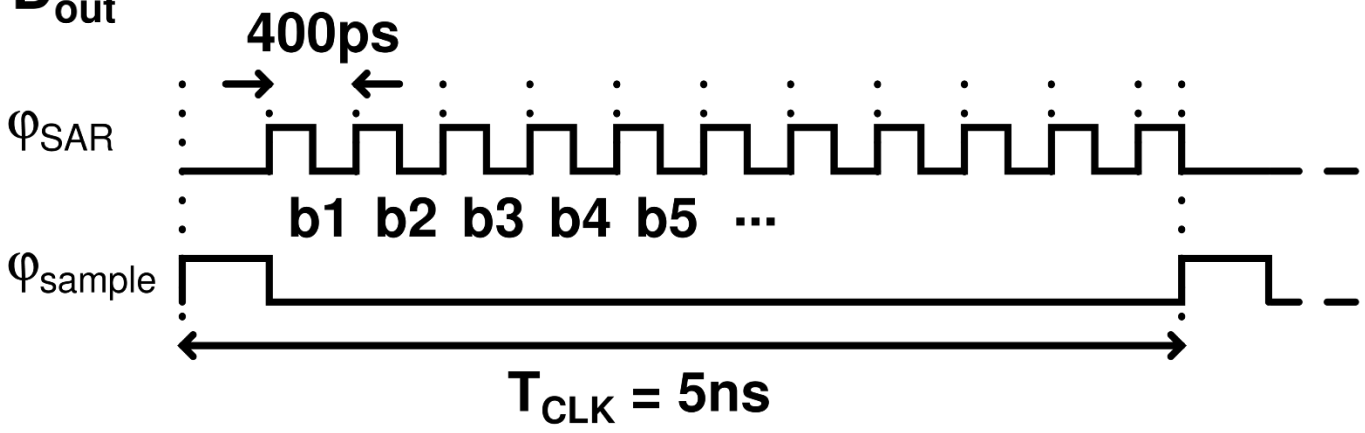
- Example: 12bit 200MS/s $1V_{P-P}$ SAR ADC for 5G/6G applications
- Comparator speed (CLK-OUT delay) limits ADC's maximum achievable throughput

18

Comparator

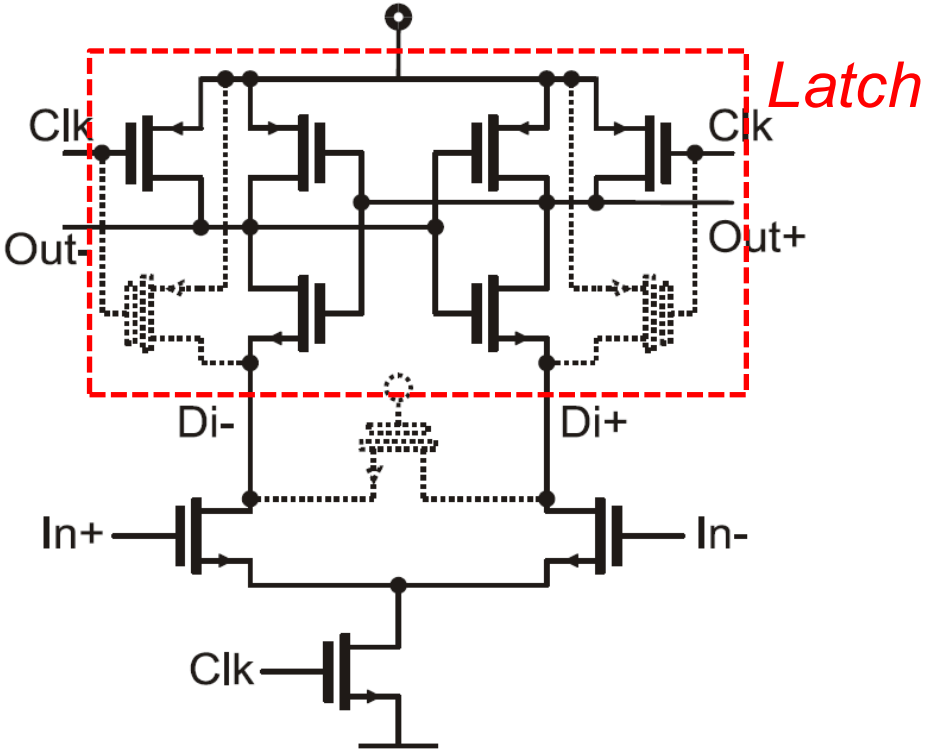


$v_{n,in} < 140\mu V_{rms}$



- Dominates ADC's power (50%) for low-noise applications
- Speed and noise also critical for applications in serial-links, memories
- Desired : Low-power, low-noise, high-speed comparison

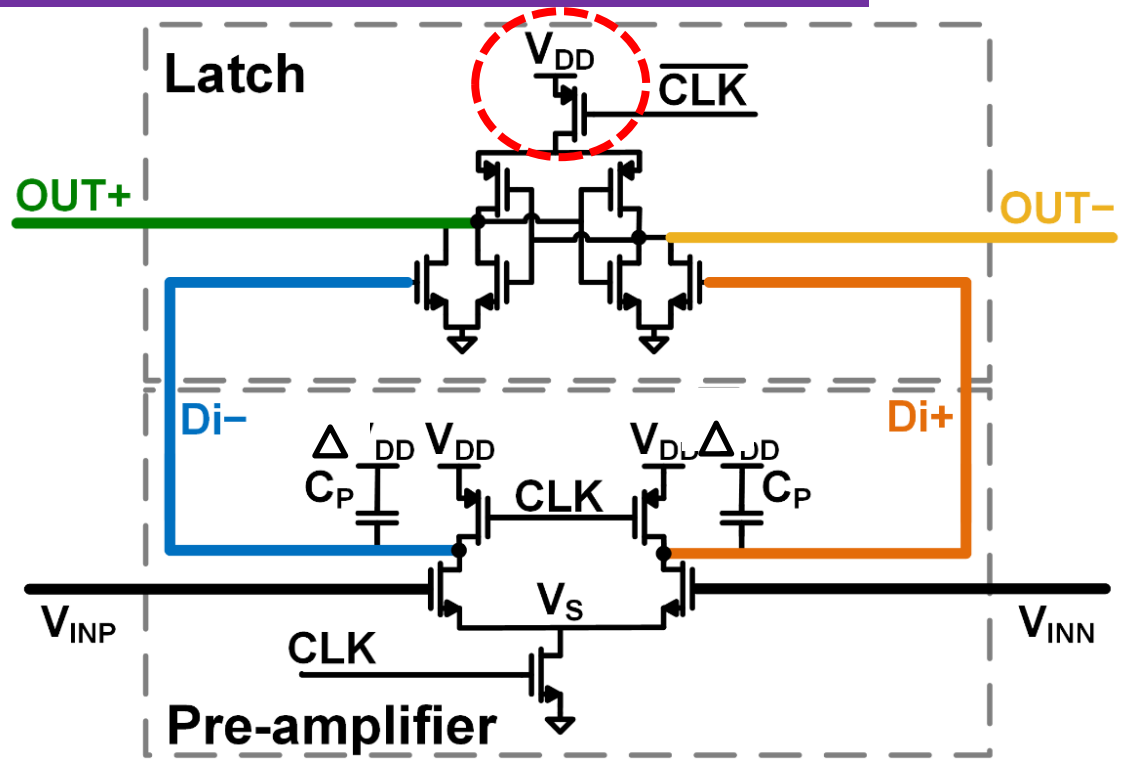
StrongARM Latch



[*Kobayashi, VLSI 1992*]

- ☺ No static power
- ☺ Fast latching owing to strong positive feedback
- ☹ Low voltage headroom
- ☹ No isolation between latch and pre-amplifier

Schinkel



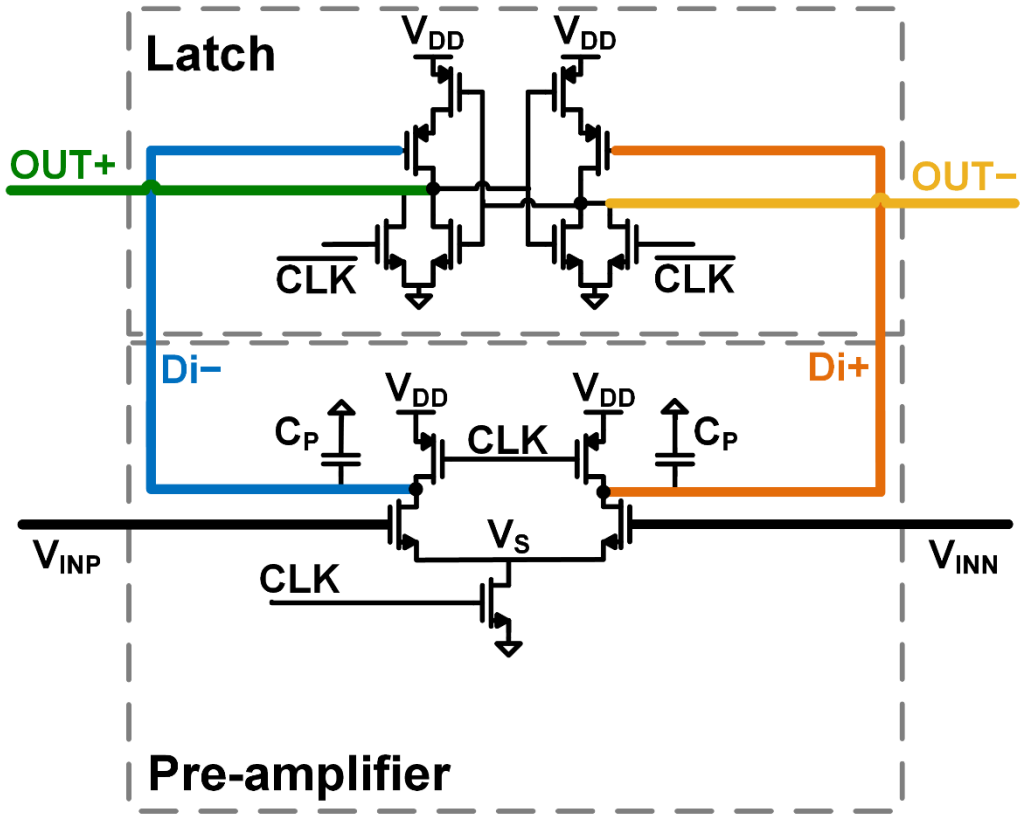
[*Schinkel, ISSCC 2007*]

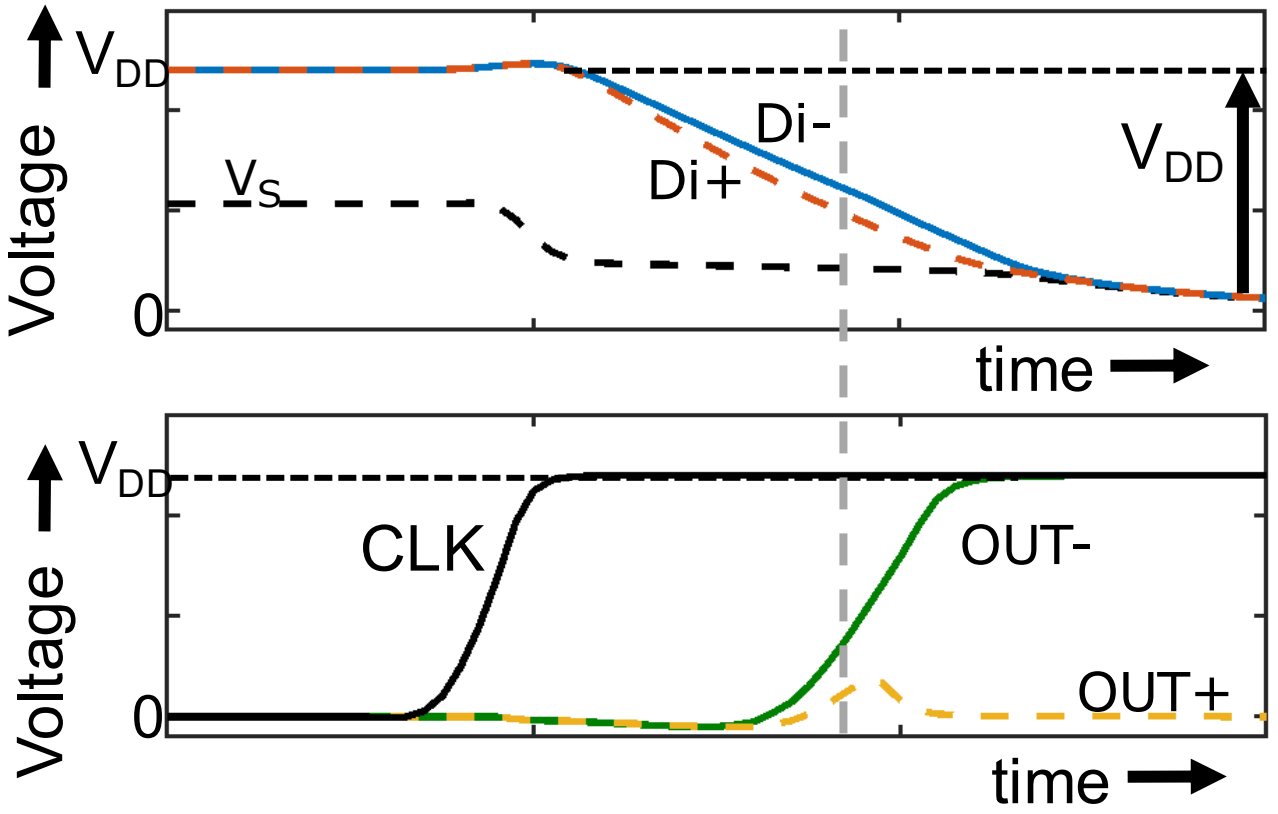
☺ Isolation : Separate pre-amplifier and latch

☺ Wide input common-mode range

☹ Latch “ON” with pre-amplifier, non-optimal for power-noise trade-off

Elzakker's Comparator



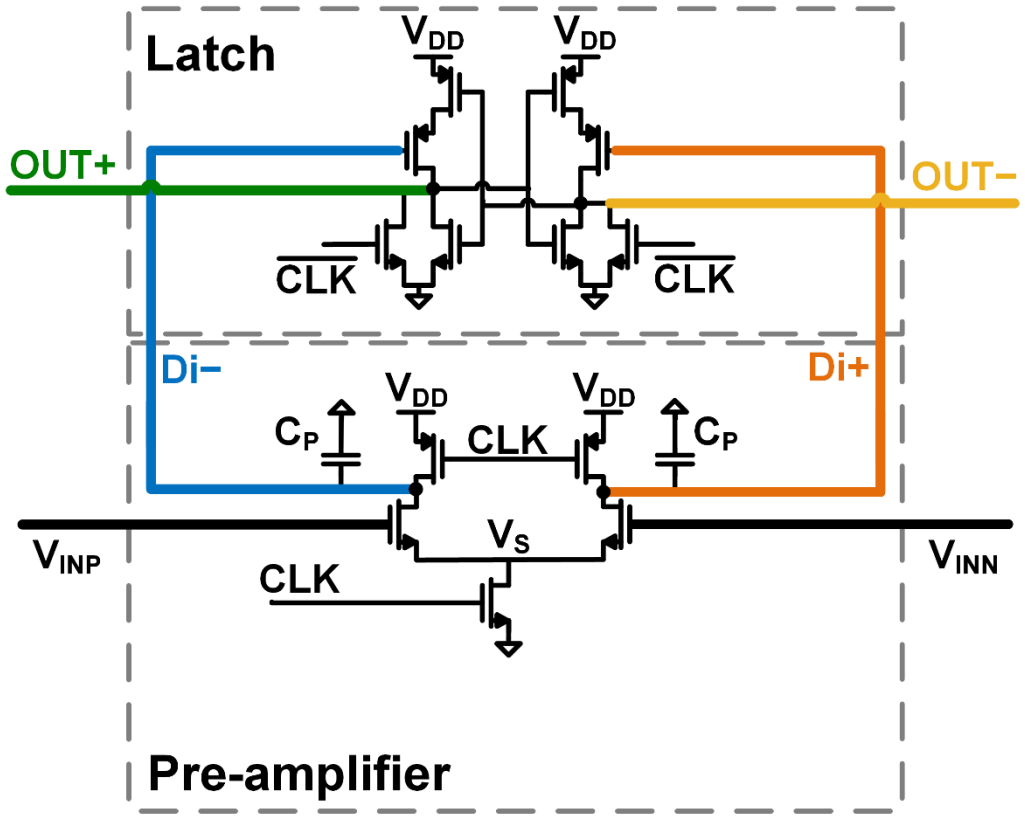


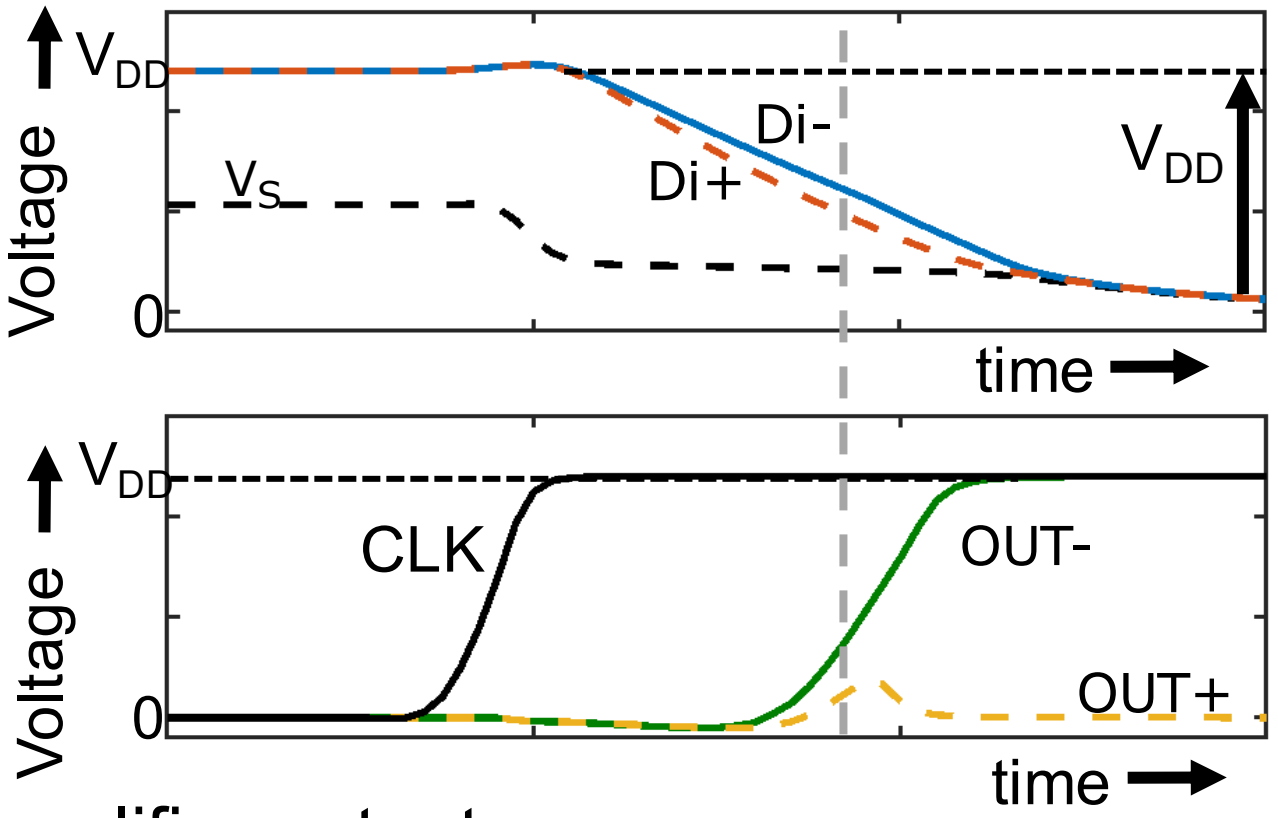
☺ Isolation : Pre-amplifier and latch

[*Elzakker, ISSCC 2008*]

☺ Pre-amplifier generates "gain" before latch turns "ON"

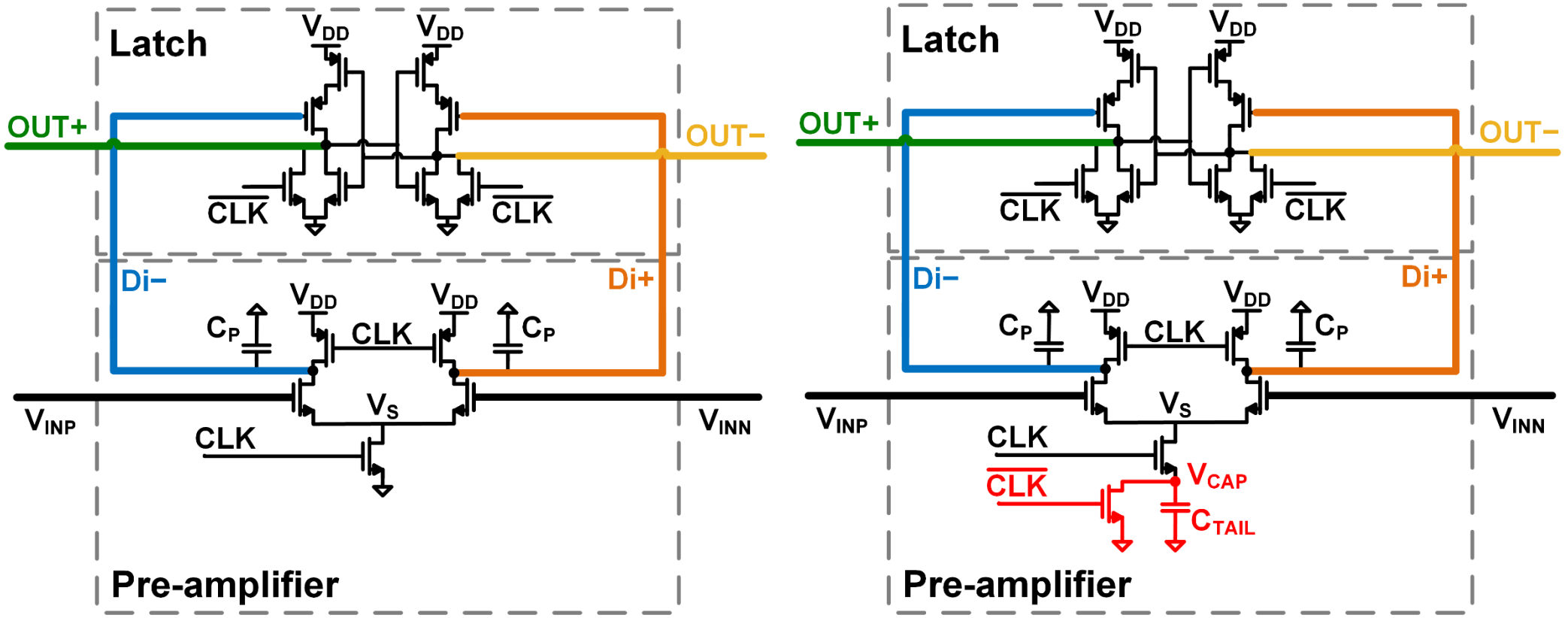
Elzakker's Comparator





- ☹ Complete discharge of pre-amplifier output
- ☹ $E_{diss,pre\text{-}amplifier} = (2 \cdot C_P \cdot V_{DD} \cdot V_{DD}) \approx 80\%$ of comparator's energy

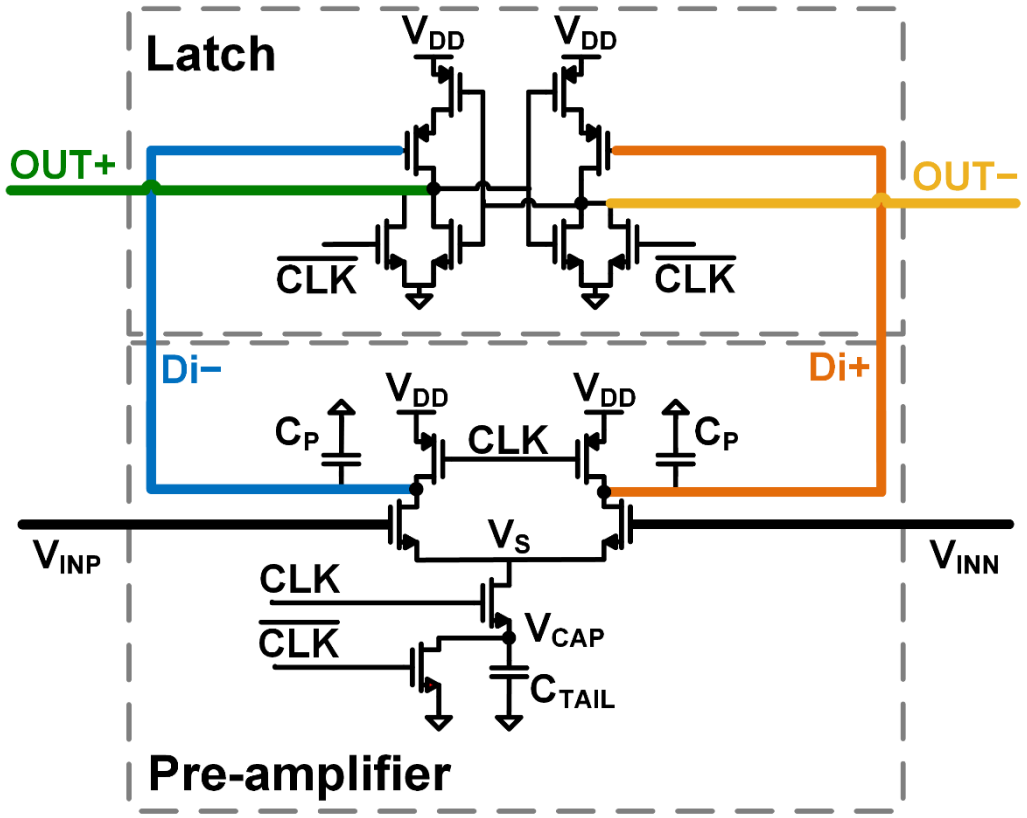
Dynamic Bias Comparator

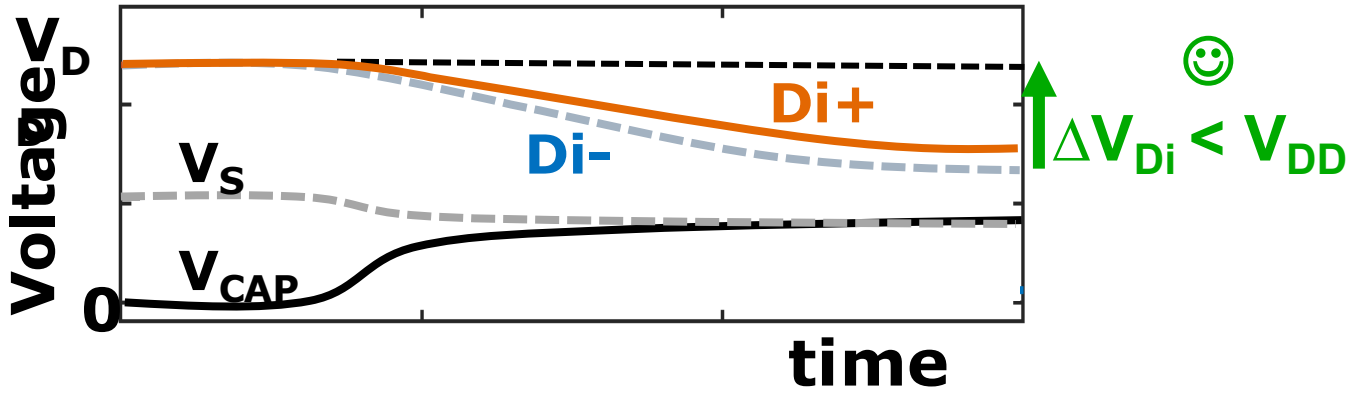


- Switched capacitor (C_{TAIL}) in pre-amplifier's tail
- Reset: V_{CAP} = 0V

[*Bindra, JSSC 2018*]

Dynamic Bias Comparator

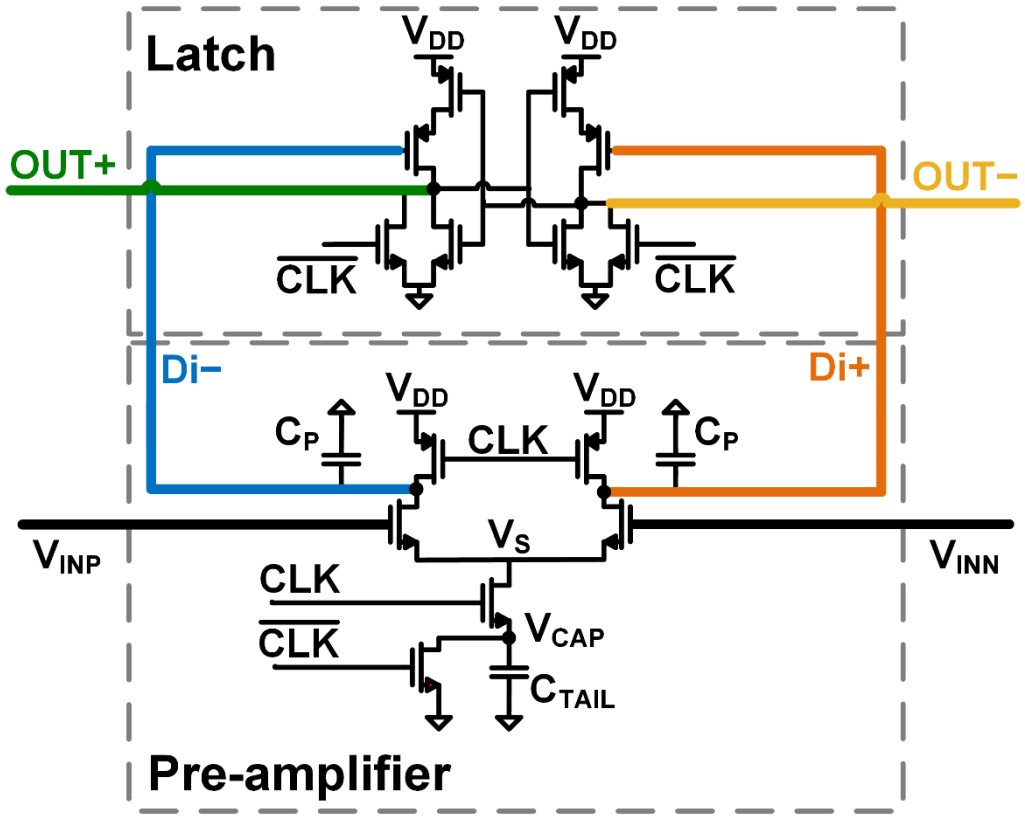


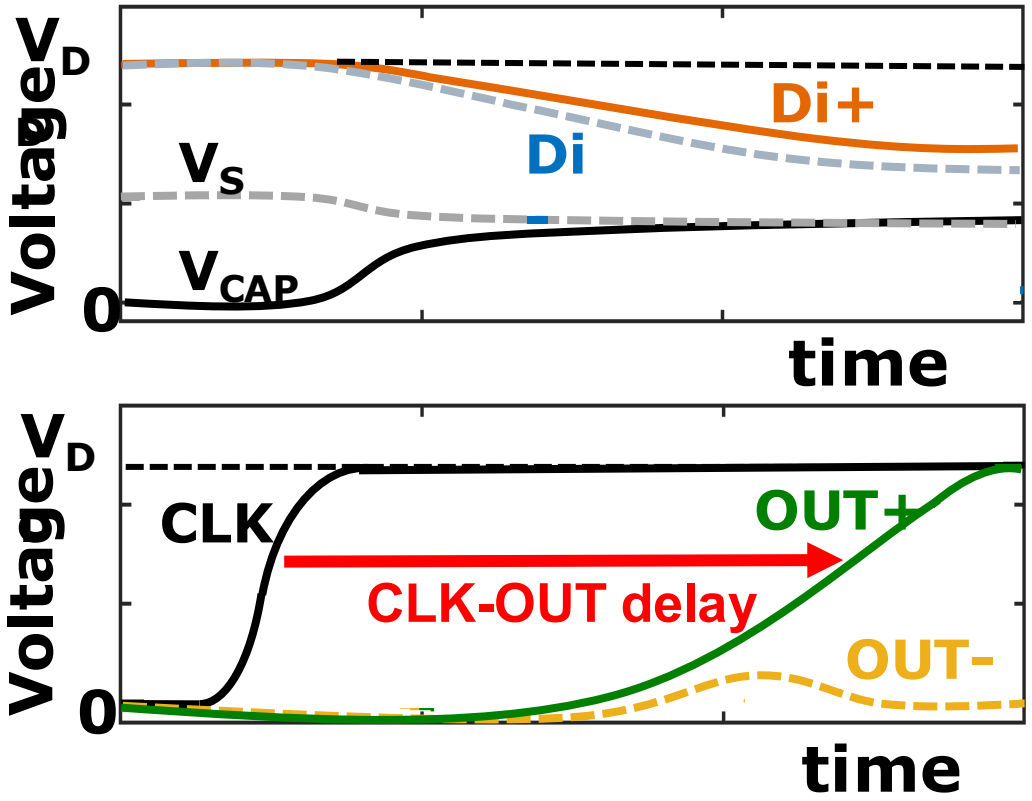


- ☺ Amplification: Input diff pair dynamic biased
- ☺ Prevents complete discharge of pre-amplifier

25

Dynamic Bias Comparator

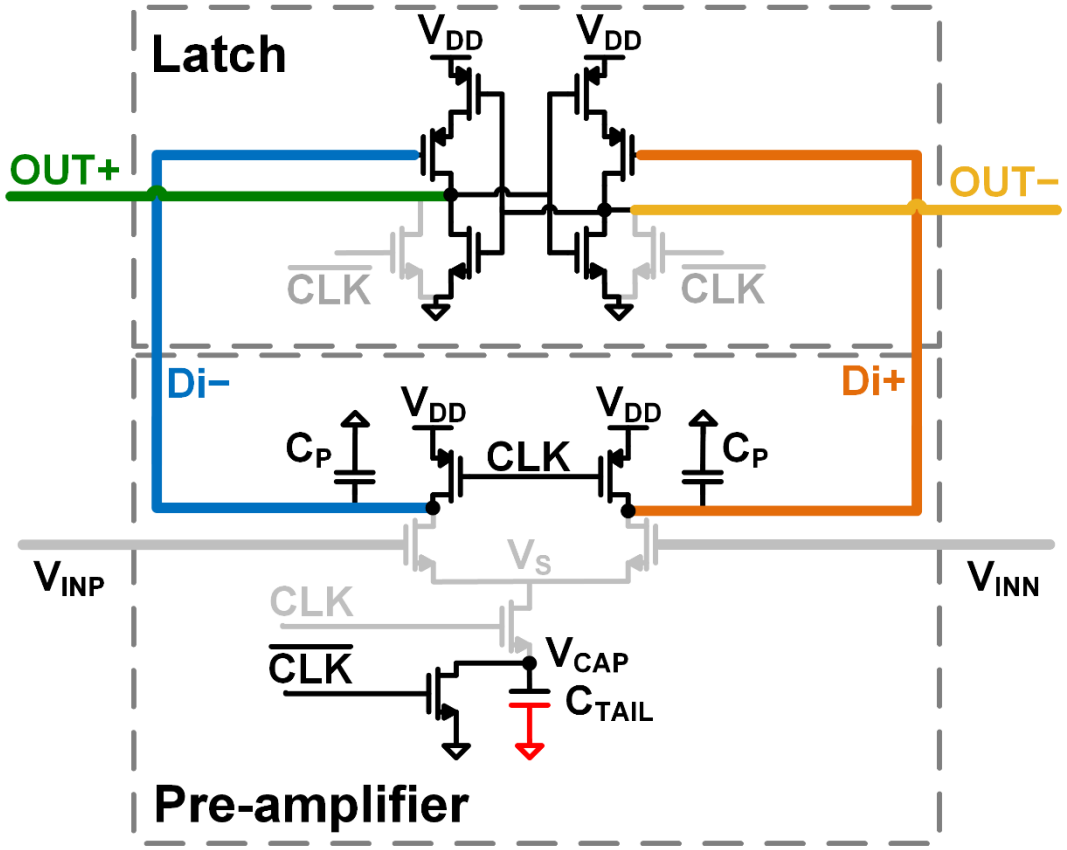




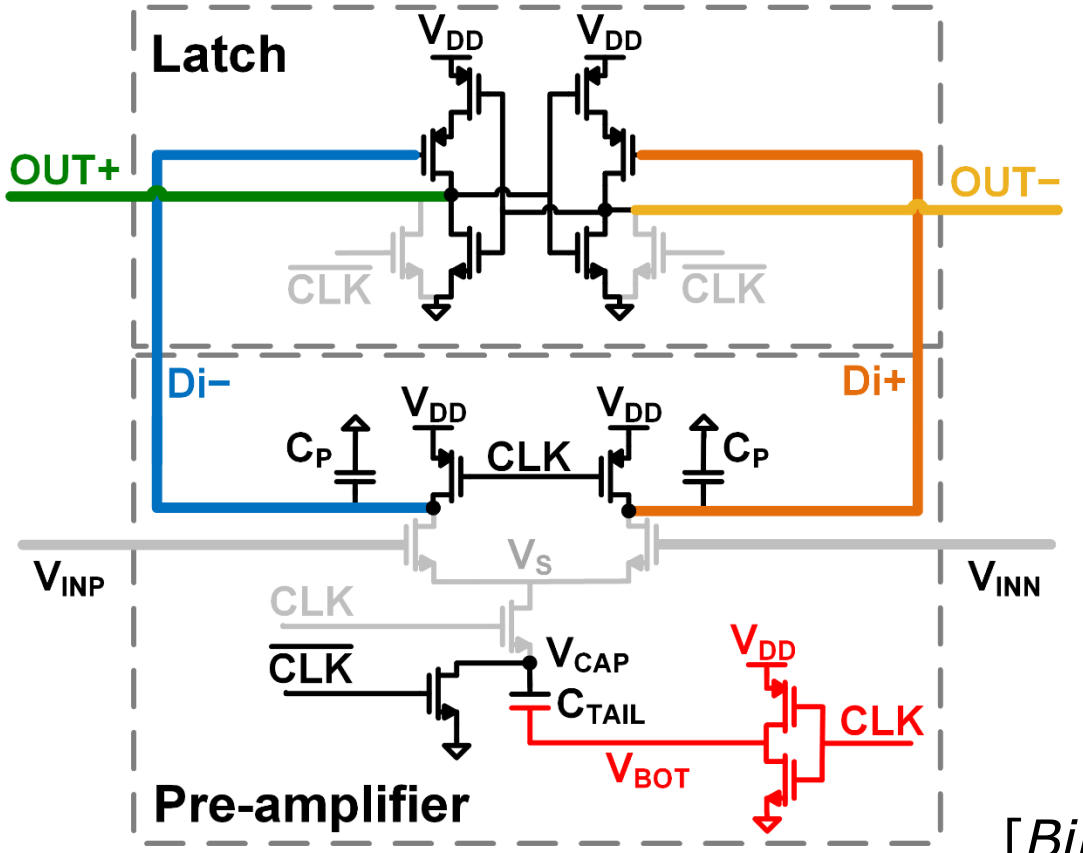
☹ V_{CAP} rises for dynamic bias ➡ reducing speed

26

Dynamic Bias Pre-amplifier w/ charge-pump



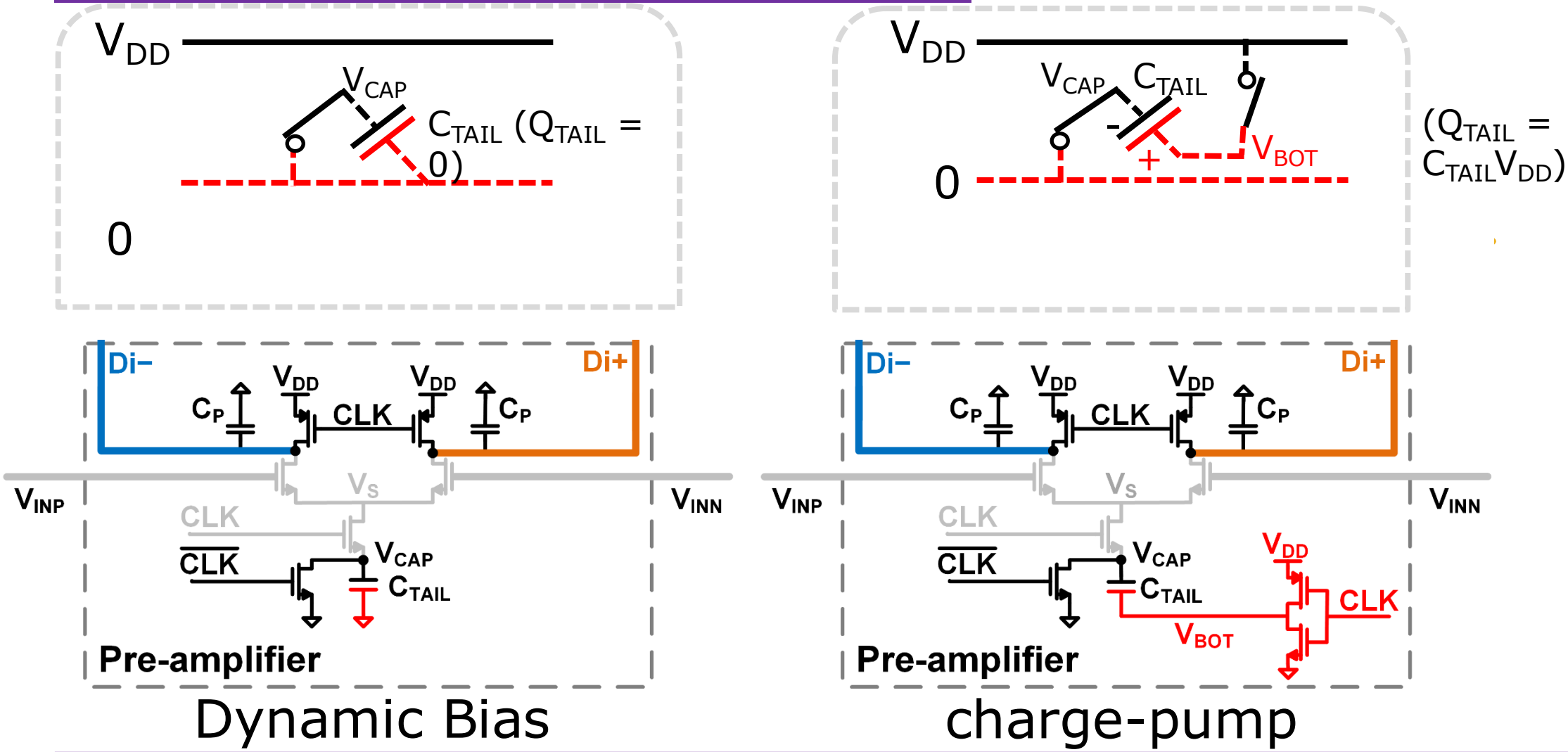
Dynamic Bias



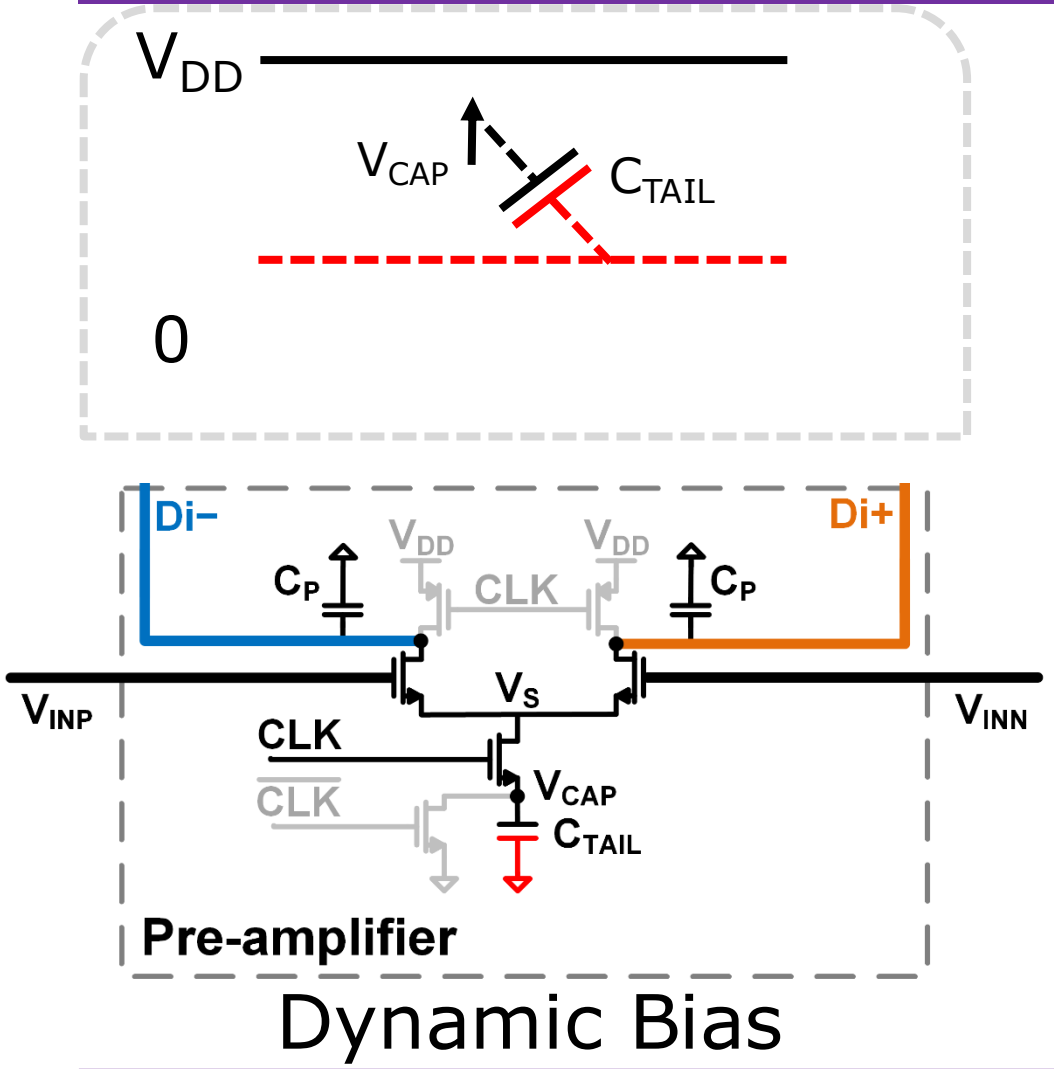
charge-pump

[*Bindra, ISSCC 2022*]

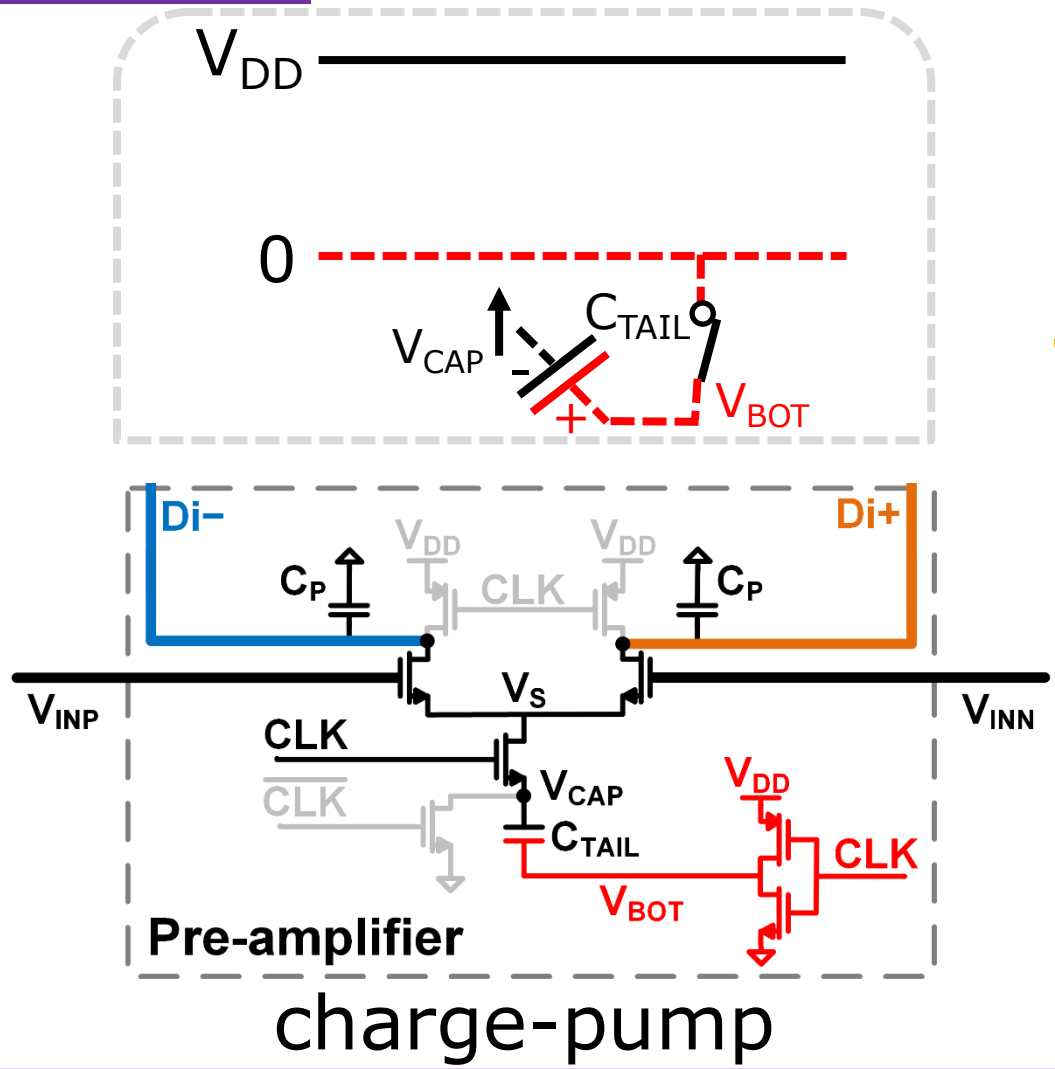
Dynamic Bias Pre-amplifier w/ charge-pump



Dynamic Bias Pre-amplifier w/ charge-pump



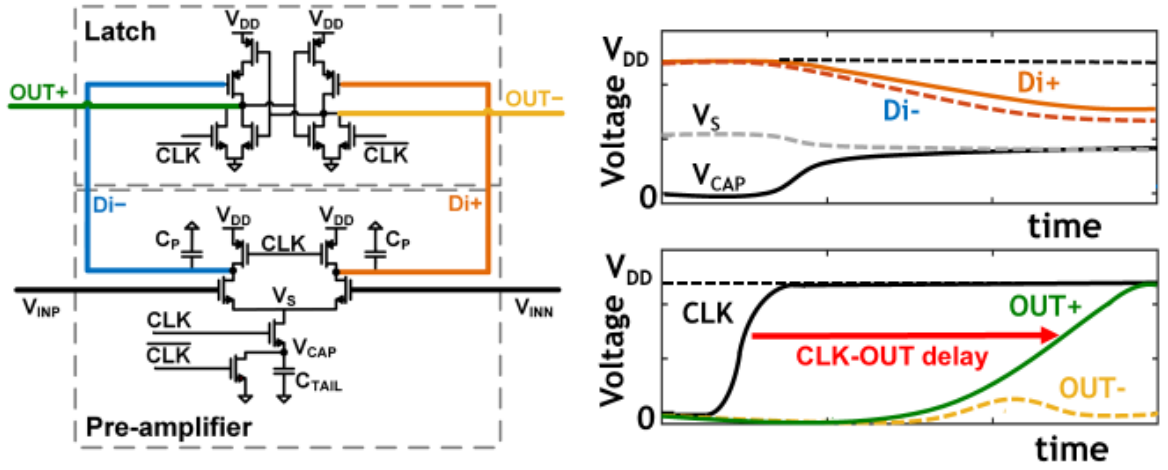
Dynamic Bias



charge-pump

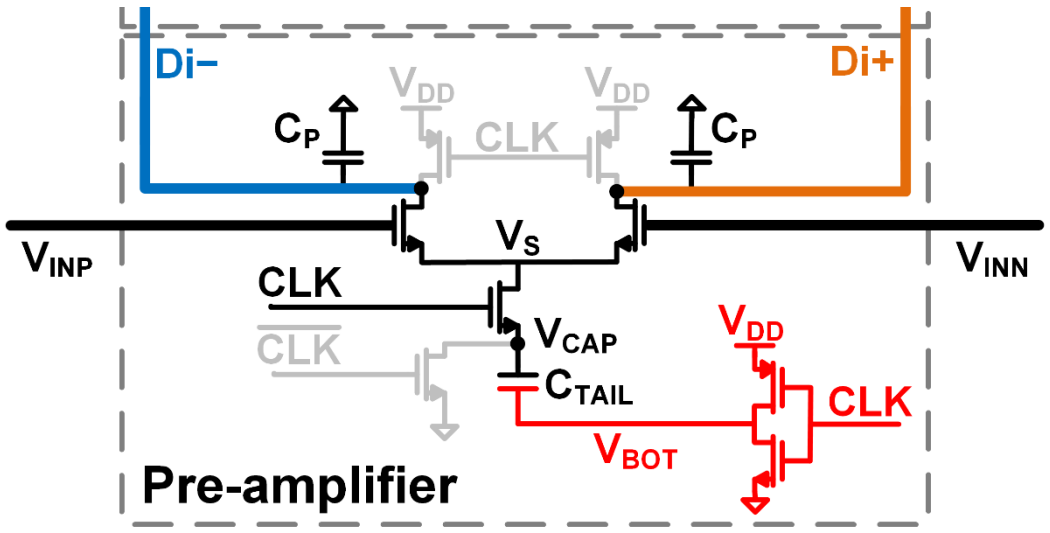
Dynamic Bias Pre-amplifier w/ charge-pump

Dynamic Bias Comparator



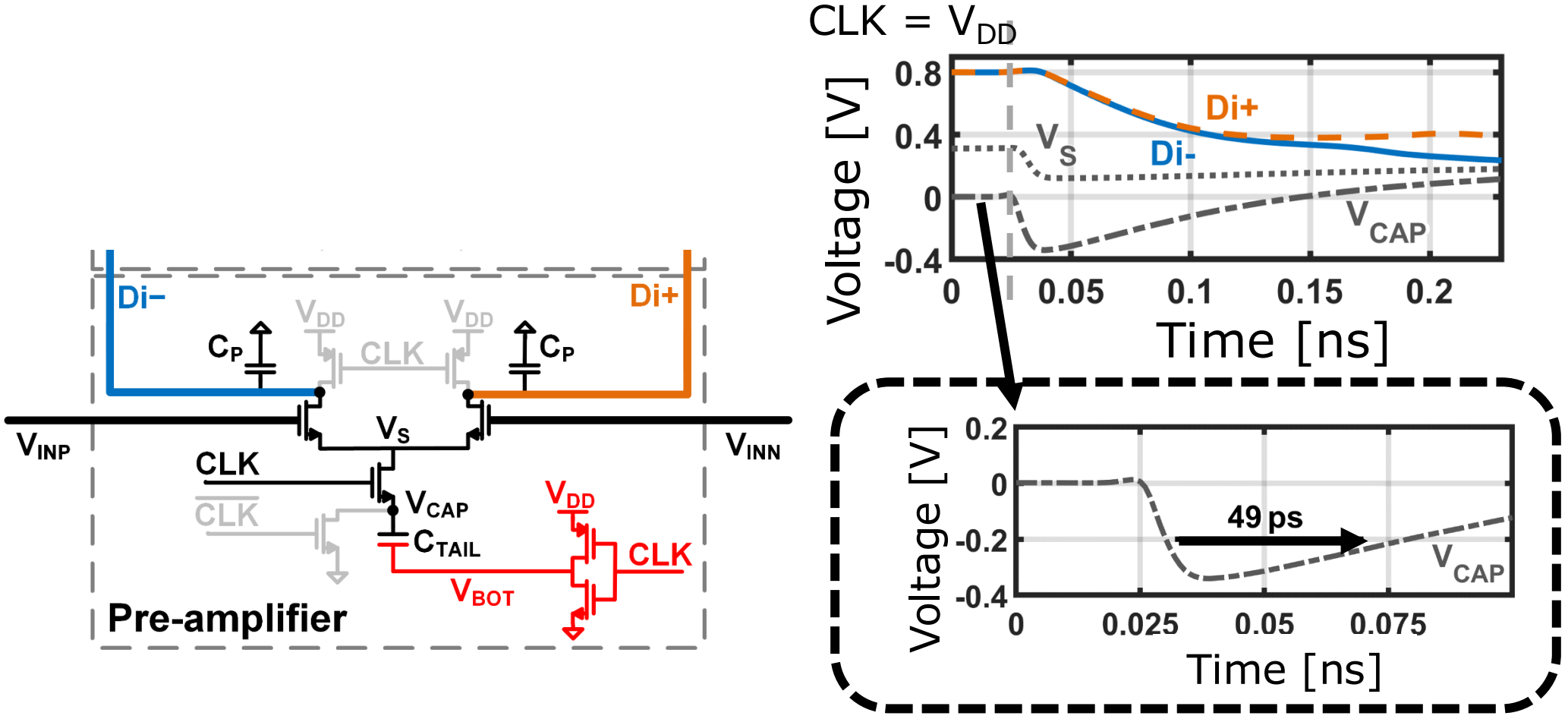
☹ V_{CAP} rises for dynamic bias ➡ reducing speed

charge-pump



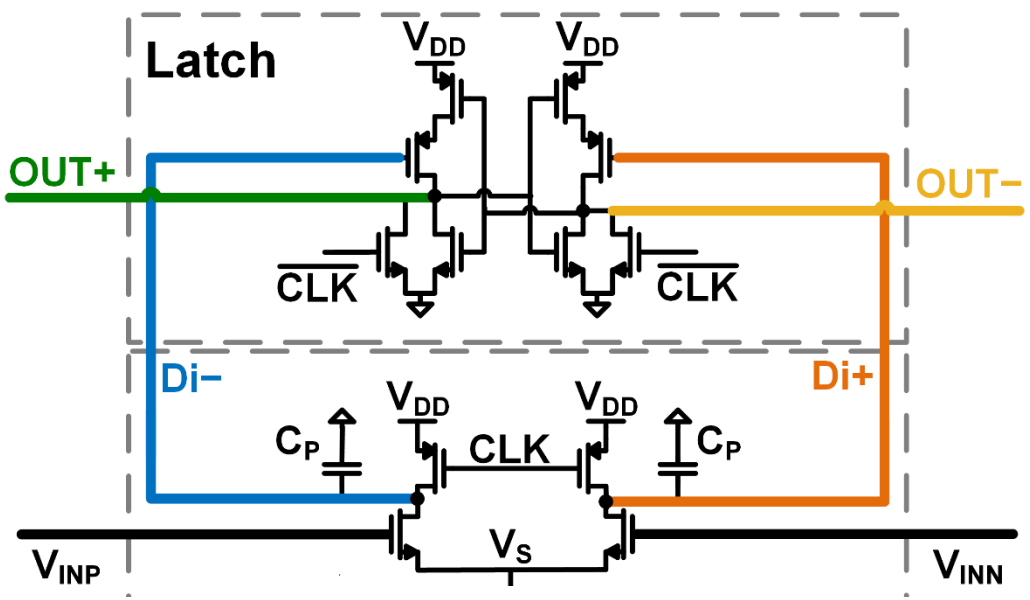
☺ CLK = V_{DD} ➡ Tail charge-pump decreases V_{CAP} below 0V

Dynamic Bias Pre-amplifier w/ charge-pump

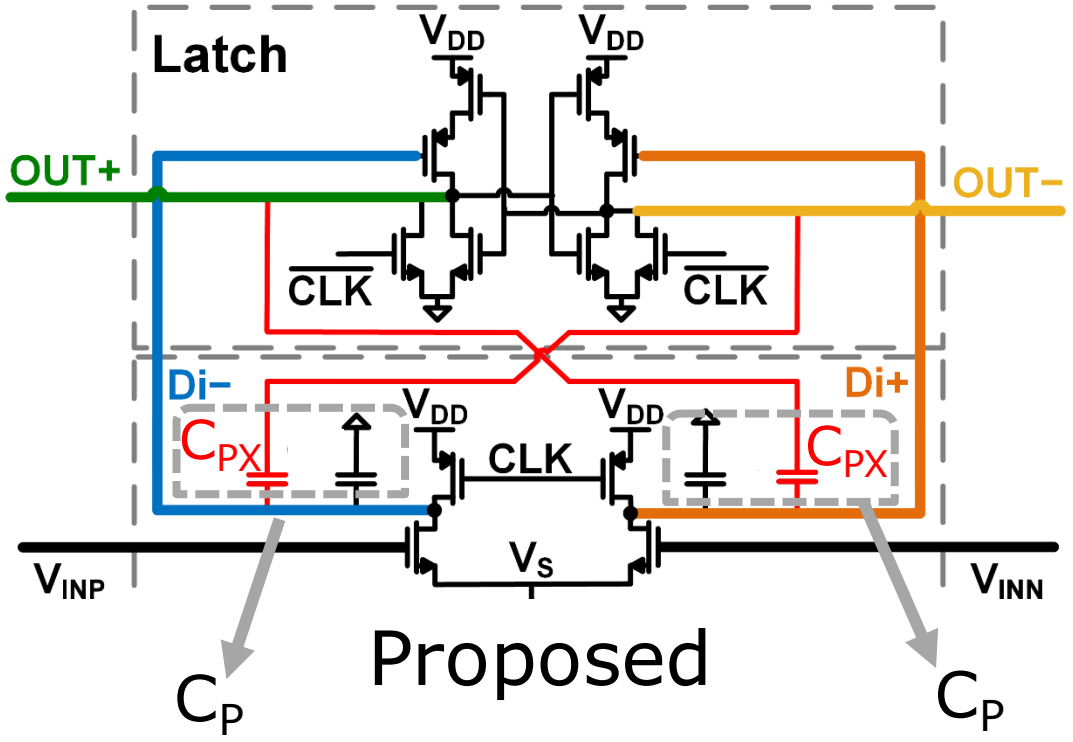


- Charge transfer from C_P to C_{TAIL} ➡ $0 > V_{CAP} > -V_{DD}$

Cap Neutralization across Latch



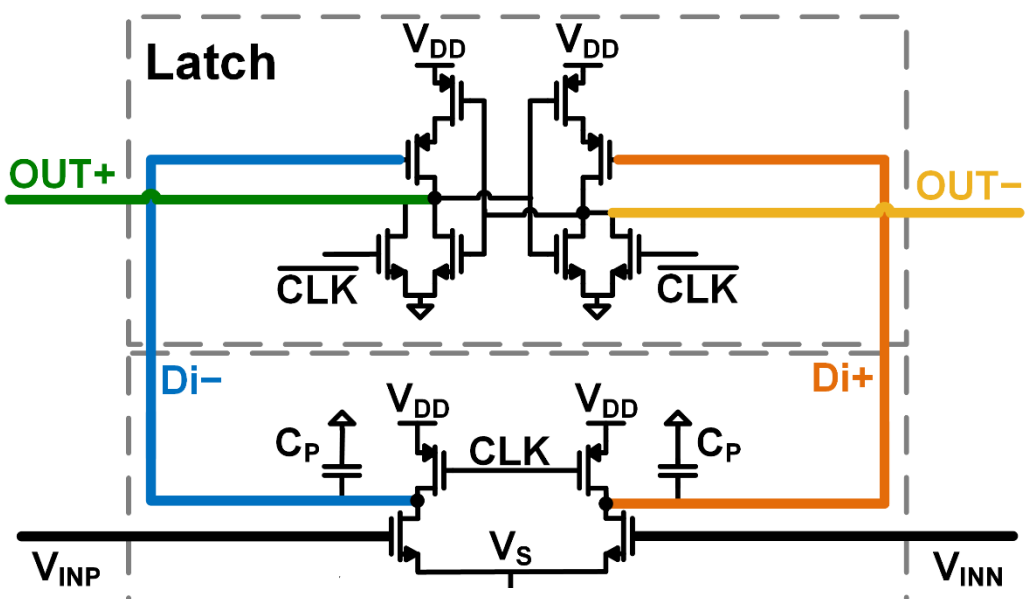
Dynamic Bias



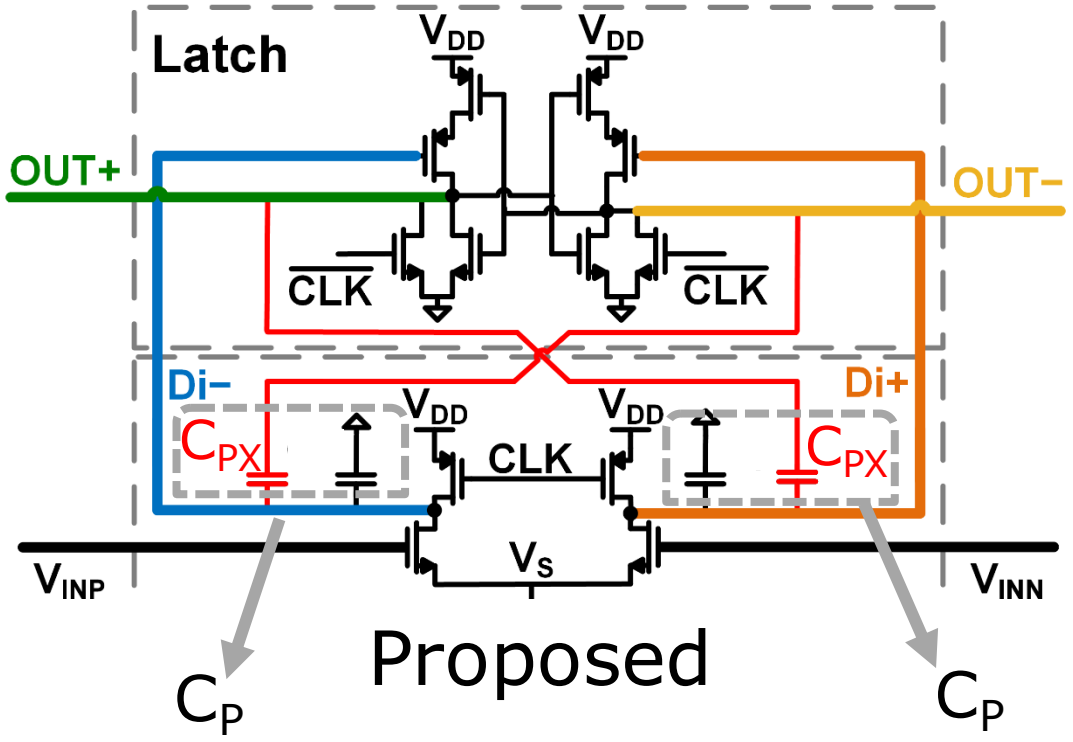
Proposed

- No (explicit) feedback from latch to the pre-amplifier
- C_{PX} constitute feedback from latch to the pre-amplifier (Neutralization)

Cap Neutralization across Latch



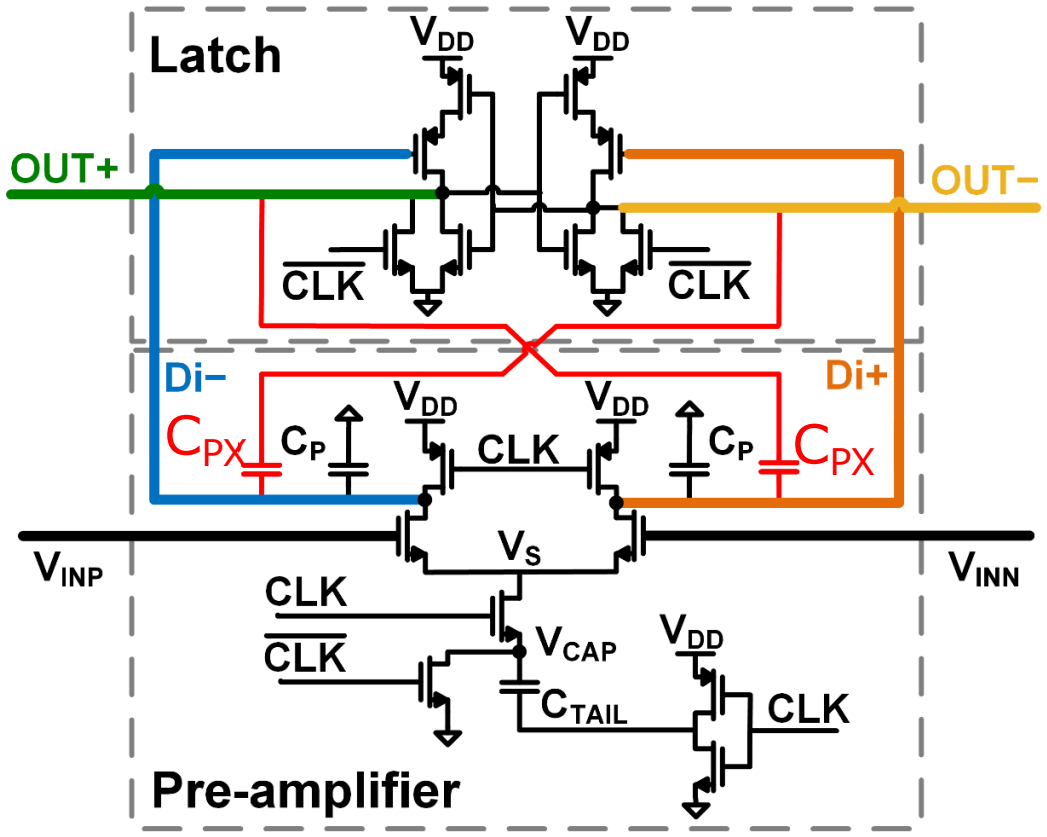
Dynamic Bias

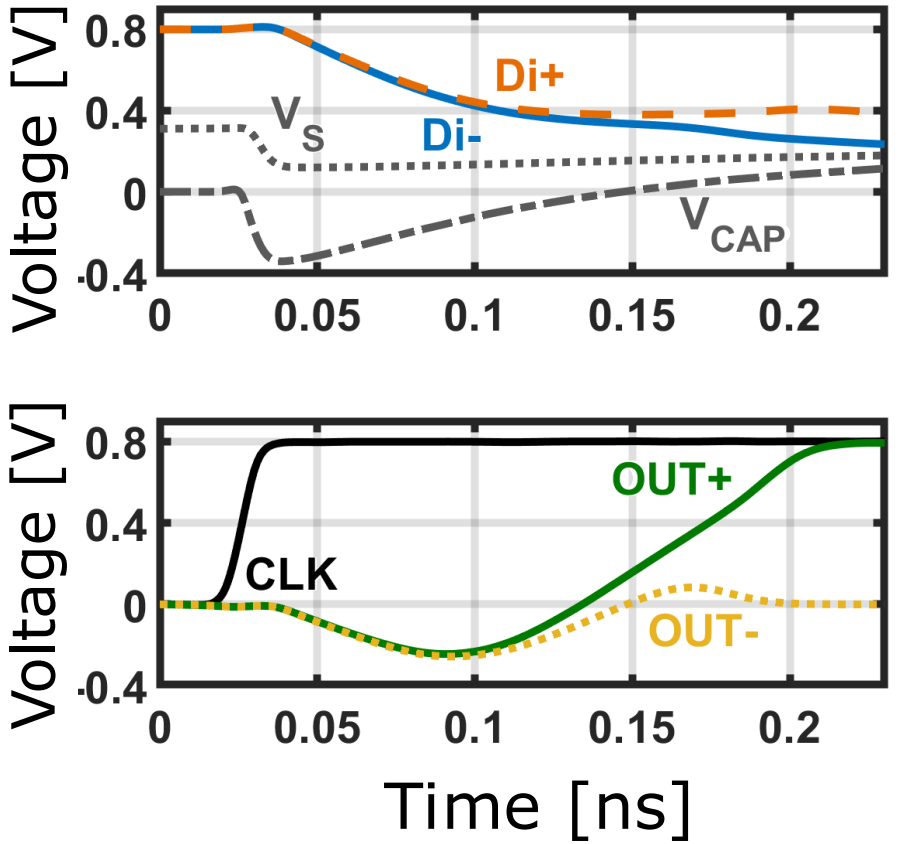


Proposed

- C_{PX} is part of the (noise) integration capacitance (40% of C_P)
- C_{PX} is reset via latch (CLK = ‘0’)

Cap Neutralization across Latch

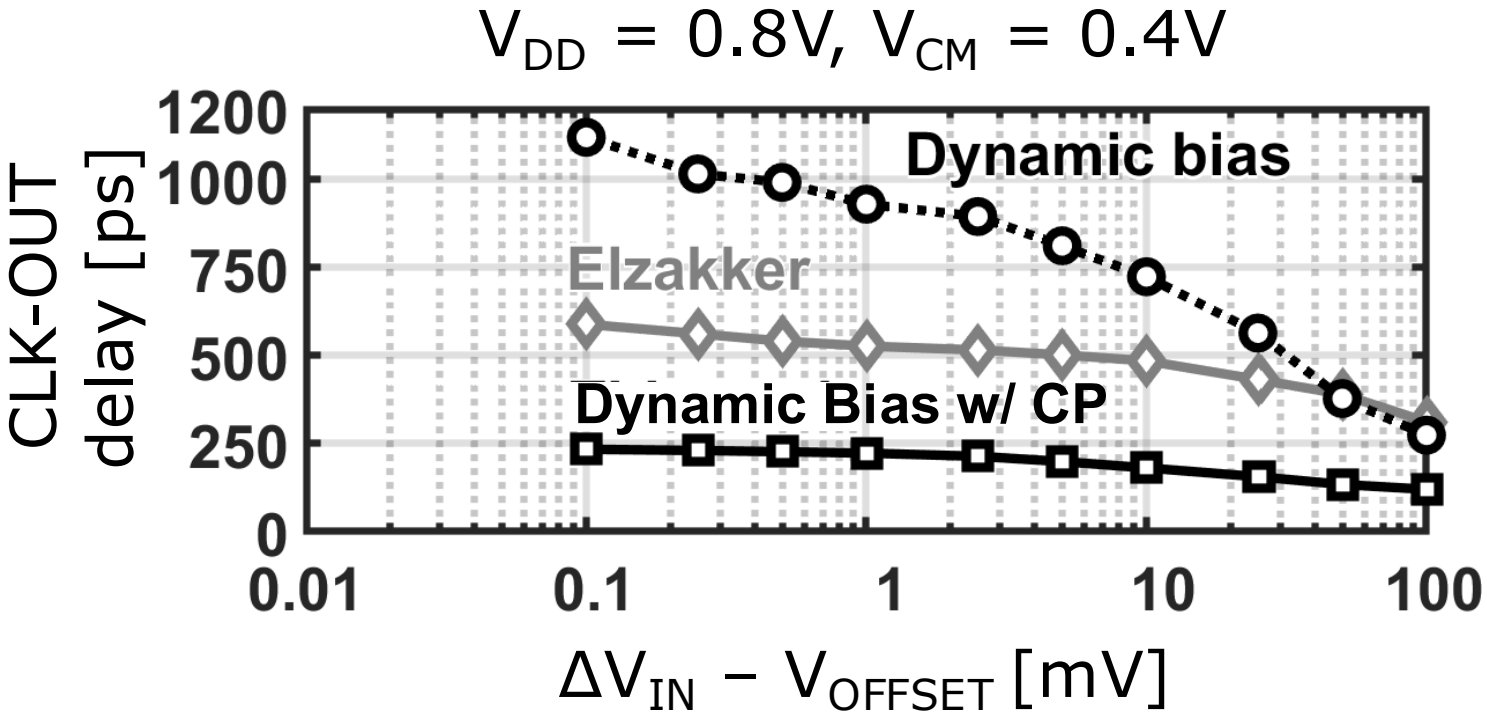




- Latch output couples back through capacitors C_{PX}

☺ Increases the latch regeneration speed

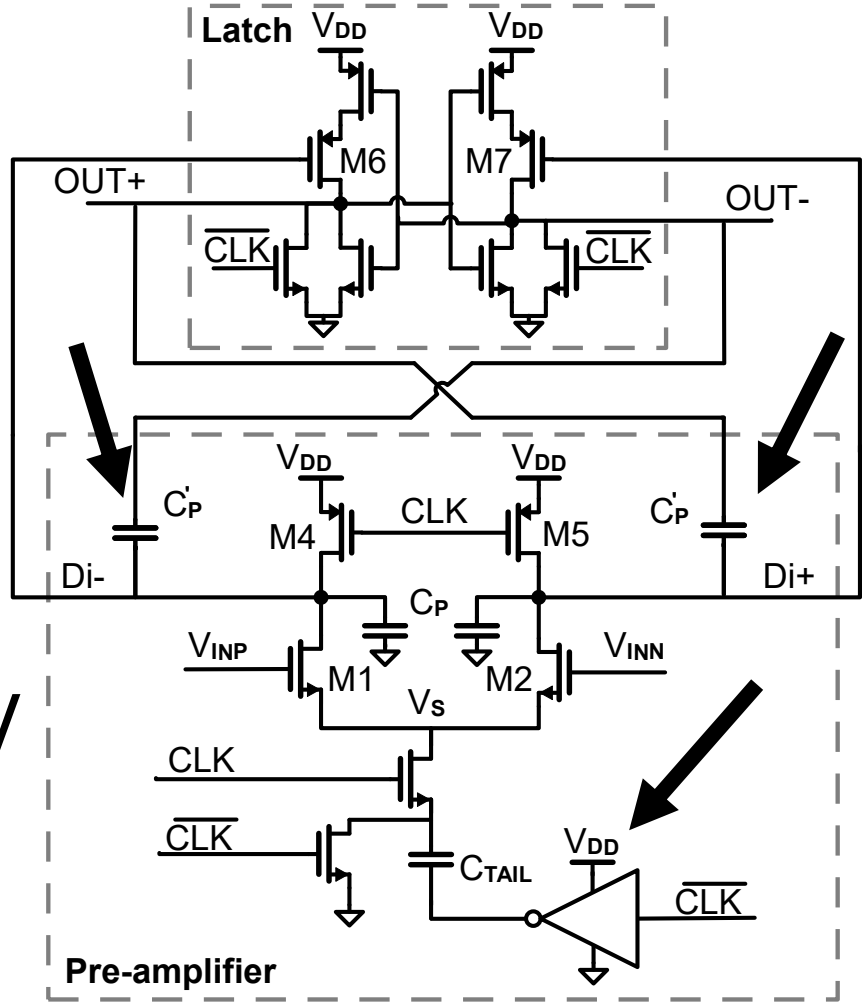
Simulation: CLK-OUT delay v/s ΔV_{IN}



- Compared for same input referred noise 0.2mV, all in 22nm FDSOI
- ☺ Smallest CLK-OUT delay for entire ΔV_{IN} range

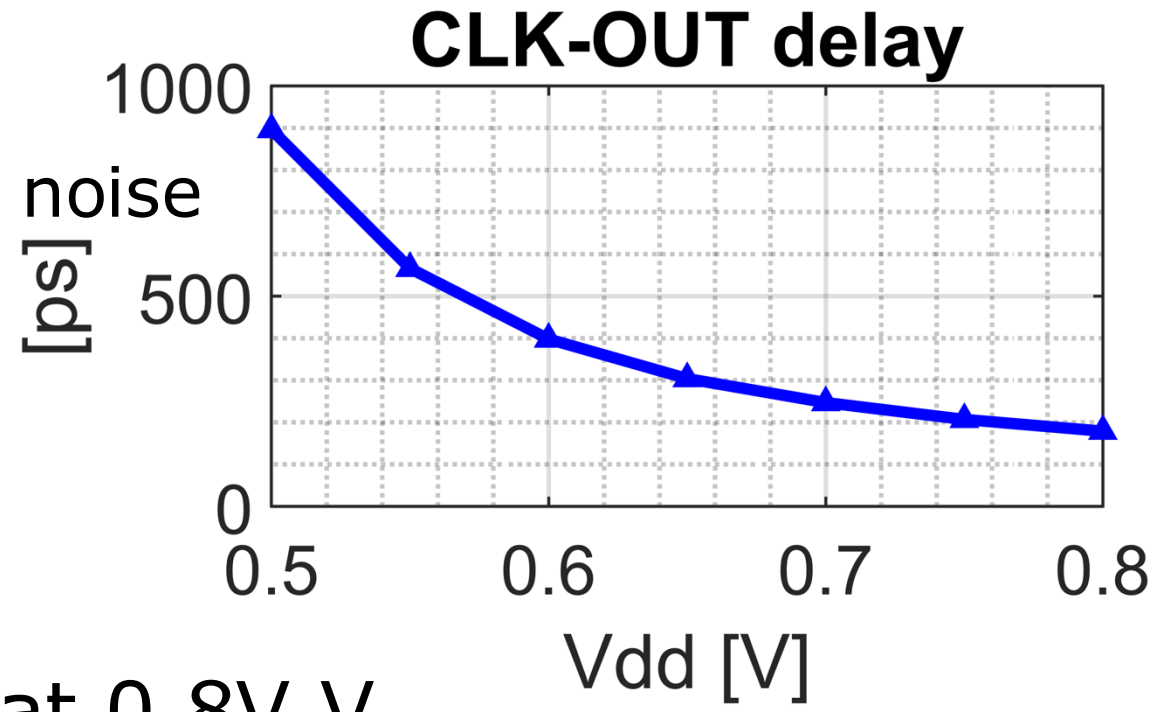
Dynamic bias comparator w/ CP

- Energy efficient ☺
  - 75 fJ/comparison @ 0.2mV noise
- Maximized speed of double tail architecture
  - Charge pump
  - Capacitive neutralization
- CLK-OUT delay of 280 ps at 0.8V V_{DD}



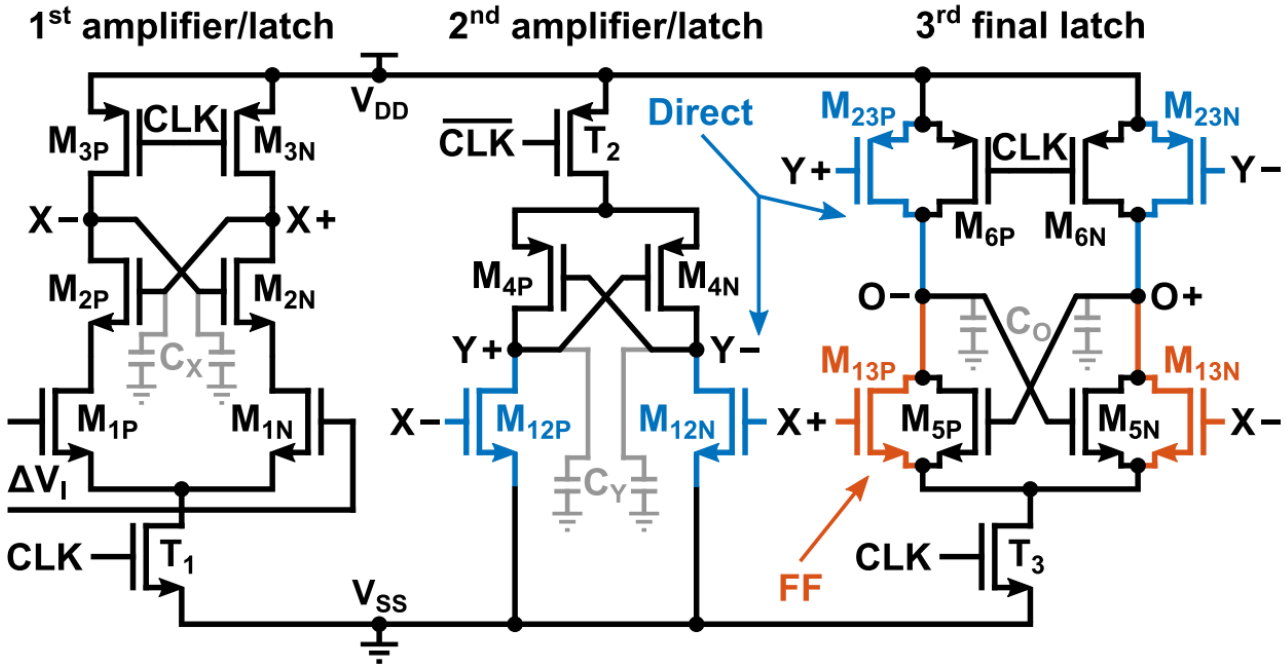
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- Maximized speed of double tail architecture
  - Charge pump
  - Capacitive neutralization
- CLK-OUT delay of 280 ps at 0.8V V_{DD}
  - Simulated > 500 ps at 0.5V V_{DD} ☹



Bindra, ISSCC '22

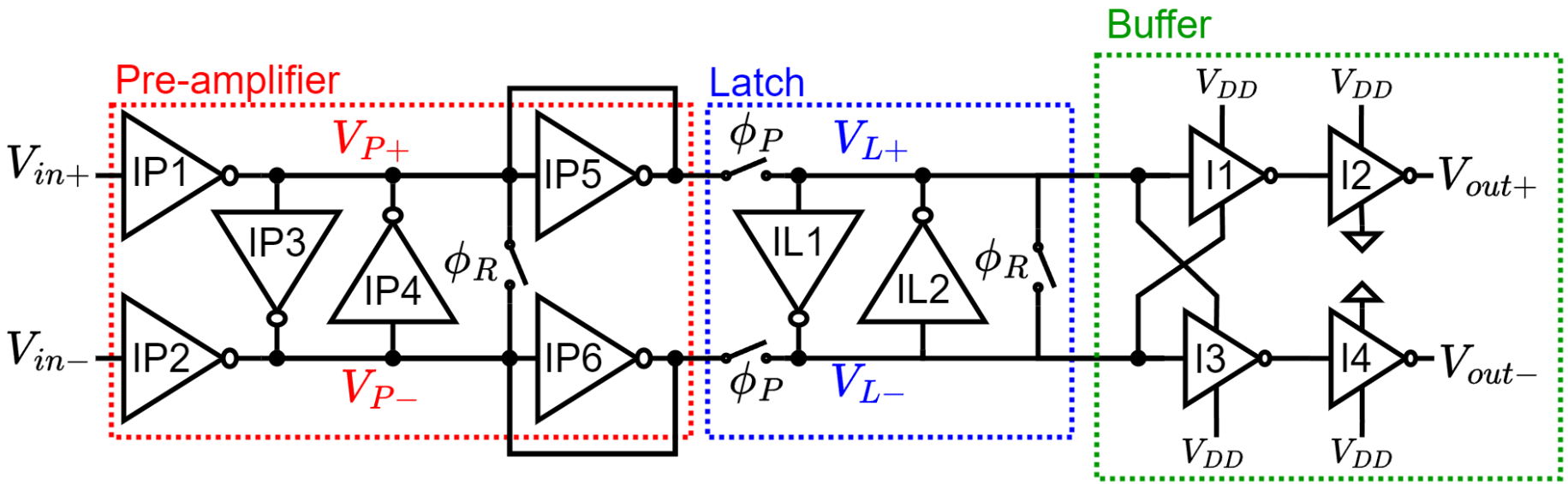
TLFF comparator



- Fast operation, low CLK-OUT delay ☺
  - 27 ps at 1V V_{DD}, 70 ps at 0.6V V_{DD}
- Low power efficiency ☹
  - 163 fJ/comparison @ 0.9mV noise

A. T. Ramkaj, TCAS I '22

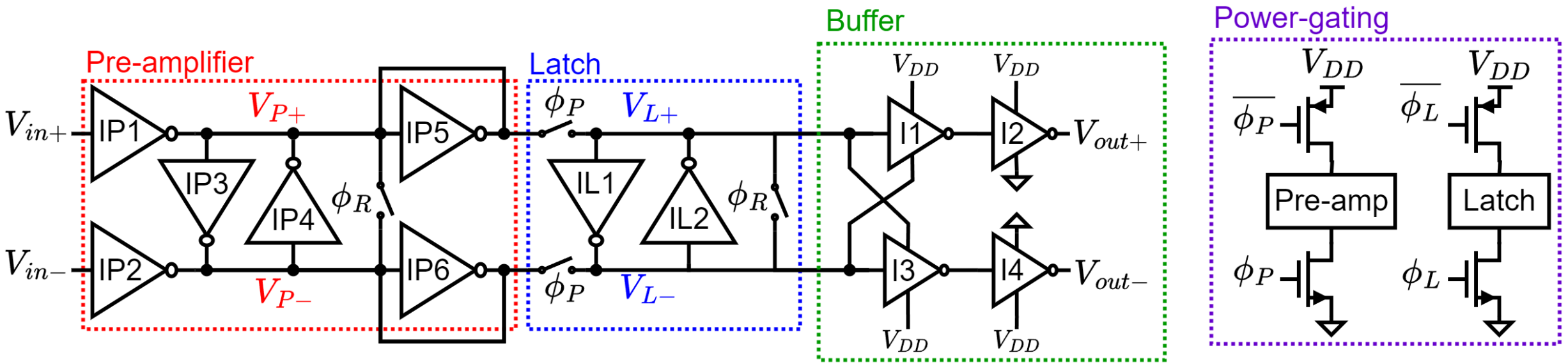
Inverter-based comparator



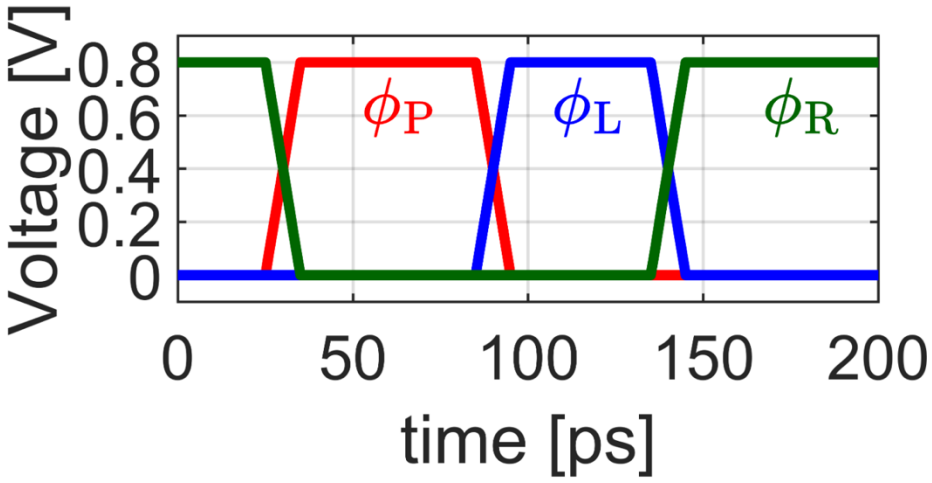
- Architecture only utilizes inverters
  - Current re-use + good supply scaling

Cents et. al, VLSI `26

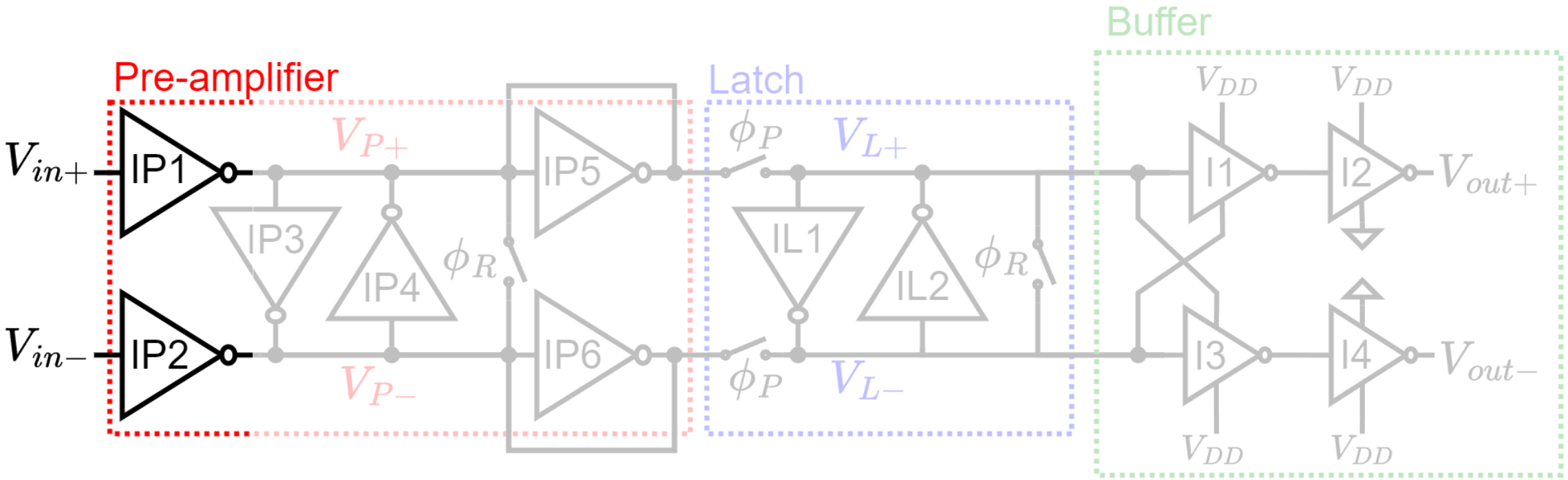
Inverter-based comparator



- Pre-amplifier + latch + output buffer
  - Power gated
  - Controlled by separate clock phases

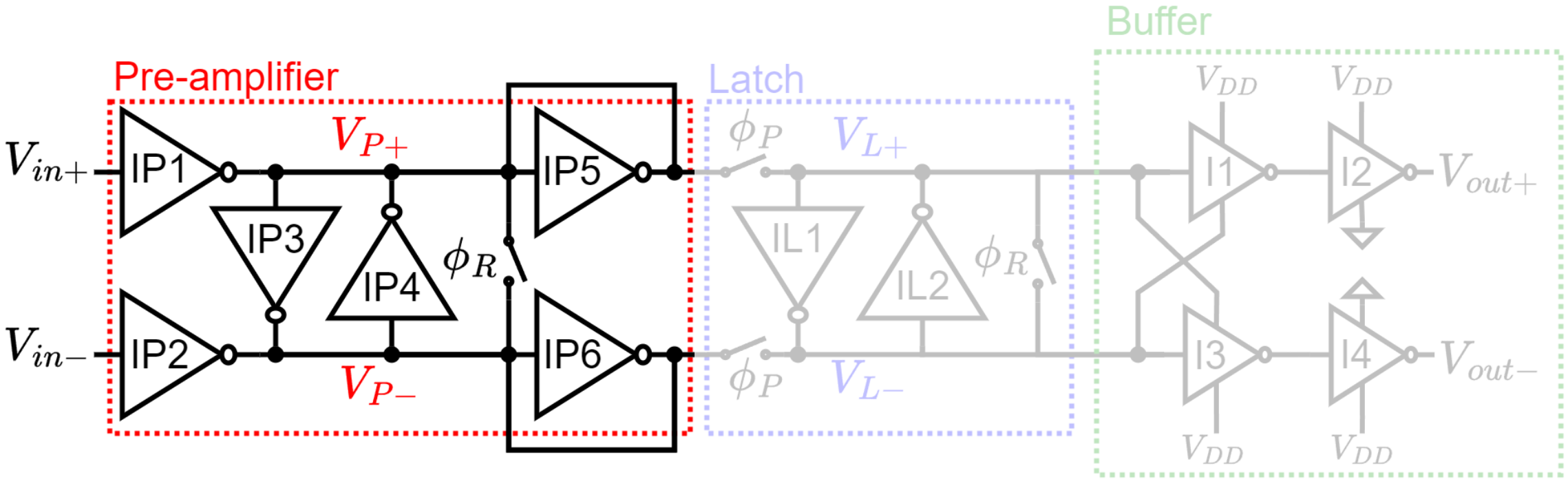


Inverter-based comparator: pre-amplifier



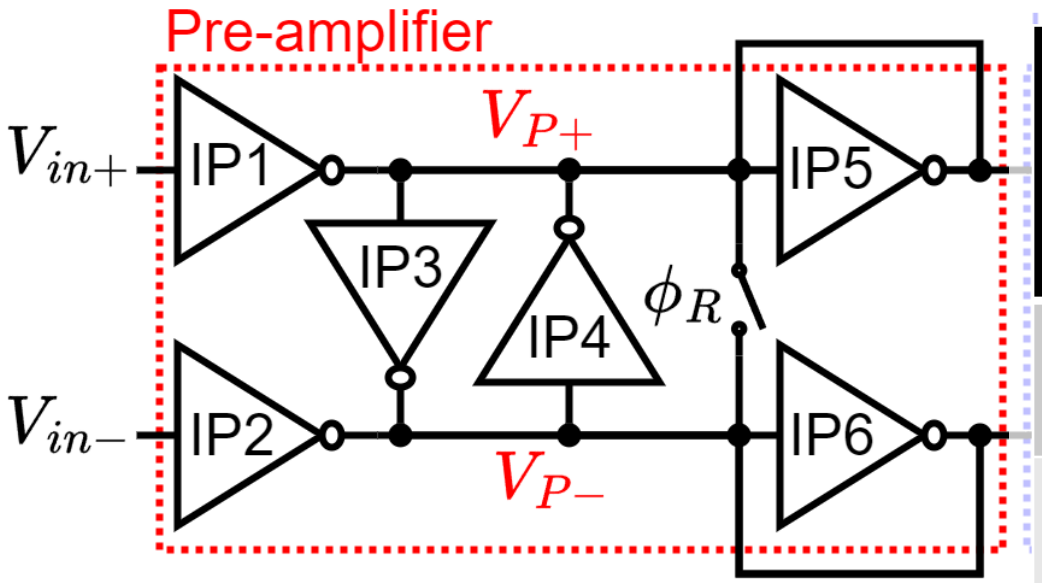
- Basic inverter stage: small V_{CM} range ☹
- Inverter-based OTA [Nauta, JSSC '92]
  - IP3-IP6 added: improved VCM range + improved gain

Inverter-based comparator: pre-amplifier



- Sum of IP3-IP6 → reduced CM gain
  - Wider VCM input range
  - CM gain designed ≈ 3

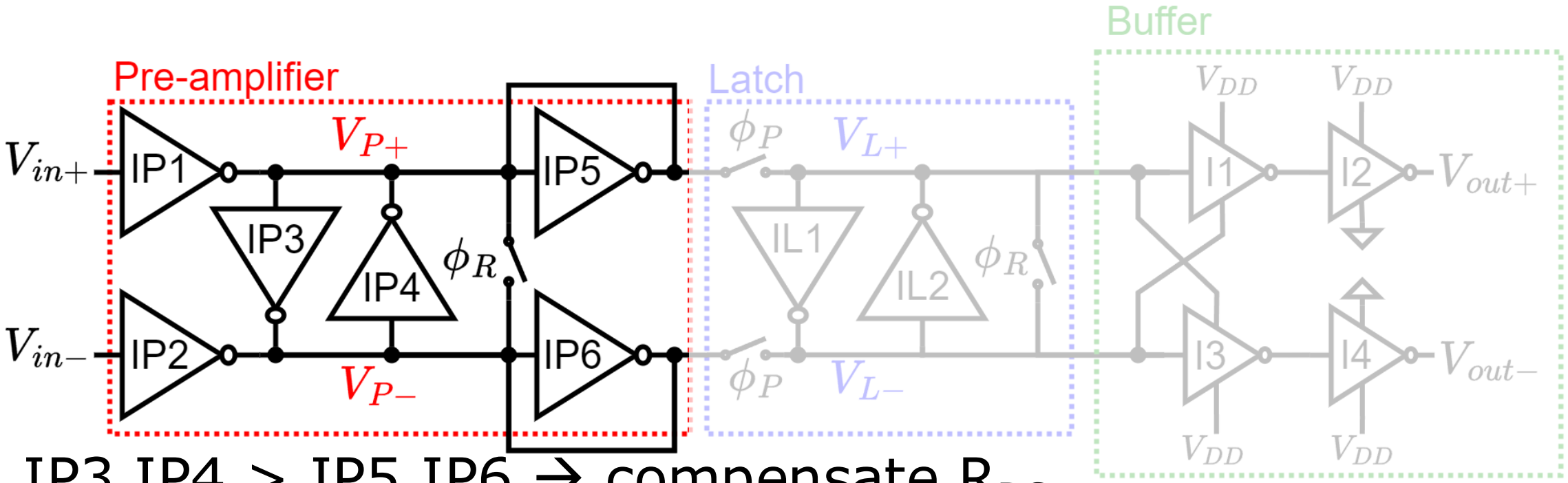
Inverter-based comparator: pre-amplifier



Buffer

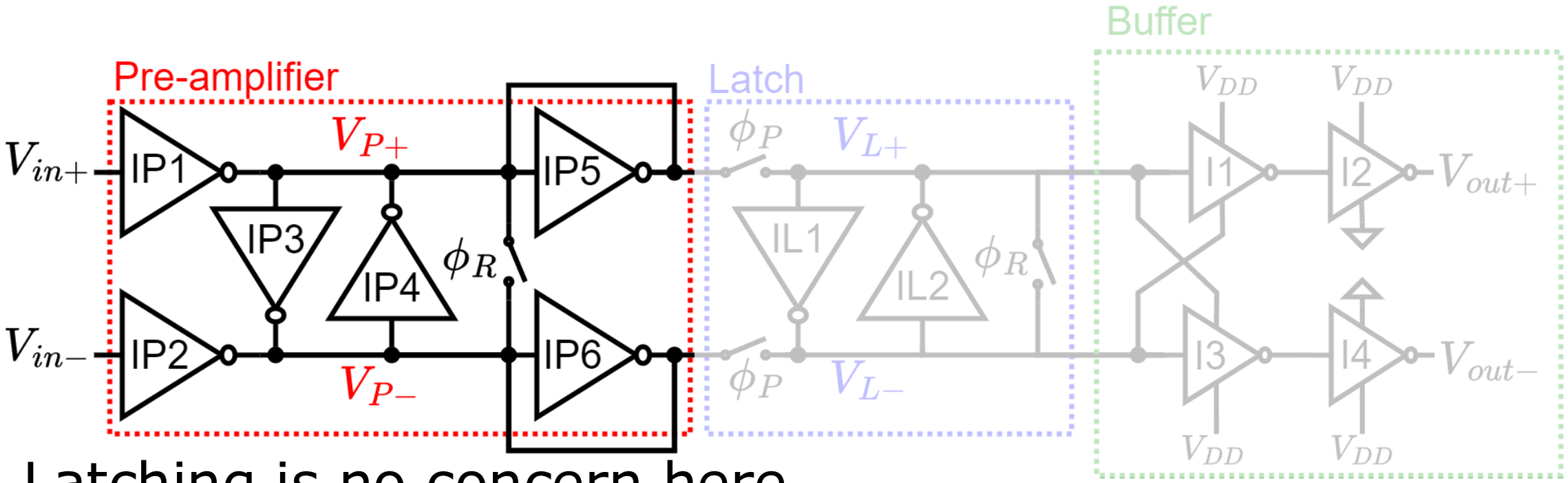
Output node	Common resistance	Differential resistance
V_{P+}	$\frac{1}{g_{m,IP5}+g_{m,IP4}}$	$\frac{1}{g_{m,IP5}-g_{m,IP4}}$
V_{P-}	$\frac{1}{g_{m,IP6}+g_{m,IP3}}$	$\frac{1}{g_{m,IP6}-g_{m,IP3}}$

Inverter-based comparator: pre-amplifier



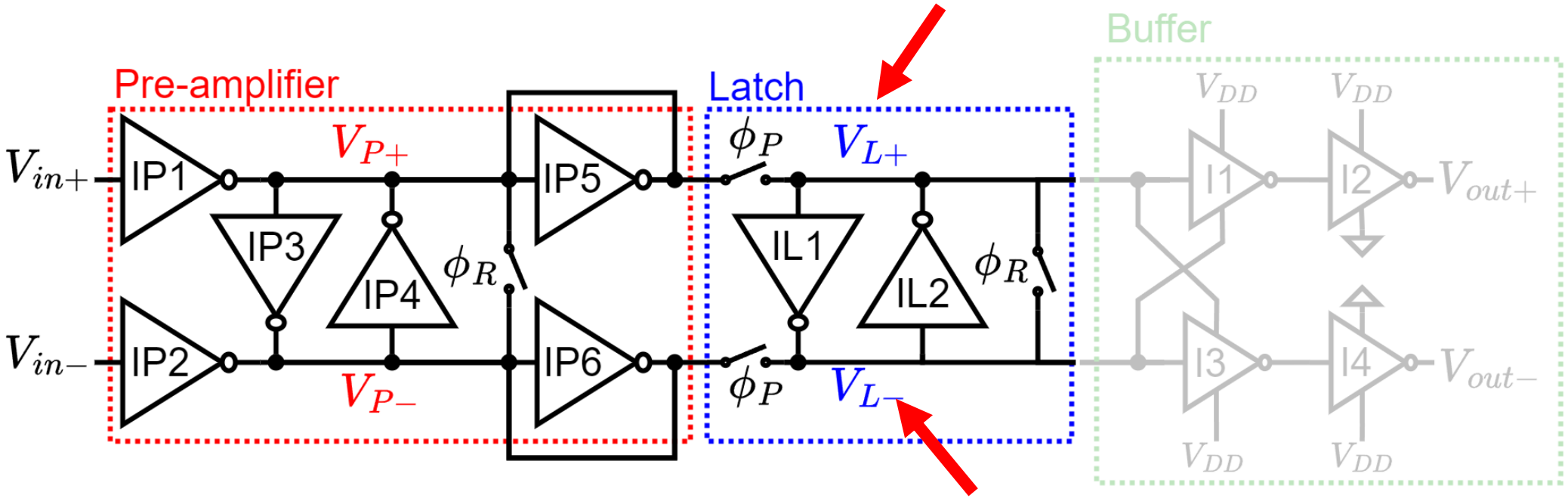
- IP3,IP4 > IP5,IP6 → compensate R_{DS}
  - Maximize voltage gain
- Partial compensation → gain limitation
- Overcompensation → latching behavior

Inverter-based comparator: pre-amplifier

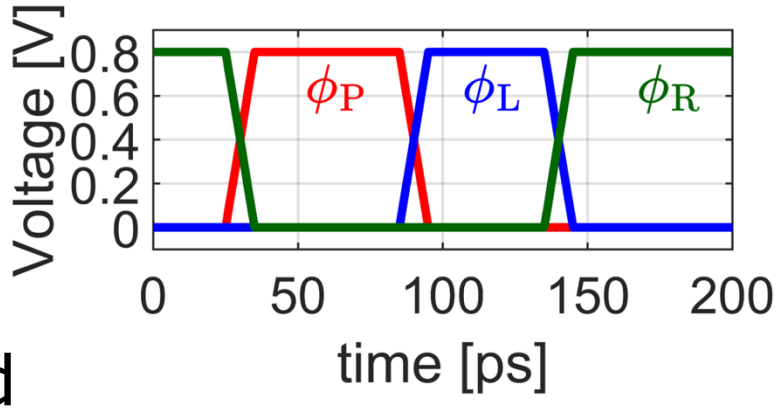


- Latching is no concern here
  - Comparator output is latched anyway
  - R_{DS} is fully compensated
- No integration caps maximizes speed

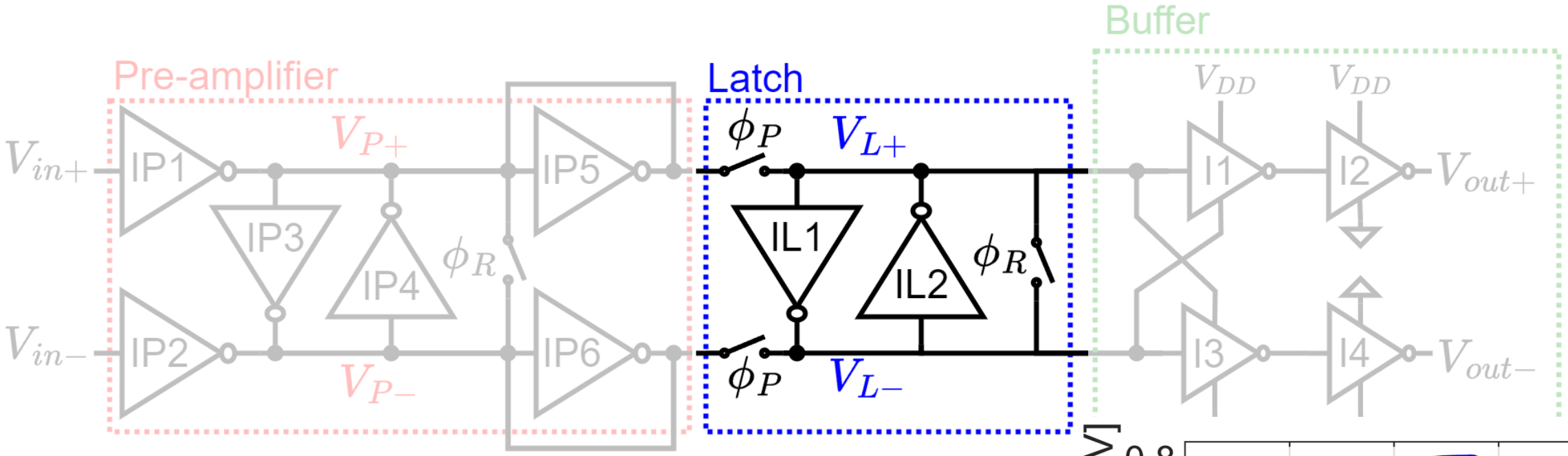
Inverter-based comparator: pre-amplifier/latch

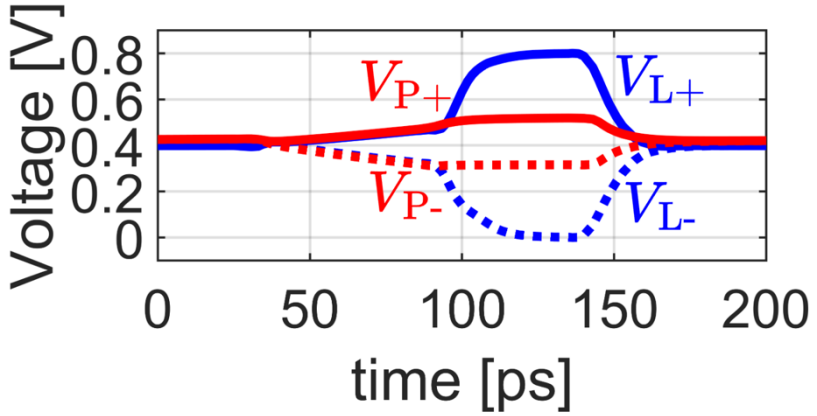


- Pre-amp connected to latch only in ϕ_P
  - Direct path V_{P+} to V_{out} → fast at large V_{in}
- Pre-amp turns off → latch turn on (ϕ_L)
  - Pre-amp not loading latch improves speed



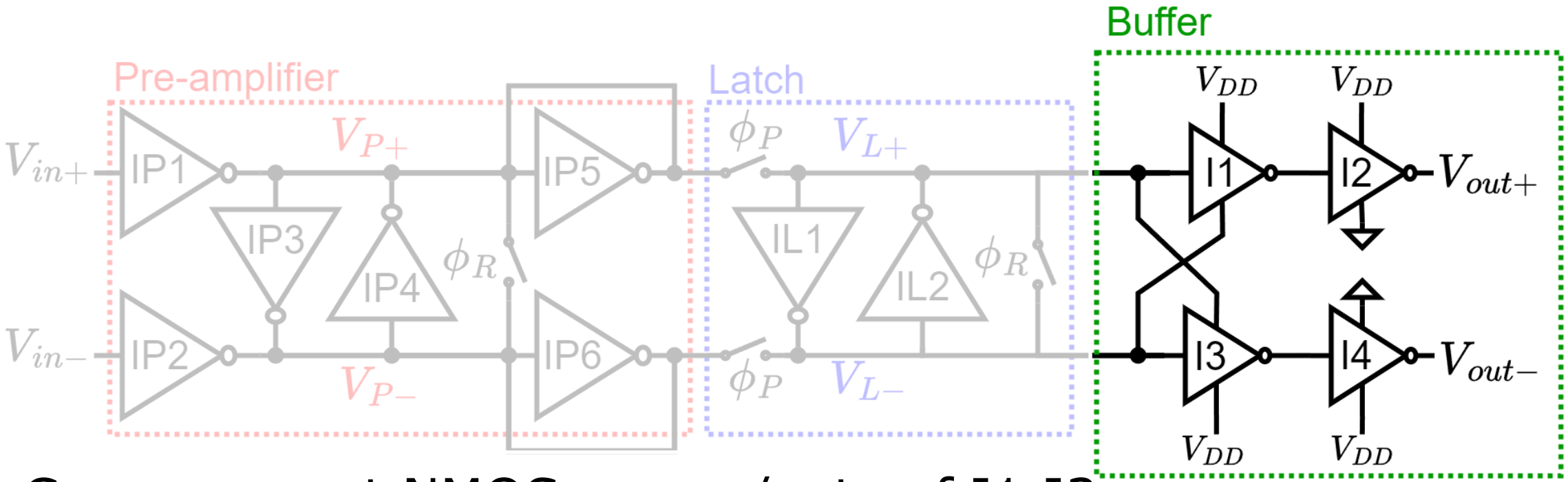
Inverter-based comparator: latch





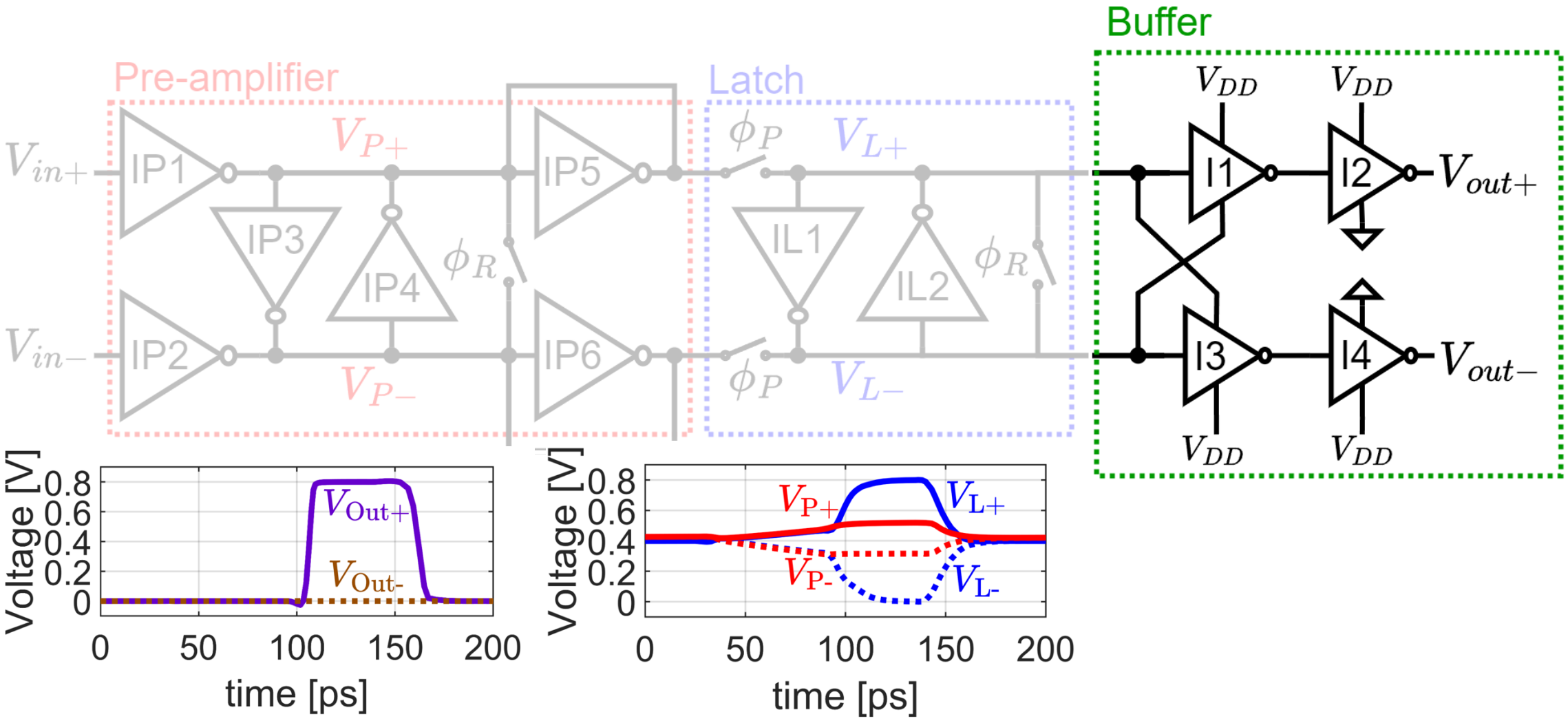
- Latch output cannot directly be used!

Inverter-based comparator: output buffer

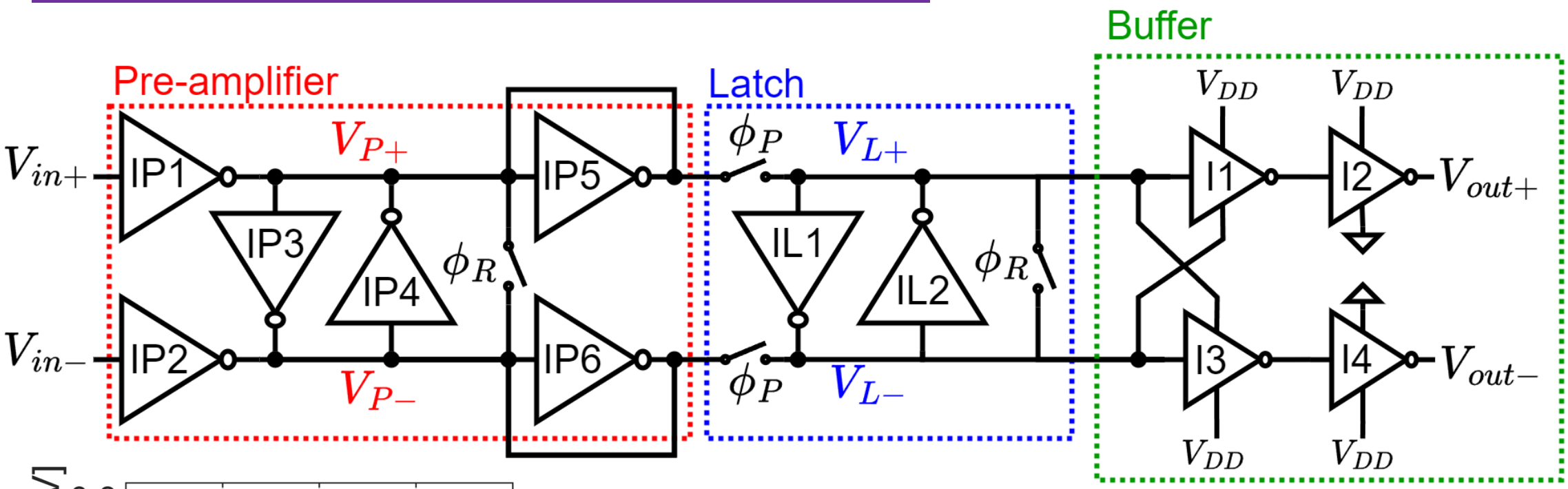


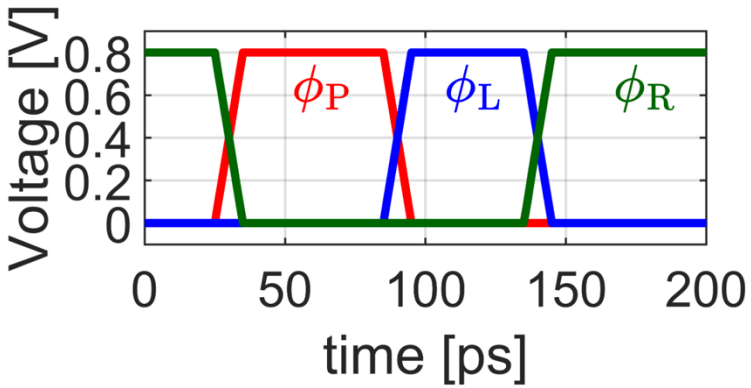
- Cross connect NMOS source/gate of I1,I3
- One NMOS always off
  - Well defined output
  - Reduced metastability

Inverter-based comparator: output buffer



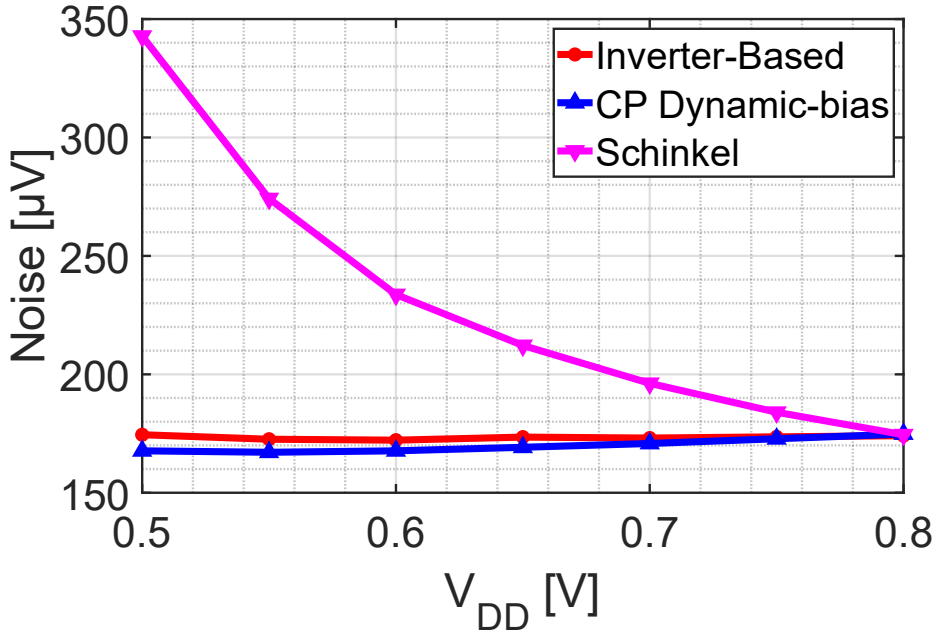
Inverter-based comparator: reset

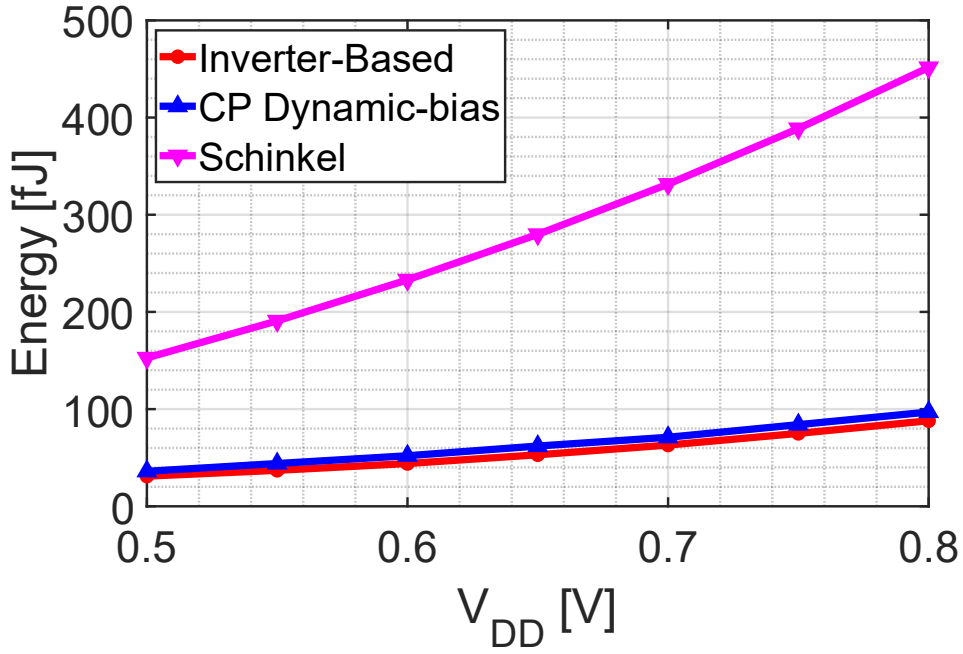




- Voltage equalization → transmission gates
- No CM discharging of parasitics
  - Reduces power and improves speed

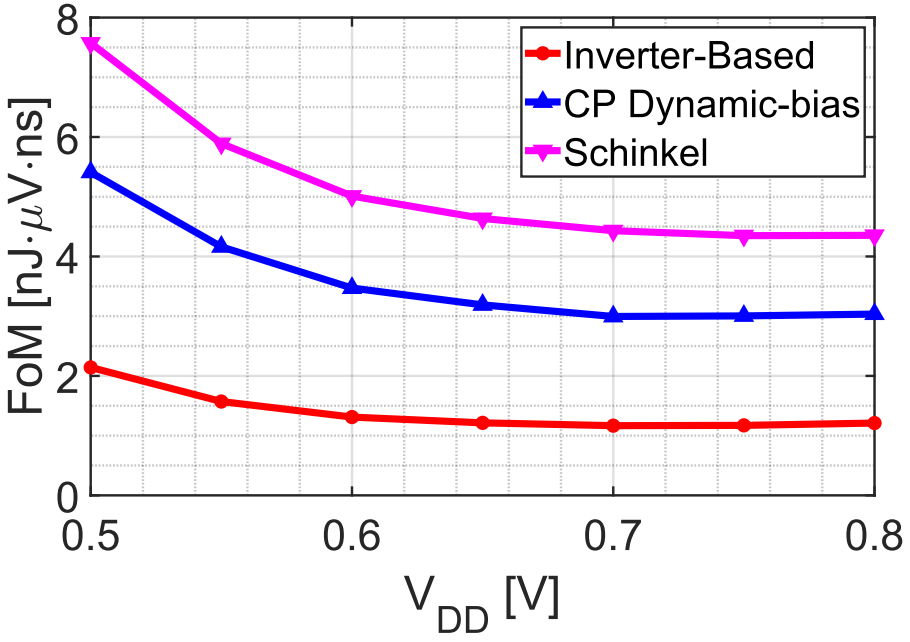
Simulation comparison

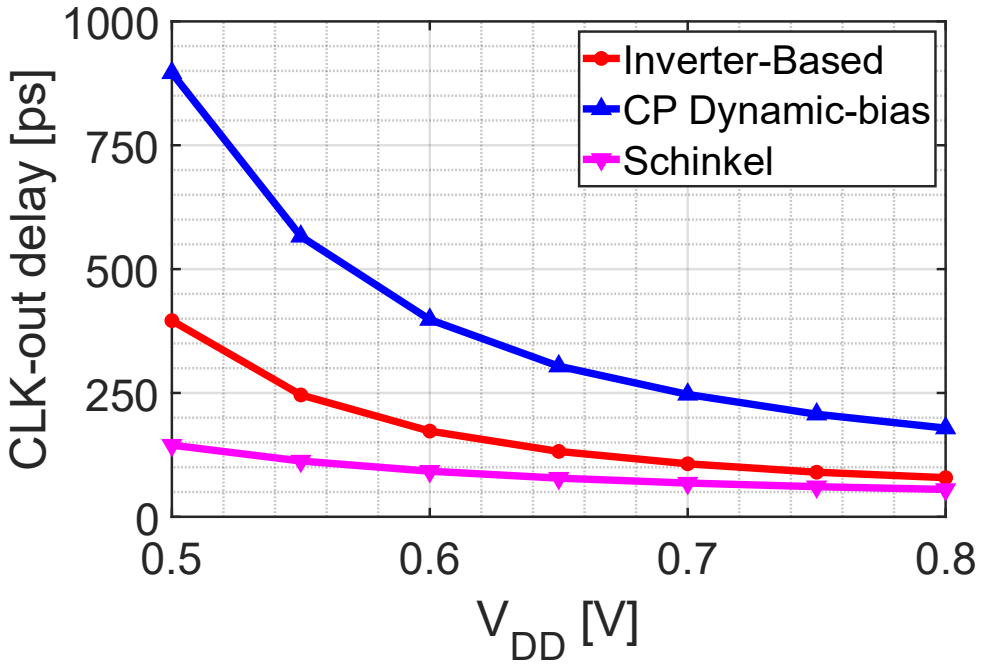




- Scaled for similar noise at nominal 0.8V V_{DD}
- Inverter-based comparator and Dynamic Bias w/ Chargepump maintain near constant IRN

Simulation comparison





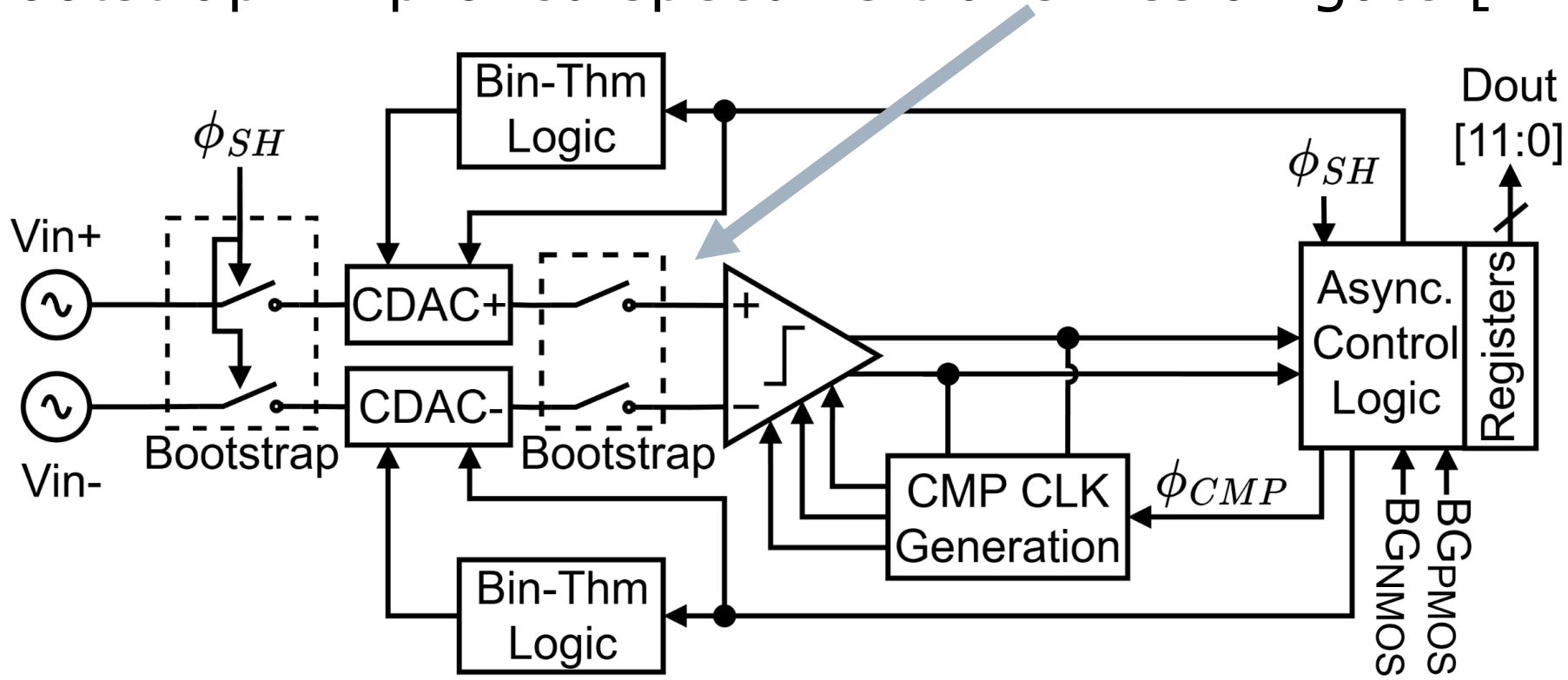
- Inverter-based comparator
  - Atleast 2x faster CLK-OUT delay than the Dynamic Bias w/ chargepump throughout V_{DD} and < 500 ps at 0.5V V_{DD}
  - 2x faster than the Schinkel comparator at lower V_{DD}

Outline

- Introduction
- Ultra-low-power SAR ADCs overview: Building blocks perspective
  - Comparator
  - Design Example
- Wide-bandwidth (GS/s) SAR ADCs
- Auxiliary blocks and its requirements
- Conclusion

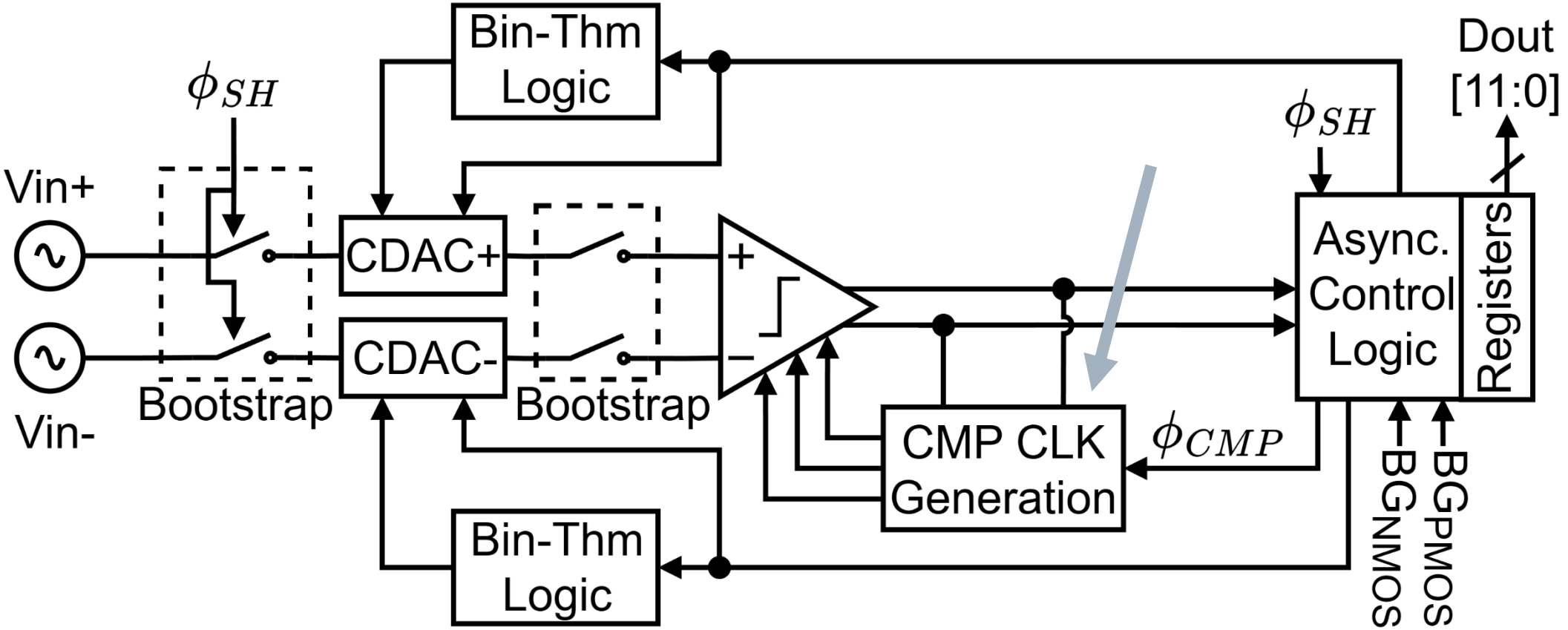
SAR ADC – A low power design example

- Comparator-DAC isolation for improved linearity
  - Bootstrap: improved speed vs transmission gate [Bindra, VLSI '19]

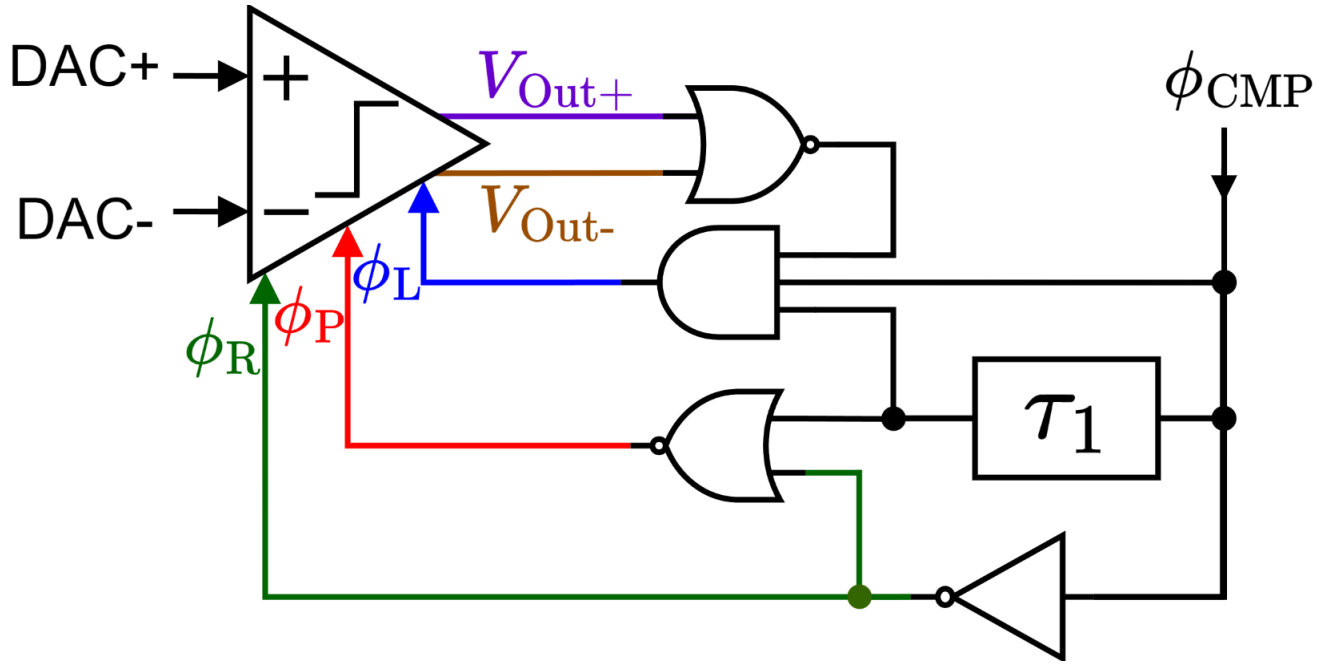


SAR ADC – A low power design example

- Comparator clock generation



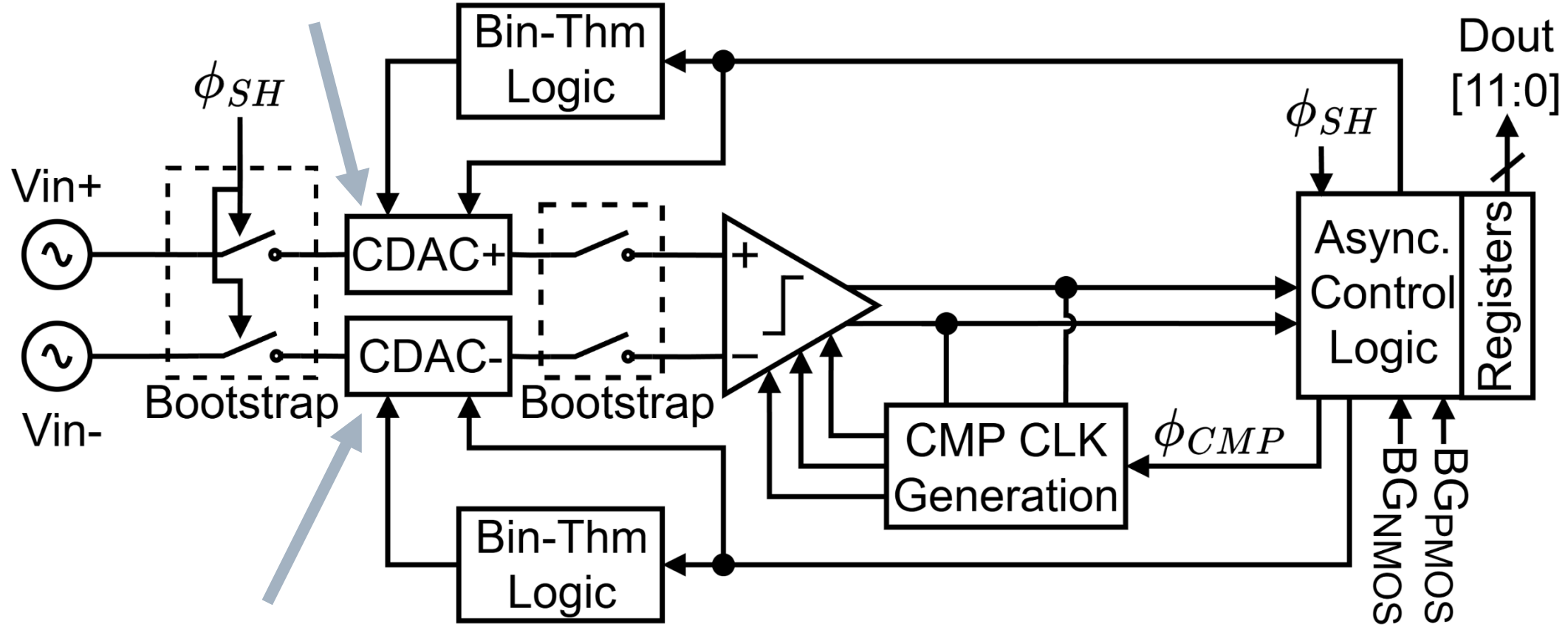
Comparator clock generation



- ϕ_{CMP}: single phase clock generated within ADC logic
  - Pre-amplifier (ϕ_P) ON for duration τ_1
  - Latch (ϕ_L) turns ON when pre-amplifier (ϕ_P) turns OFF
  - ϕ_{CMP} low → reset comparator

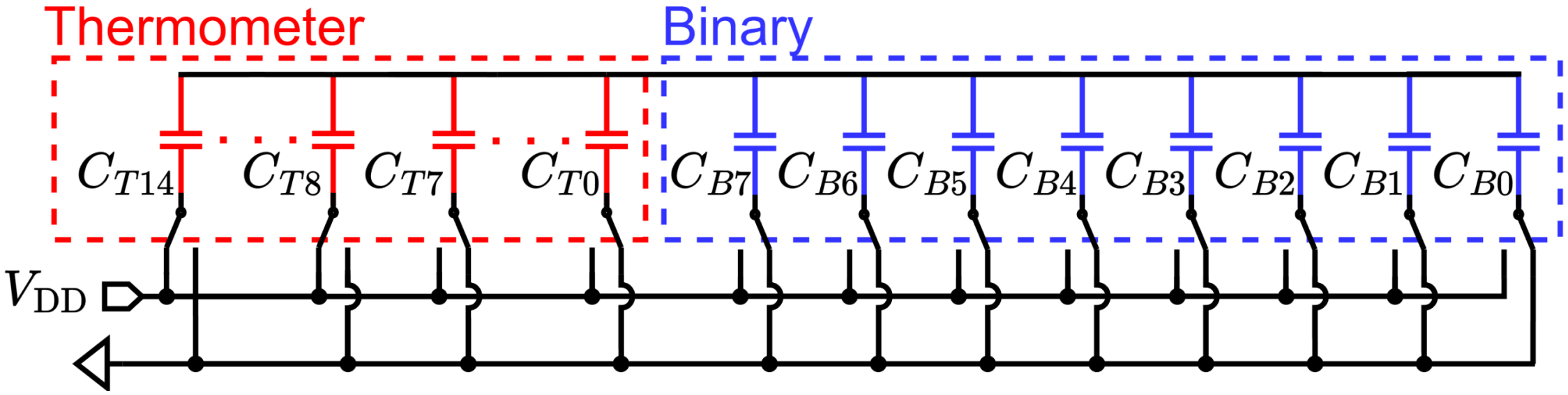
SAR ADC architecture

- DAC architecture and unit-cell



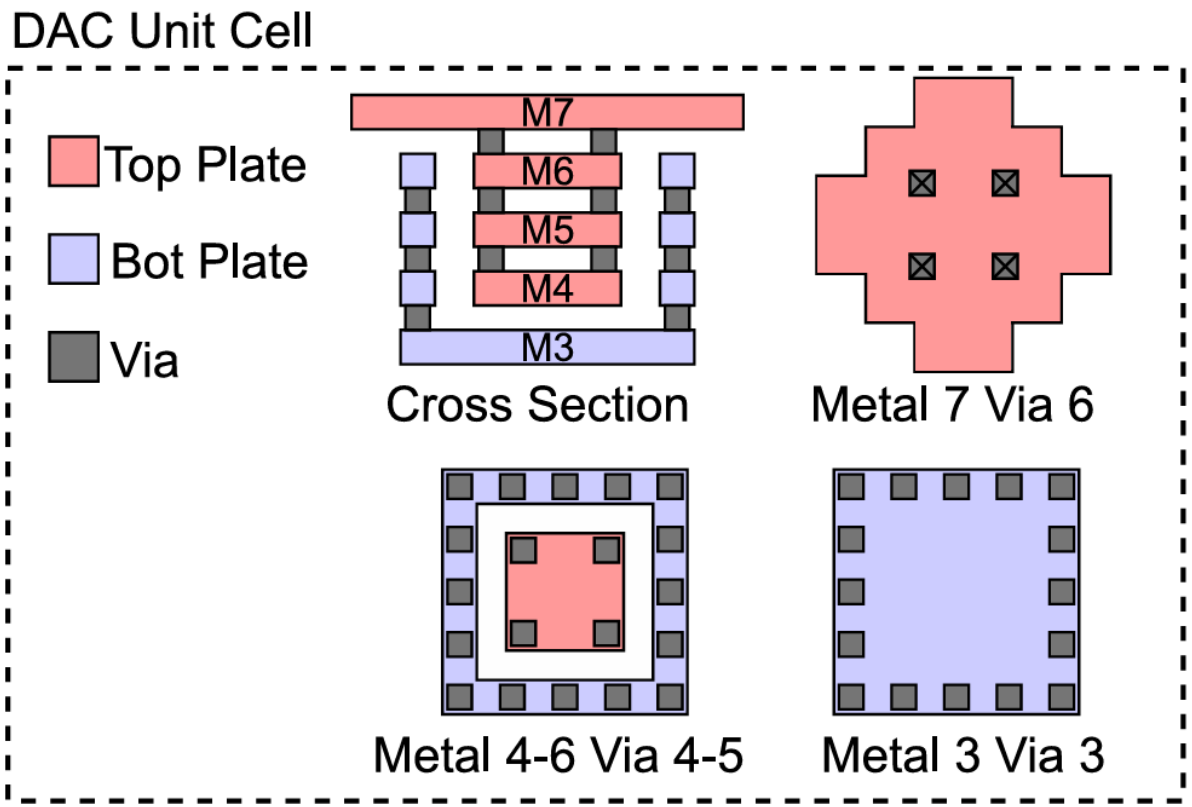
DAC Architecture

- 4-bit thermometer and 8-bit binary
  - Effective efficiency of split-DAC → 37% energy saved
  - Improved matching
- Common-mode is kept constant



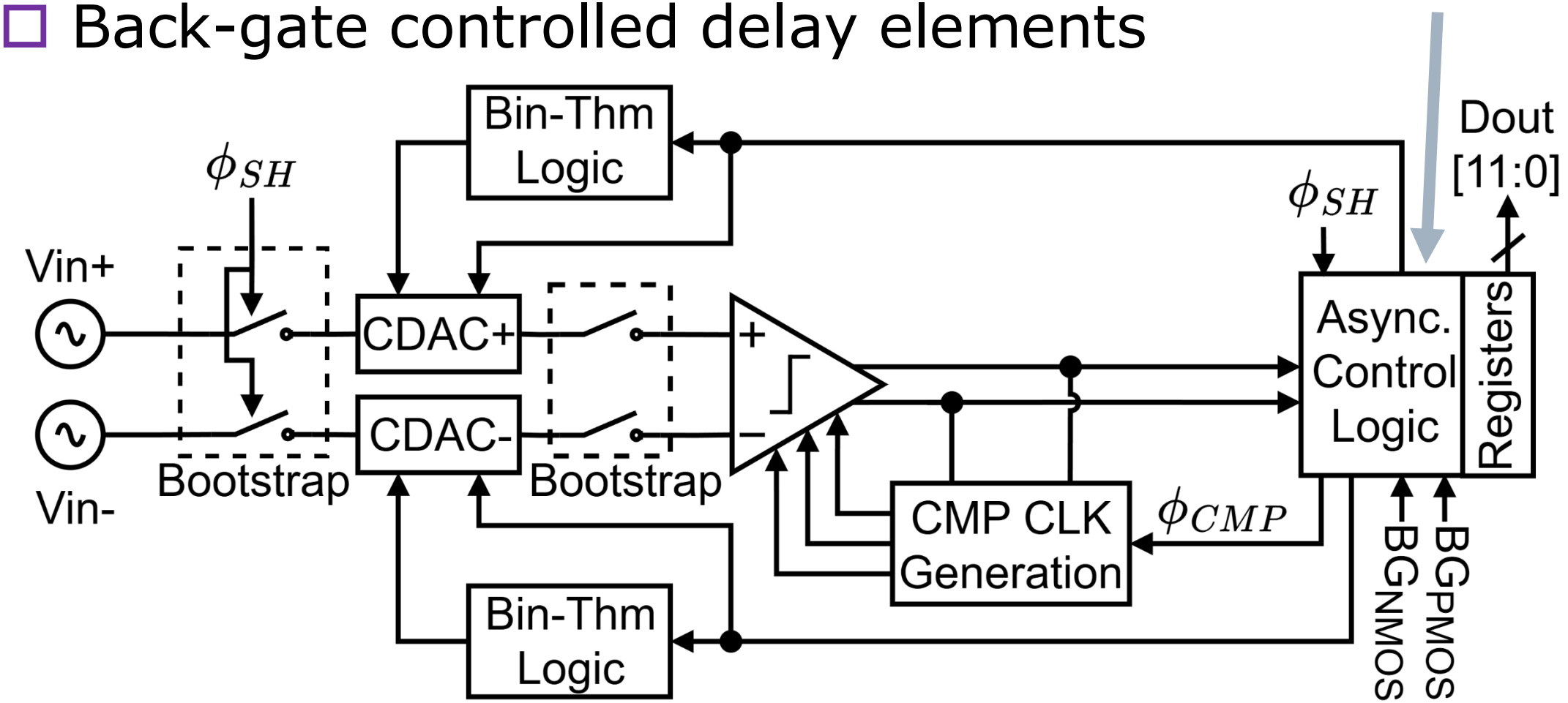
DAC unit-cell

- 290 aF unit cell
  - Top-plate isolated from substrate and routing
  - 0.56 μm^2
- 1.2 pF DAC (single-ended)
  - Close to kT/C limit at V_{DD} = 0.5V

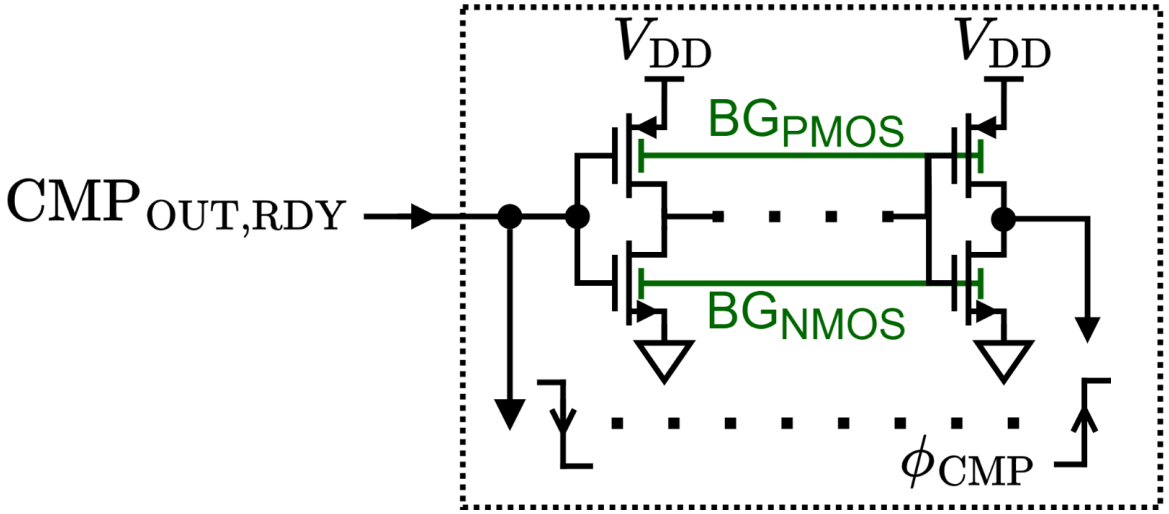


SAR ADC architecture

- Back-gate controlled delay elements



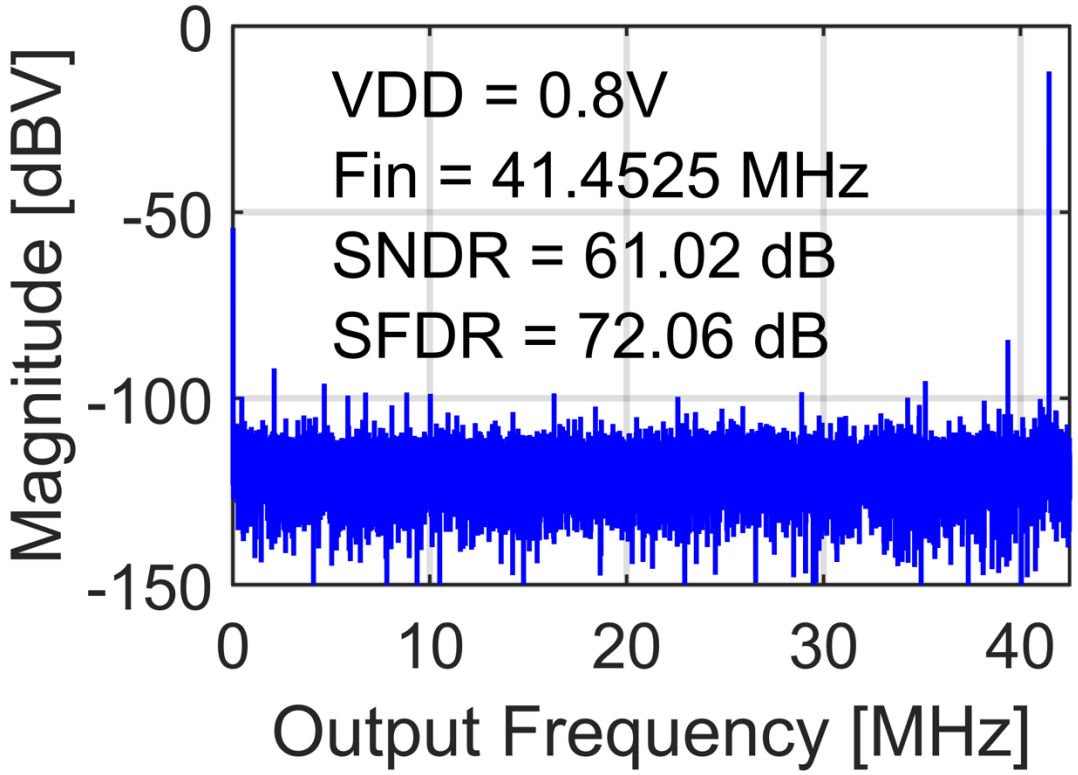
Back-gate controlled delay elements

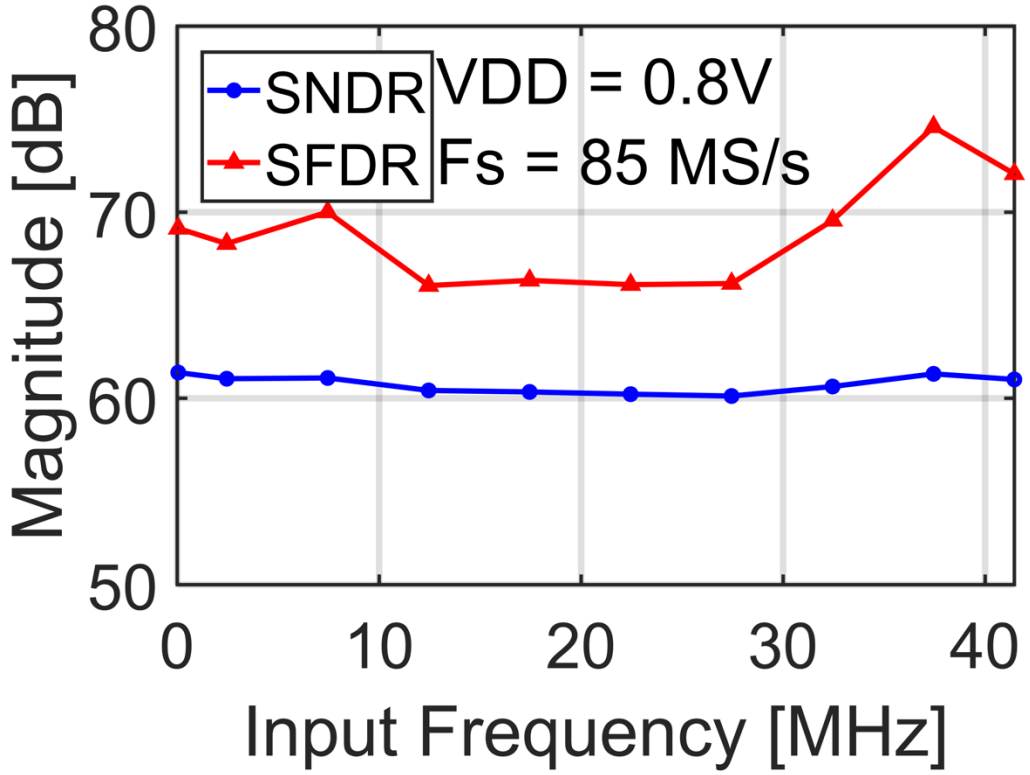


- Asynchronous control logic with tapered delay line
  - Generates delay DAC switching …. Next comparison
- Delay elements controlled by FDSOI back-gate → V_{TH} modification
- Optimization → higher Nyquist rate

Measurement results: SNDR/SFDR

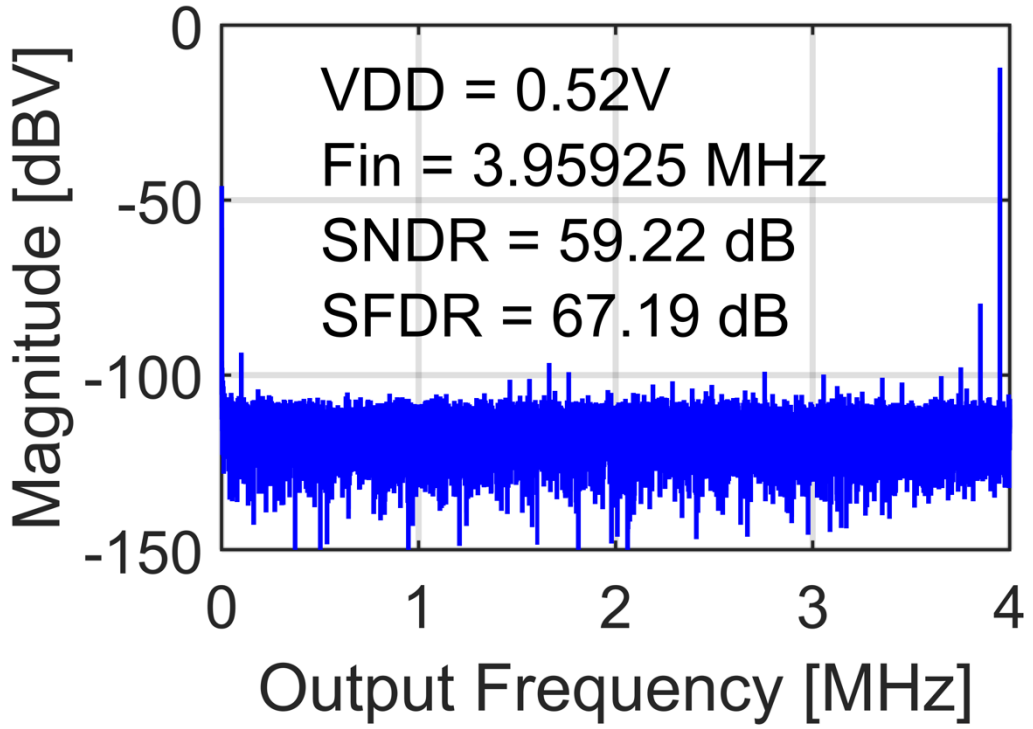
- Nominal V_{DD} (0.8V), Nominal back-gate (PMOS V_{DD}, NMOS GND)

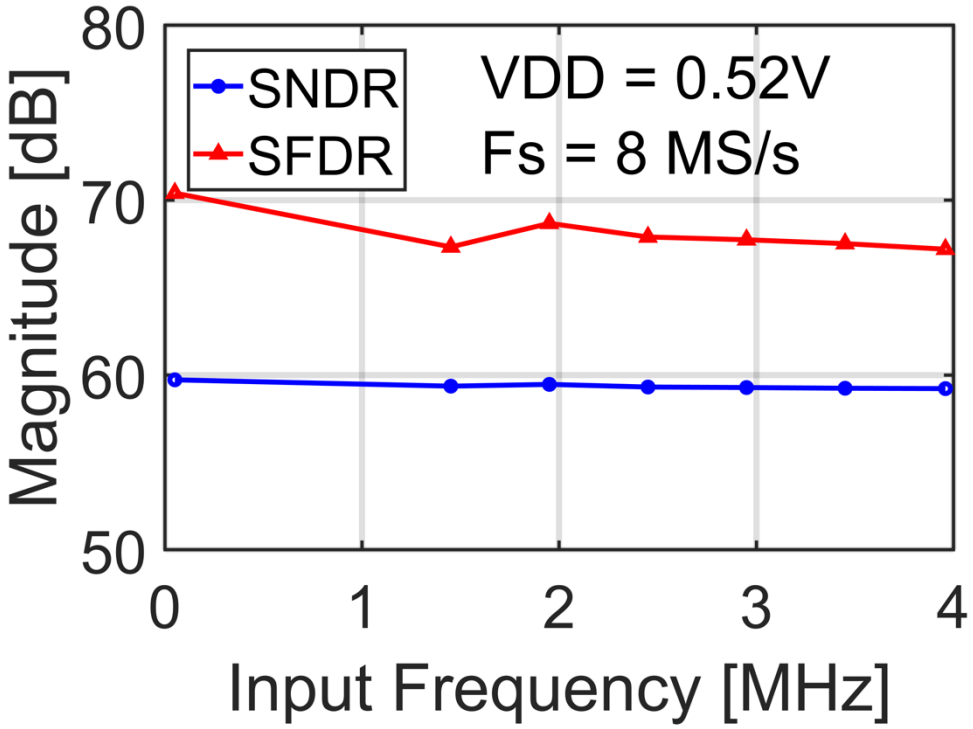




Measurement results: SNDR/SFDR

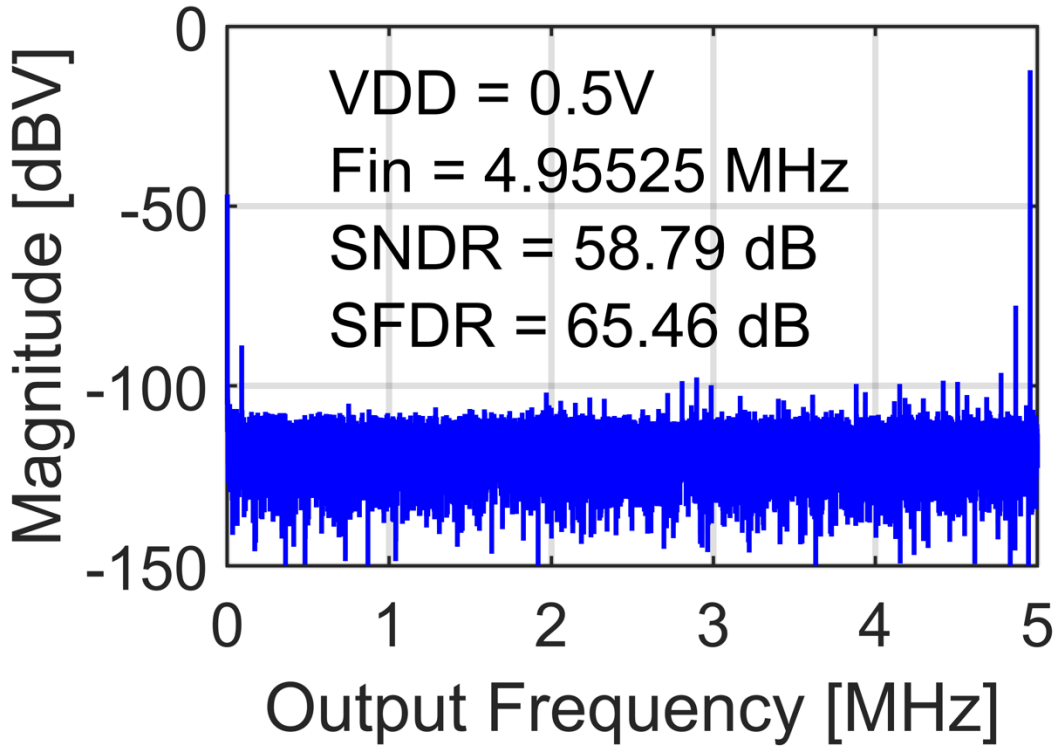
- Lowest possible V_{DD} (0.52V)
- Nominal back-gate (PMOS V_{DD}, NMOS GND)

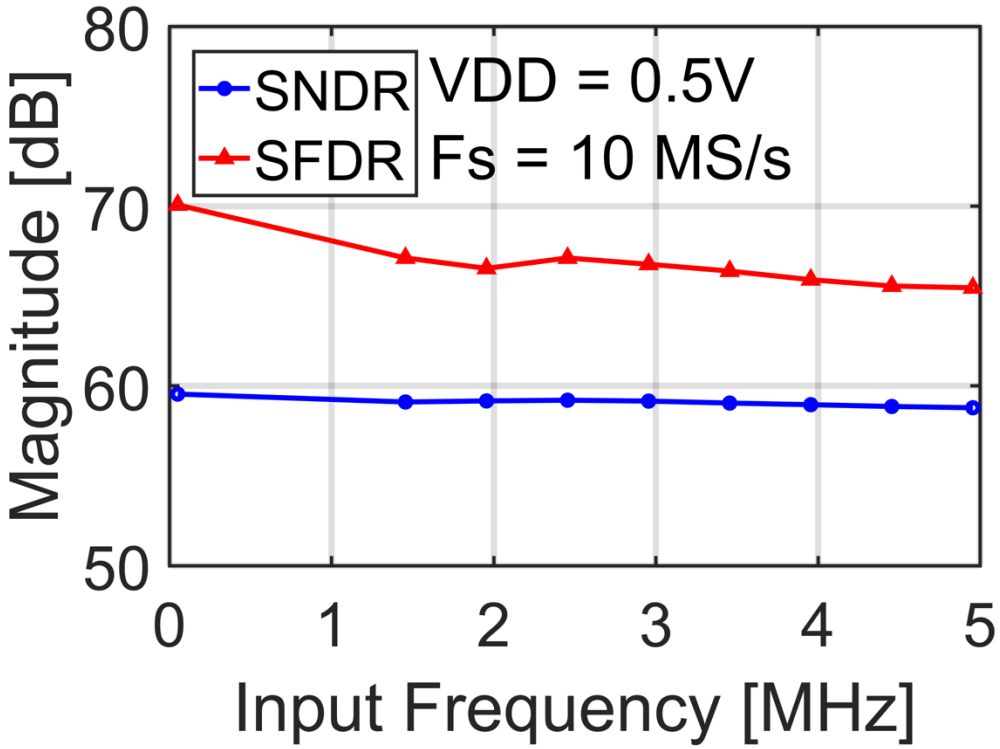




Measurement results: SNDR/SFDR

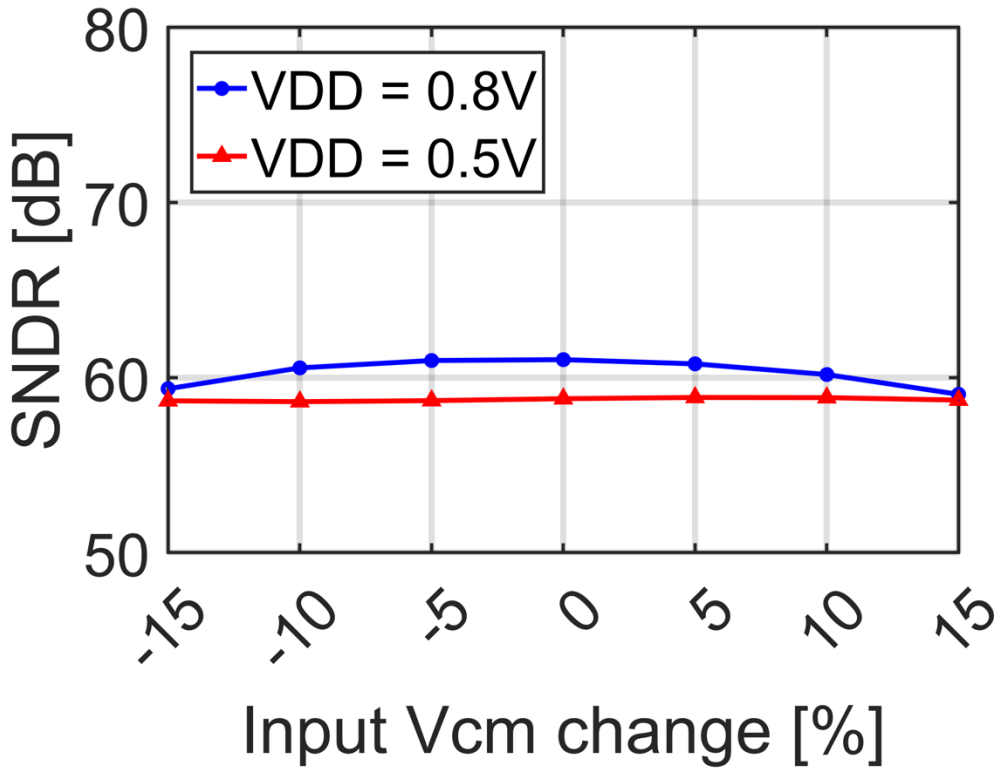
- V_{DD} = 0.5V
- Switched back-gate (PMOS GND, NMOS V_{DD})





Measurement results: V_{CM} sweep

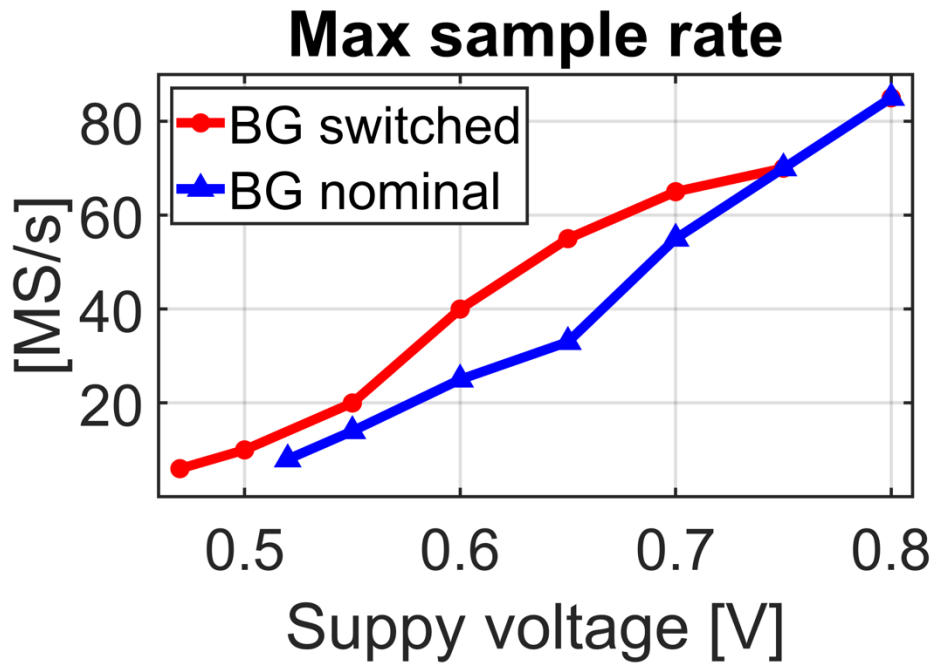
- Input V_{CM} variation of $\pm$ 15%
- 0.8V V_{DD}
  - 2.0 dB SNDR degradation
- 0.5V V_{DD}
  - 0.2 dB SNDR degradation

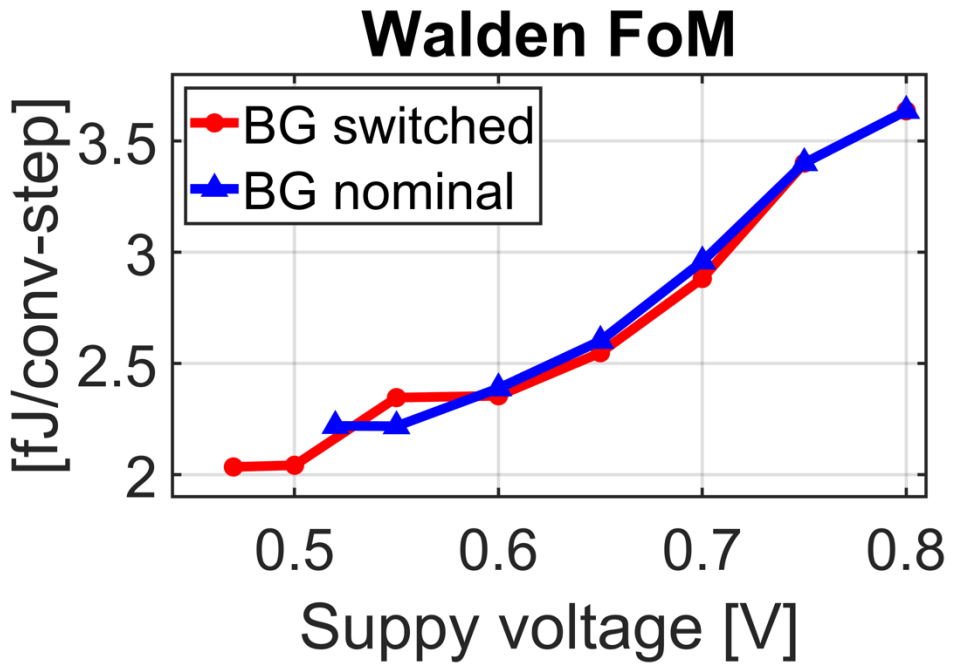


Measurement results

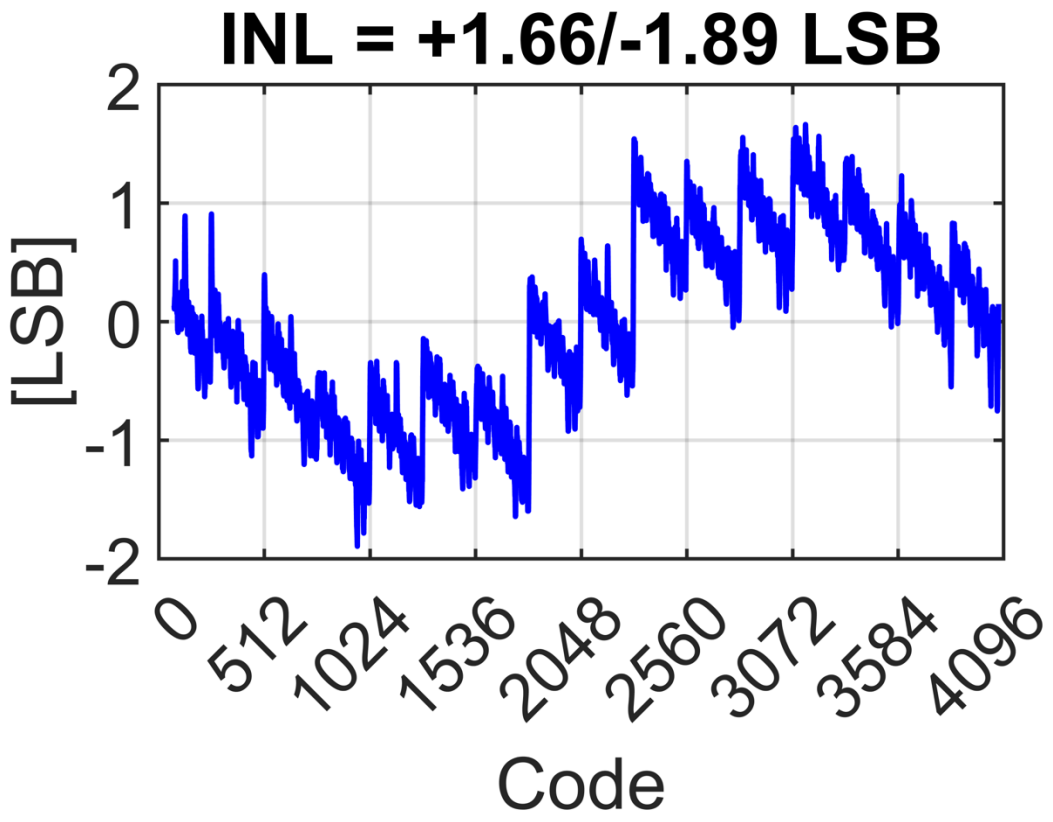
- V_{DD} Sweep
  - Nominal and switched back-gate

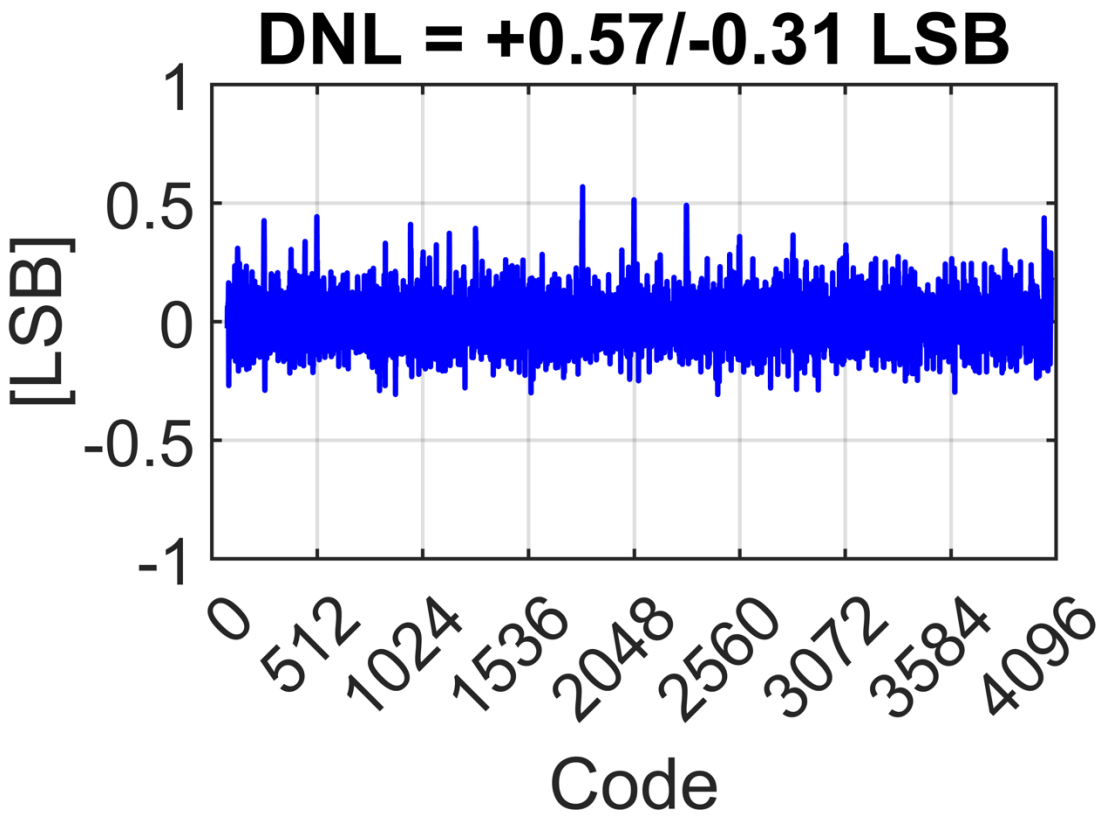
VDD [V]	0.47–0.7	0.75-0.8
Back-gate PMOS	GND	VDD
Back-gate NMOS	VDD	GND





Measurement results: INL/DNL

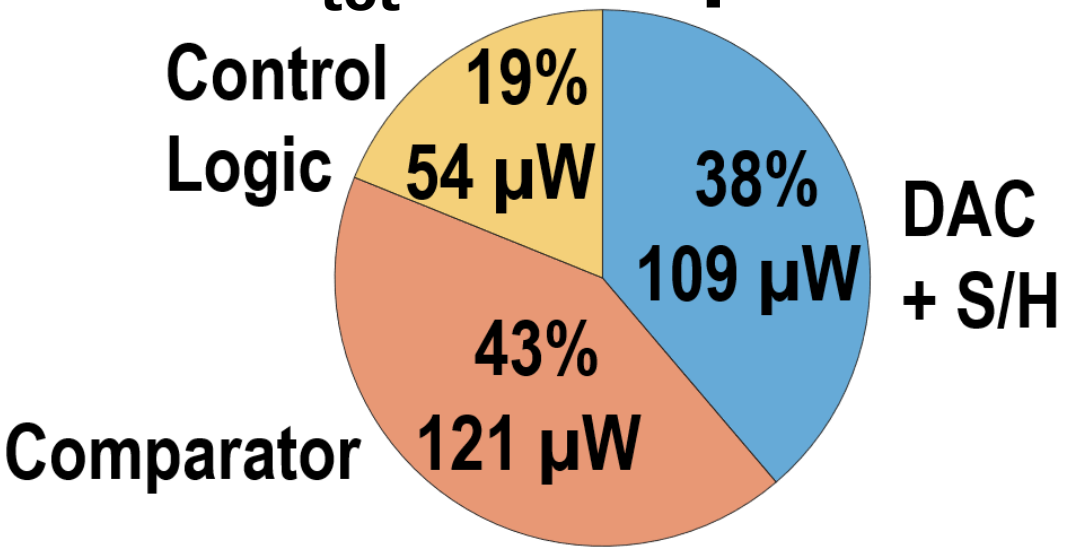




Measurement results: power breakdown

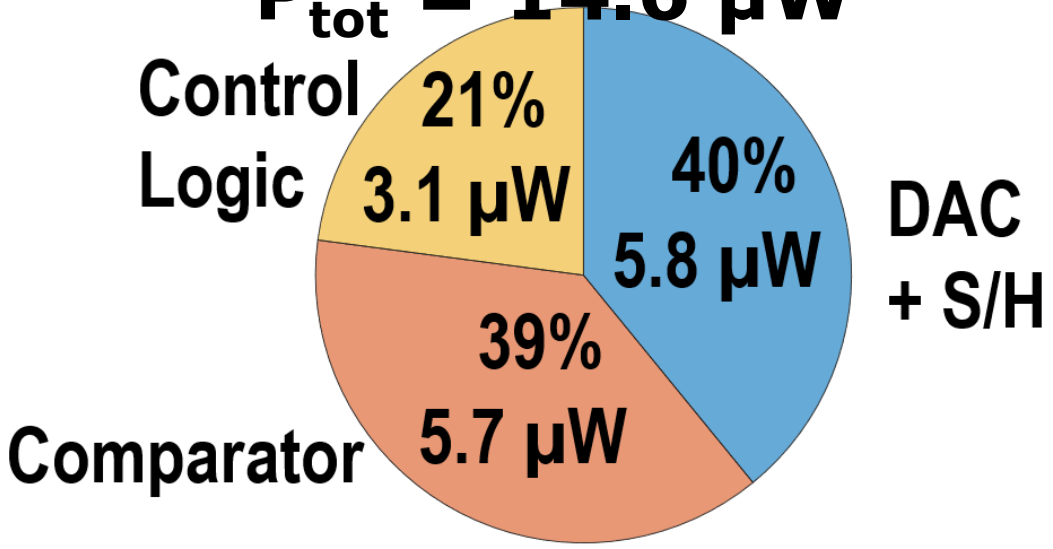
V_{DD} = 0.8V, F_S = 85 MS/s

P_{tot} = 284 µW

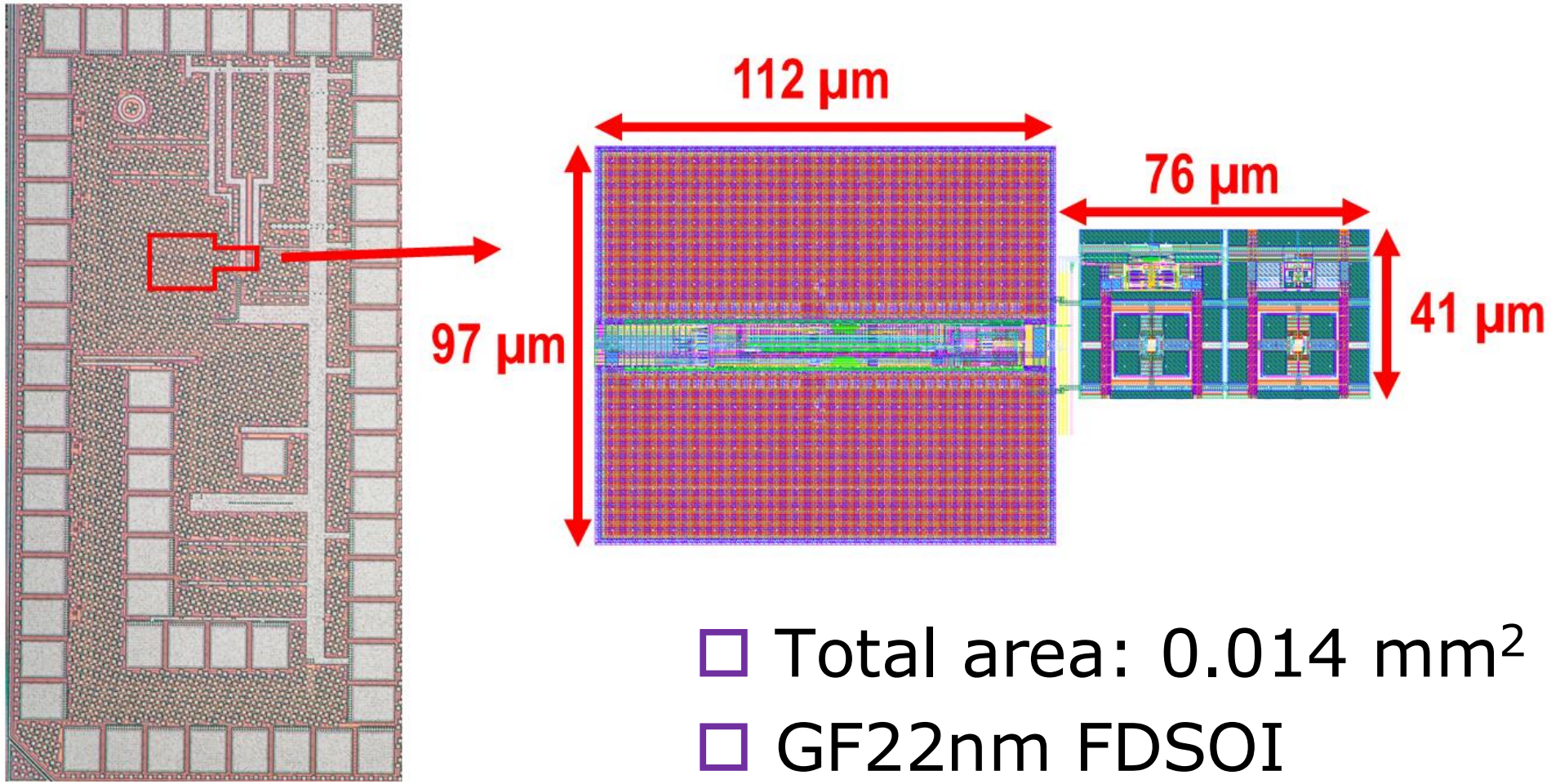


V_{DD} = 0.5V, F_S = 10 MS/s

P_{tot} = 14.6 µW



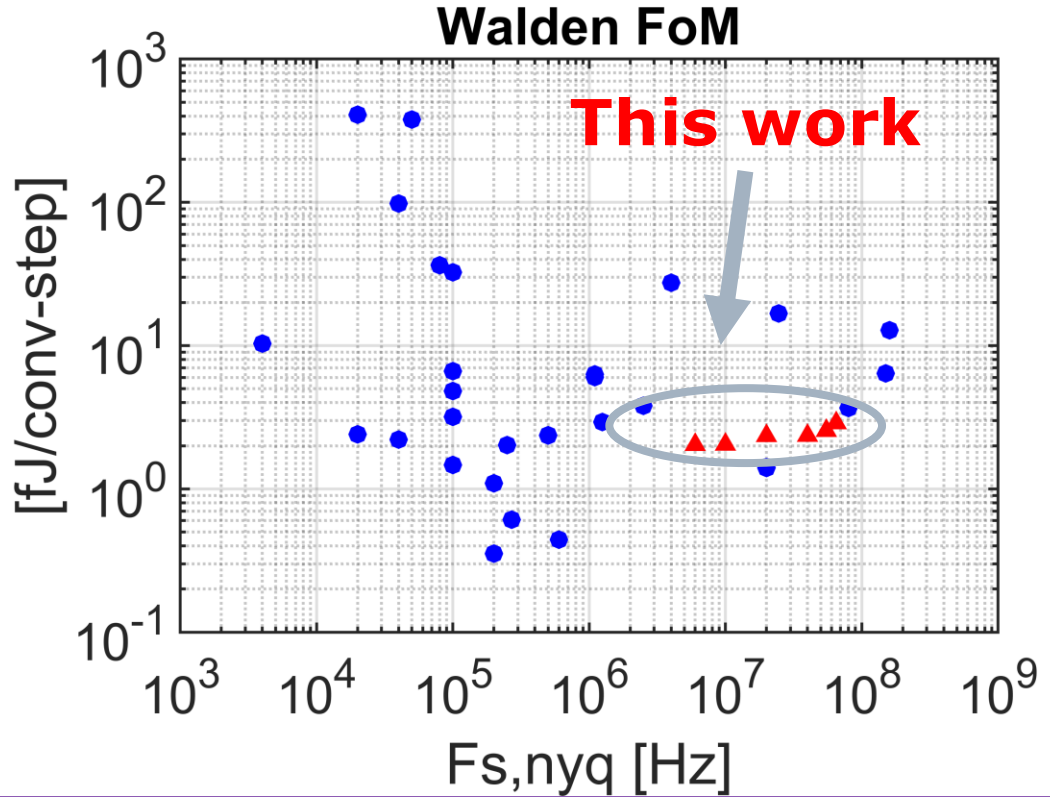
Chip micrograph and layout

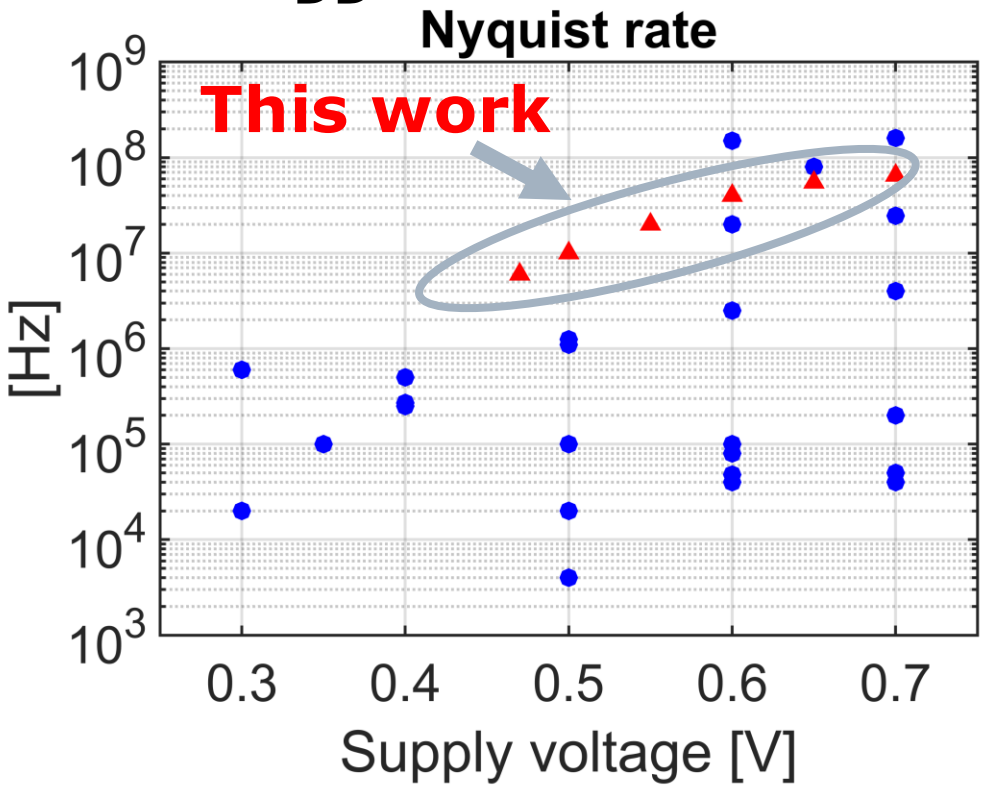


- Total area: 0.014 mm^2
- GF22nm FDSOI

Comparison

- SotA Walden FoM
- Highest reported Nyquist rate at V_{DD} < 0.6V





Summary low-power SAR ADC Design Example

- Flexible SAR ADC
  - 0.5-0.8V, @10-85 MS/s
- Inverter-based comparator architecture
  - Excellent V_{DD} scaling
- Back-gate adjustable delay element
  - Enhances V_{DD} operating range and improves Nyquist rate
- State-of-the-art FoM: 2.0-3.6 fJ/conversion-step
- Highest reported Nyquist rate for $V_{DD} < 0.6V$

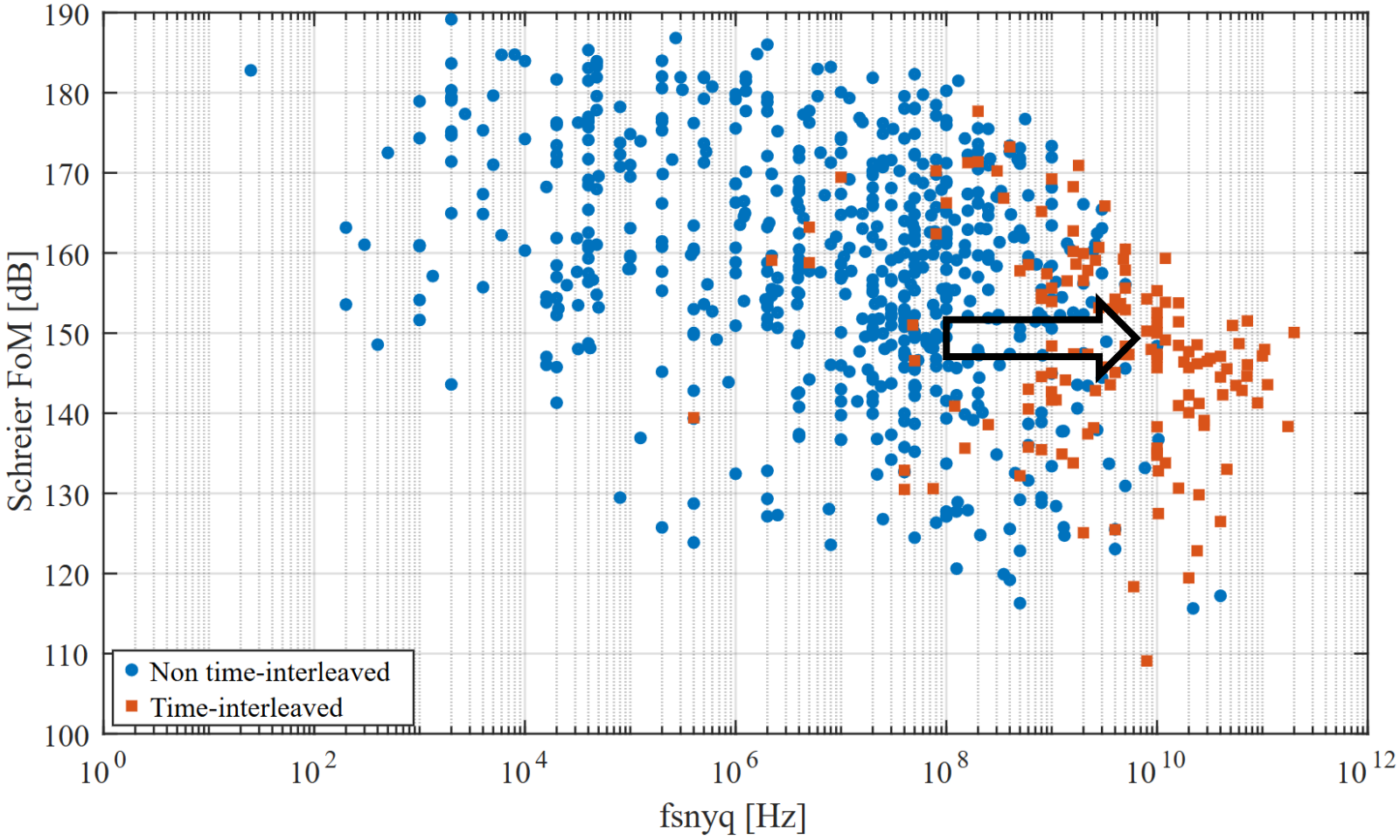
Outline

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- Ultra-low-power SAR ADCs overview: Building blocks perspective
- Wide-bandwidth (GS/s) SAR ADCs
  - Time-Interleaving
- Auxiliary blocks and its requirements
- Conclusion

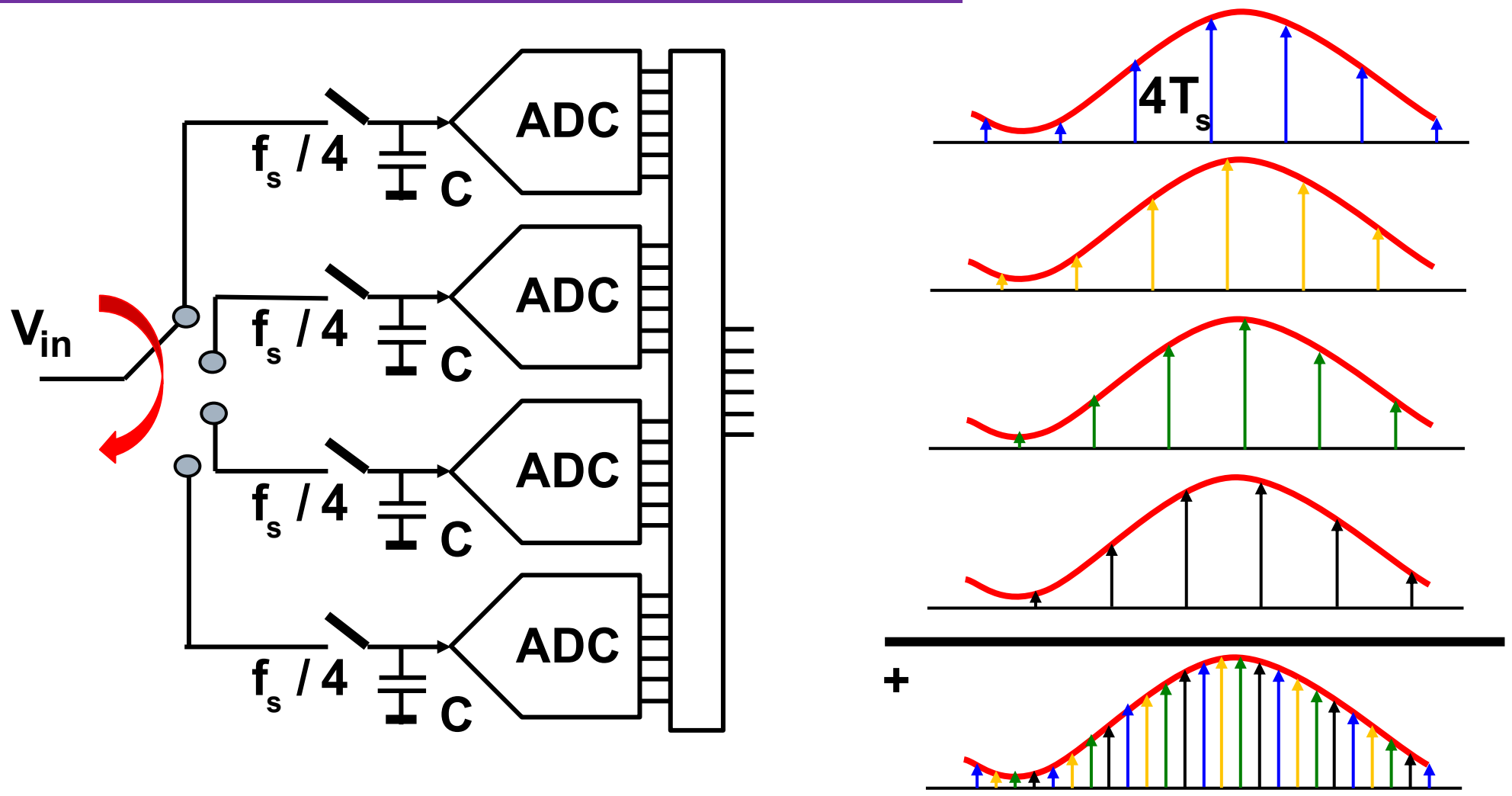
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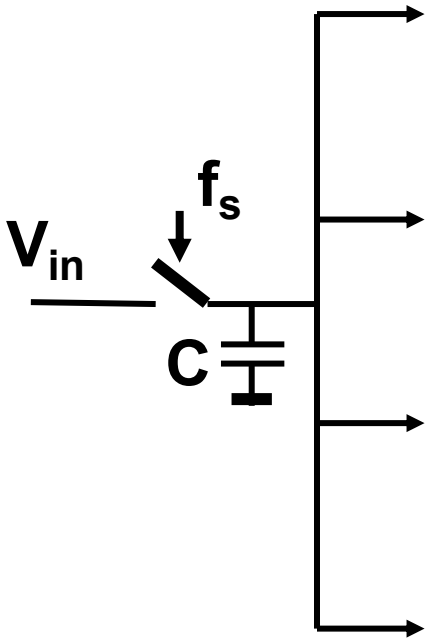
Time-interleaved ADCs

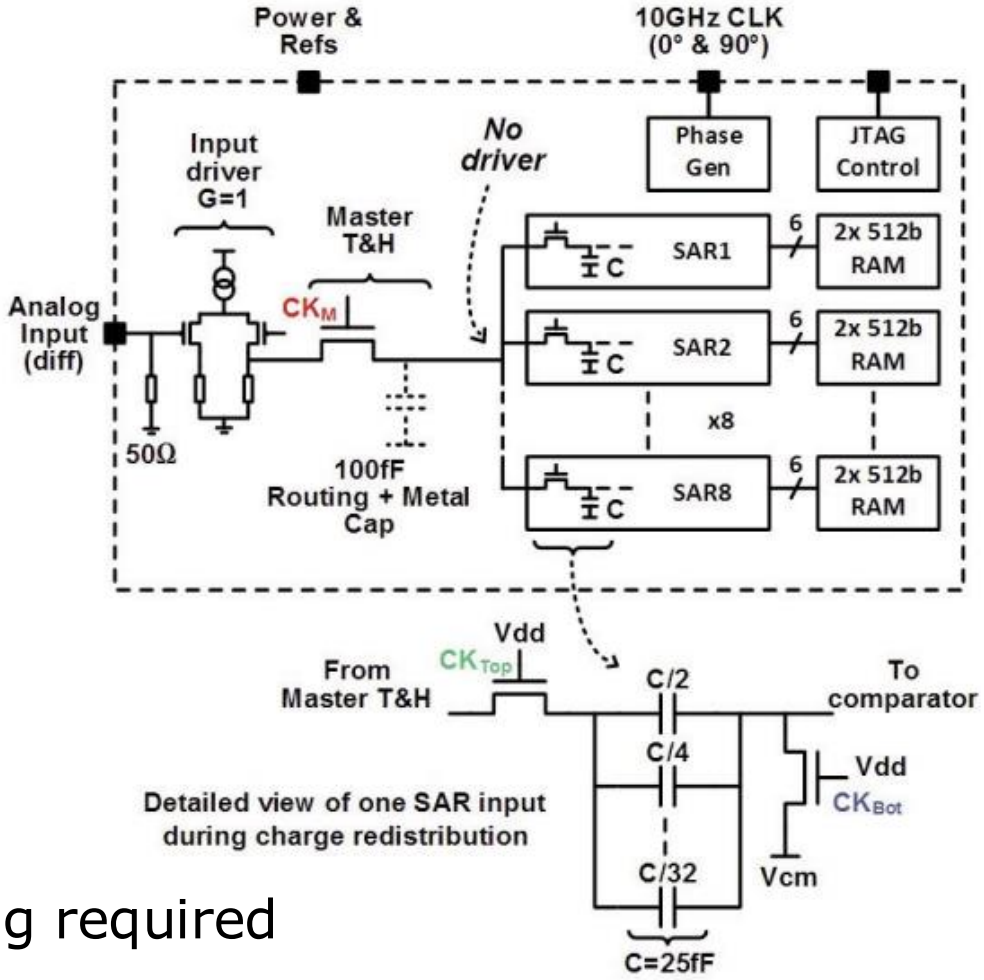


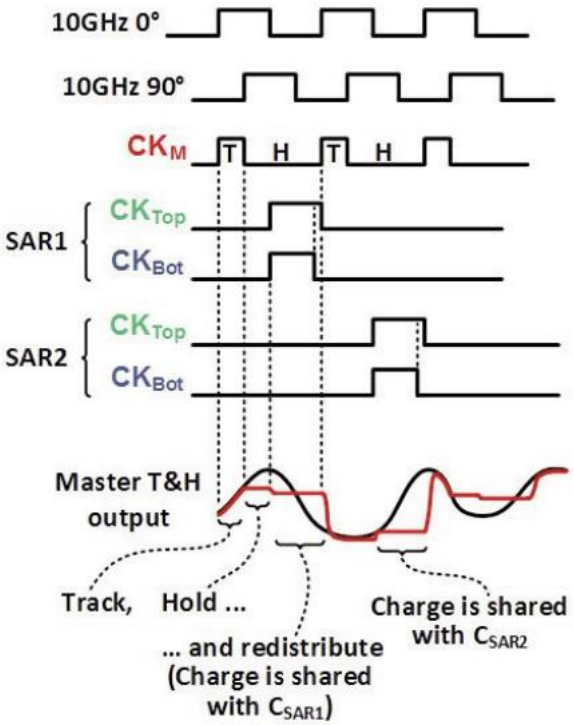
Ideal time-interleaved sampling



Master time-interleaved sampling





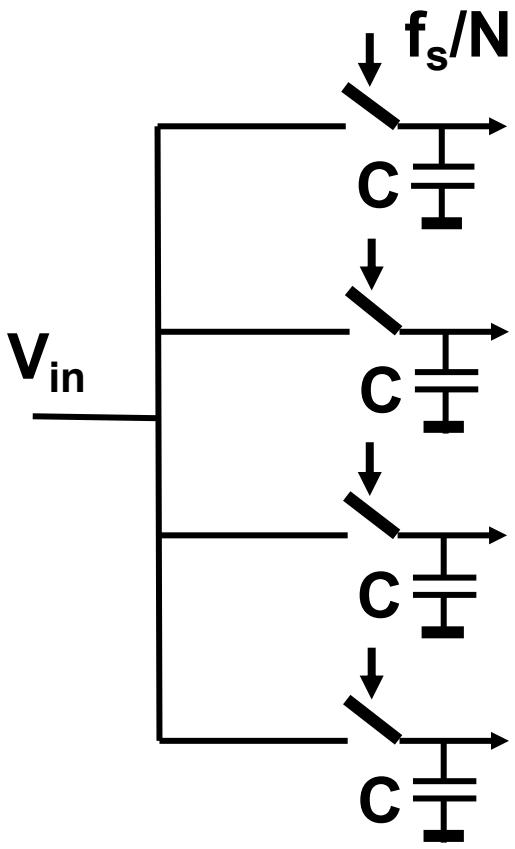


☺ Avoids timing errors

☹ Extreme fast RC settling required

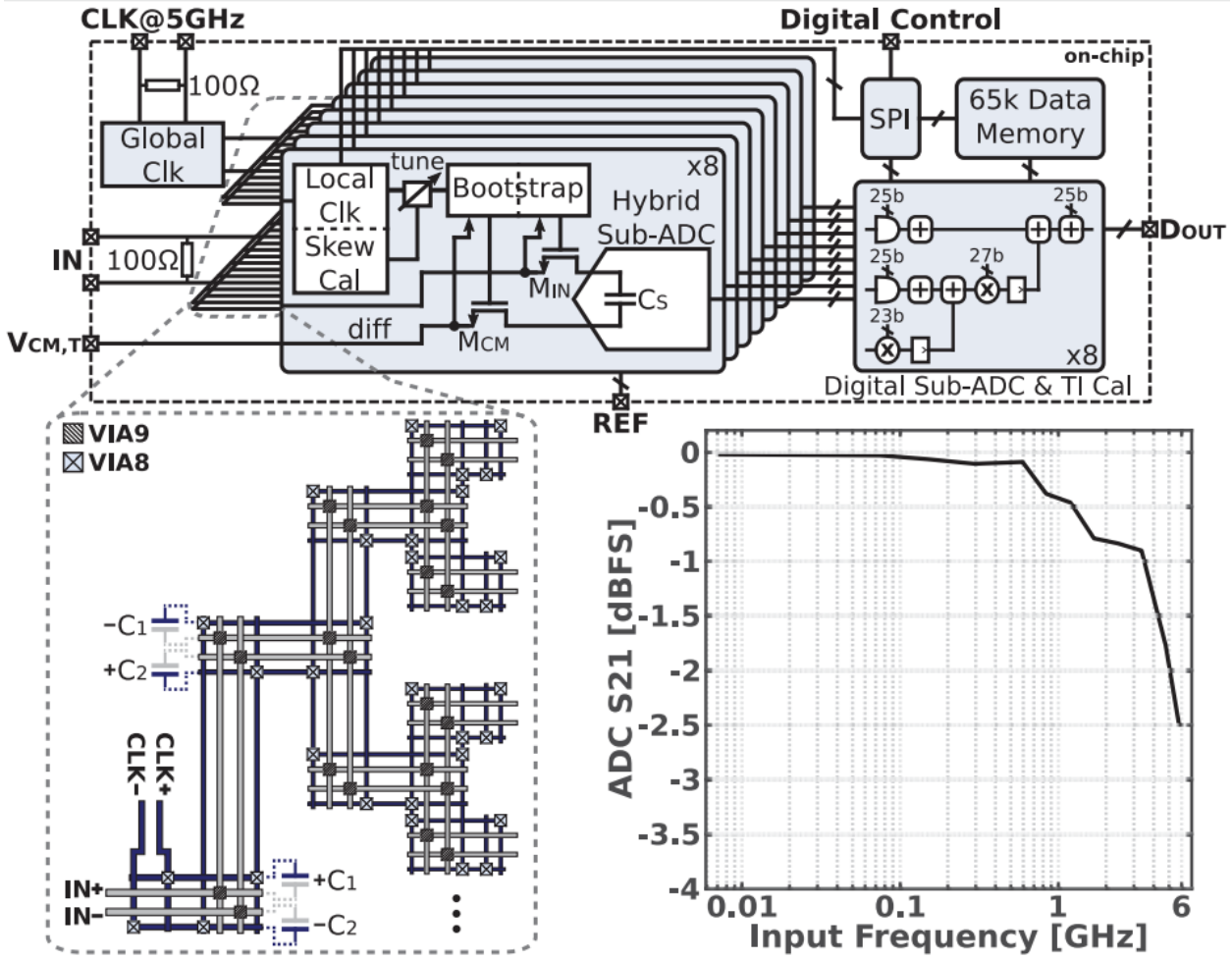
Le Tual et. al ISSCC 2014

Direct time-interleaved sampling



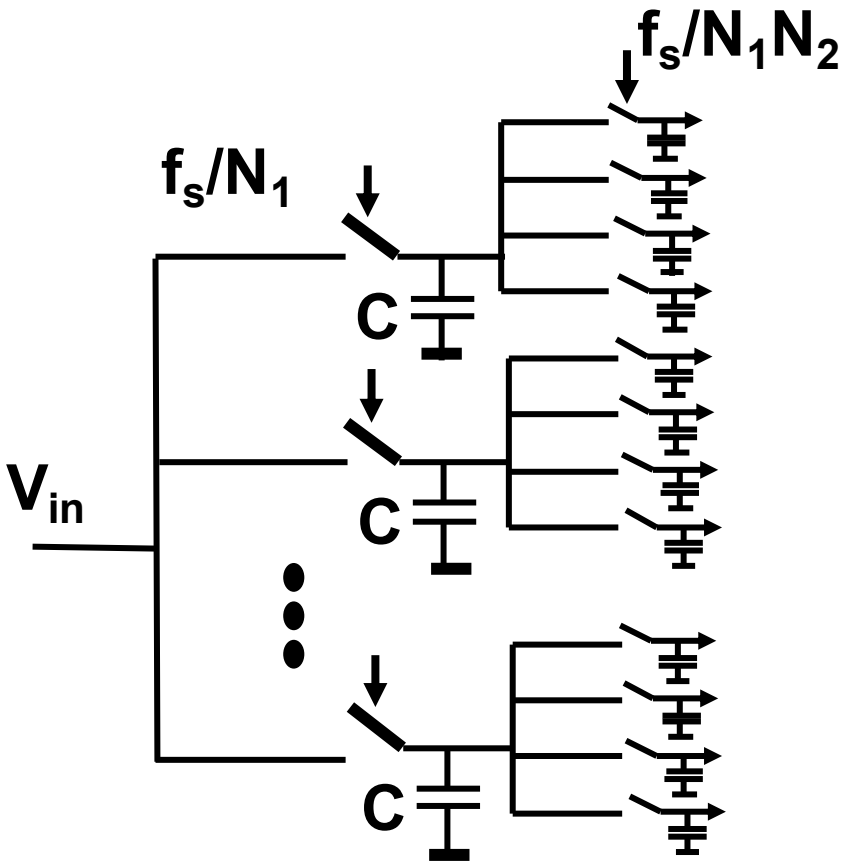
☺ Relaxed RC settling required

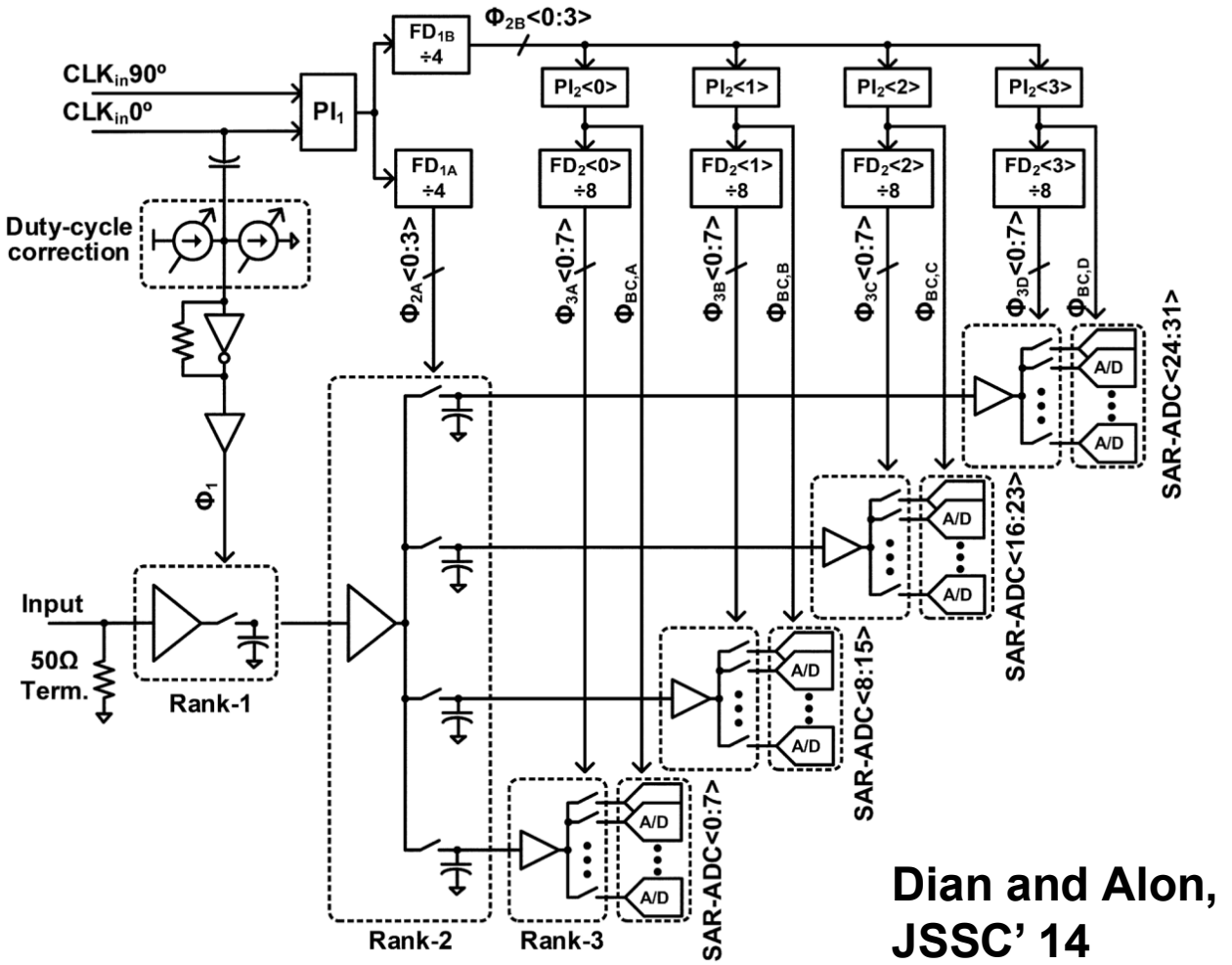
☹ Timing skew and clock sensitive



Ramkaj et. al ISSCC 2019

Hierarchical time-interleaved sampling

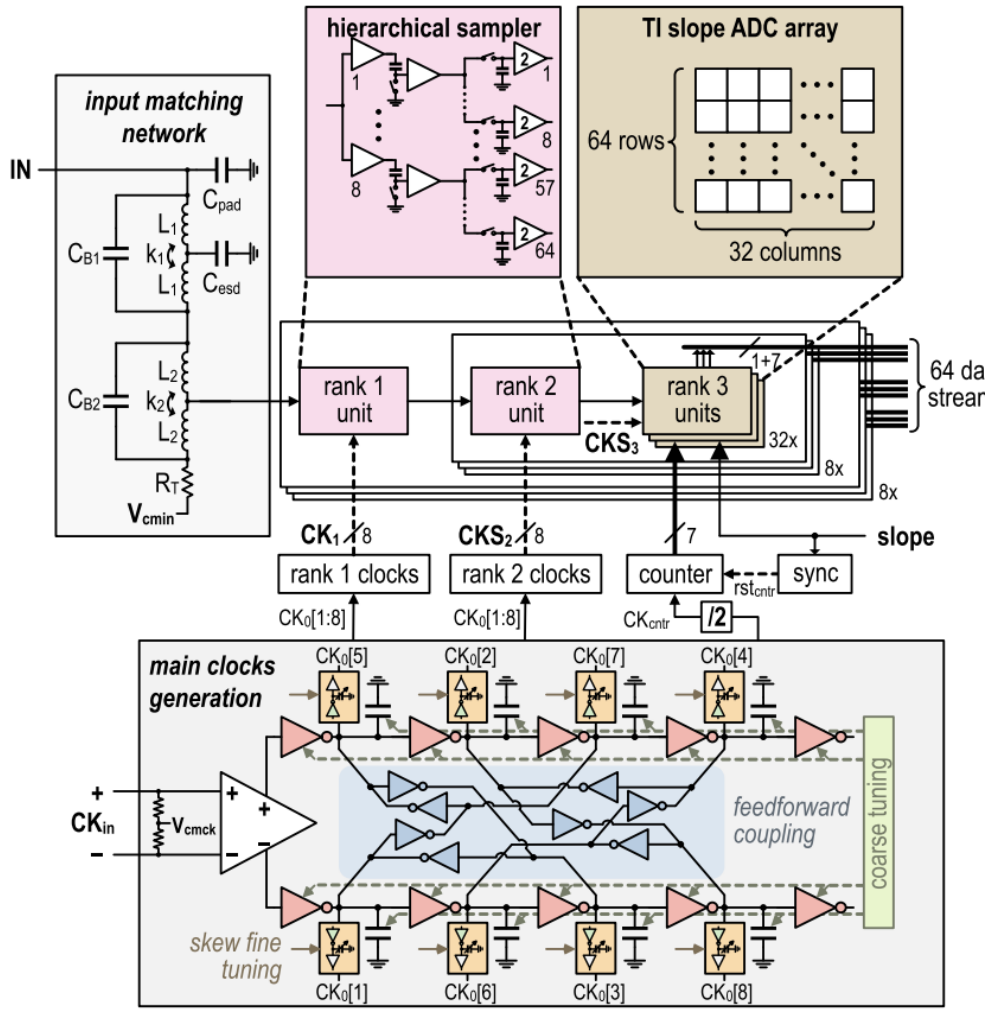




Dian and Alon, JSSC' 14

Relaxed II stage
Massive interleaving -> 100Gbps+

Hierarchical TI ADCs – Recent advancements



Demonstrates scalability of TI slope ADC
Limited 3-dB bandwidth (22GHz)

Martens et. al, ISSCC' 26

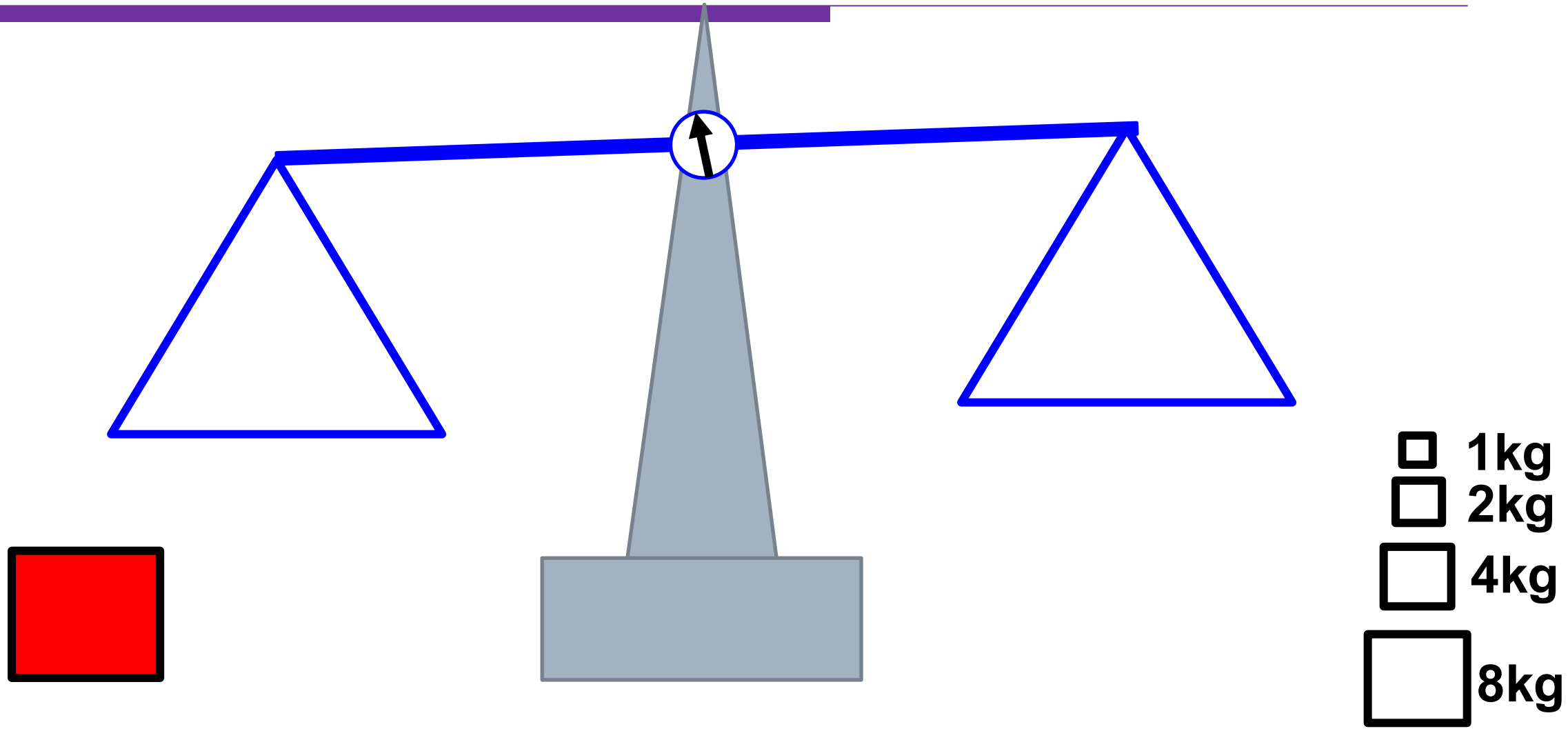
Summary – time interleaving

Architecture	Clock Jitter / Skew Sensitivity	Input Loading / Parasitic Cap.	Noise (kT/C) & Distortion	Max Scalability / Speed	Calibration Complexity
Direct Sampling	Extremely high SKEW sensitivity (N clock phases must match);	High; scales linearly with N	Low intrinsically (no resampling stages), large-N loading causes bandwidth-limited distortion	Moderate; impractical beyond tens of GS/s without buffering/matching networks	High; N-way skew calibration, grows with channel count
Master Sampling	Low SKEW (all channels inherit one held value); Single master switch's own jitter is critical	Low ; input sees only one master switch/capacitor	Moderate; one extra kT/C + nonlinearity stage from the master track-and-hold	Medium; master switch's bandwidth and settling bottleneck at extreme speeds	Low; only one critical clock/switch to characterize
Hierarchical Sampling	Medium; rank 1 sets the fundamental aperture (needs low jitter); later ranks need only alignment,	Medium-to-low loading decoupled across ranks	High; noise/attenuation accumulates across cascaded resampling stages,	Extremely high; ideal for >100+ GS/s	Medium; rank-to-rank alignment per-channel offset/gain correction,

Outline

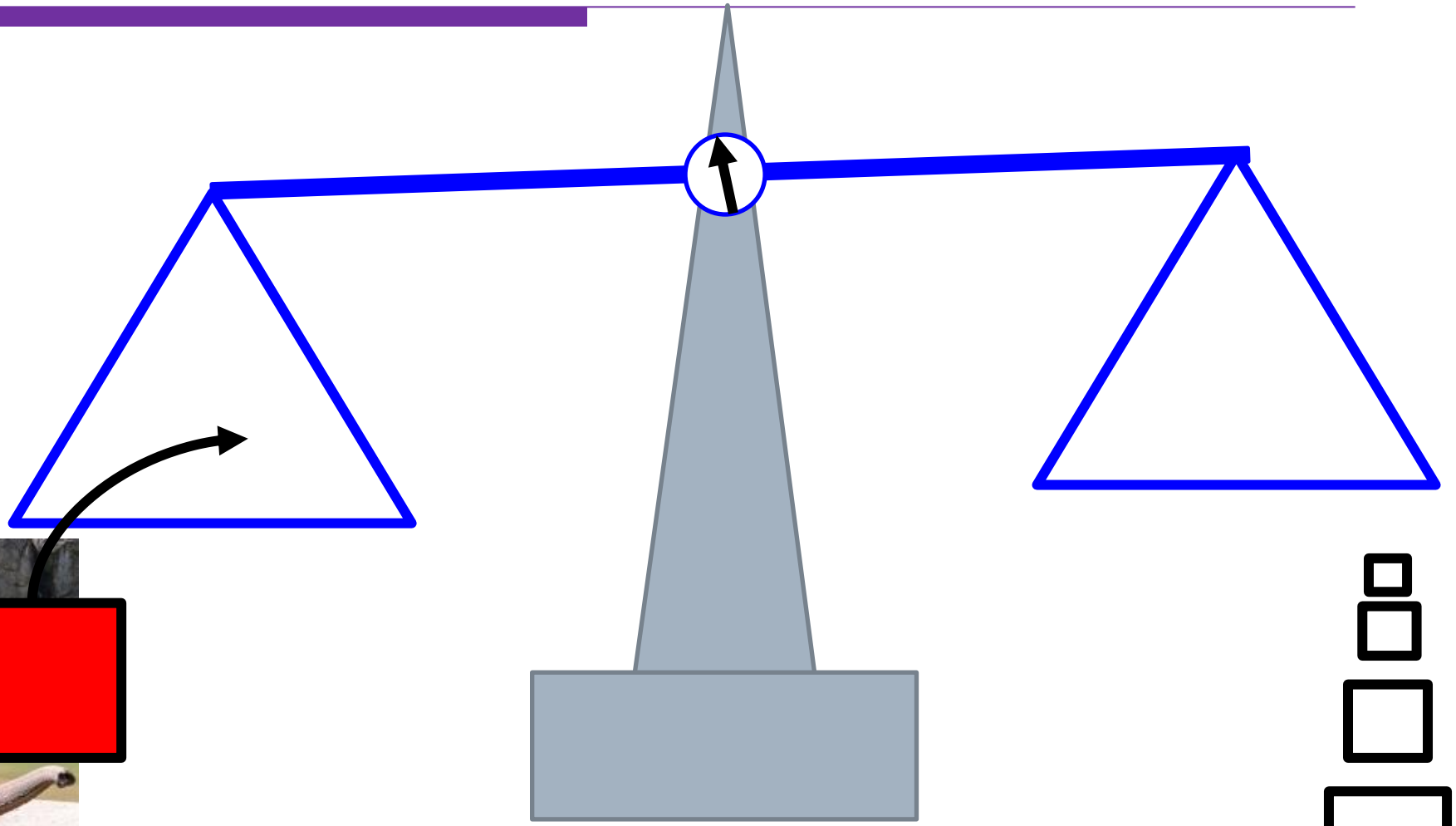
- Introduction
- Ultra-low-power SAR ADCs overview: Building blocks perspective
- Wide-bandwidth (GS/s) SAR ADCs
- Auxiliary blocks and its requirements
  - Input Driver
- Conclusion

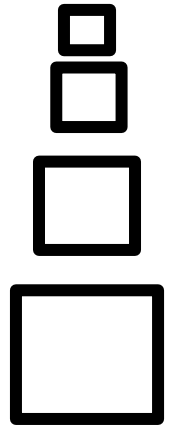
Successive Approximation - Analogy



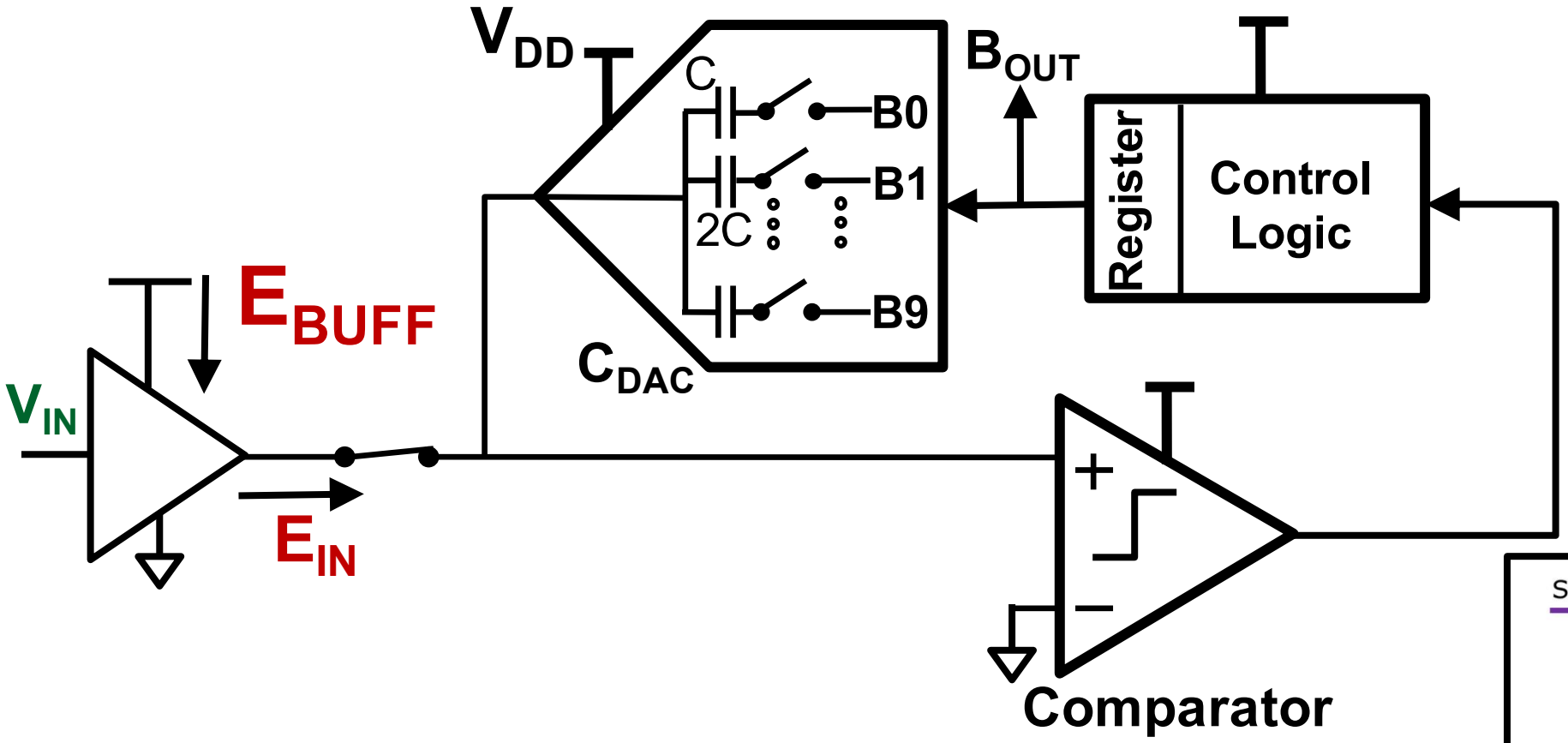
Successive Approximation - Analogy

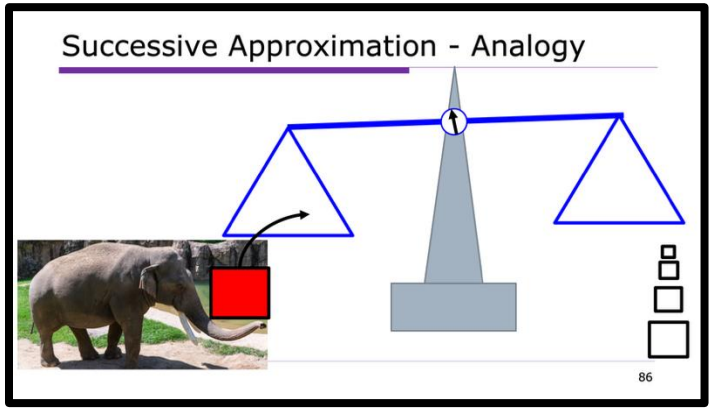






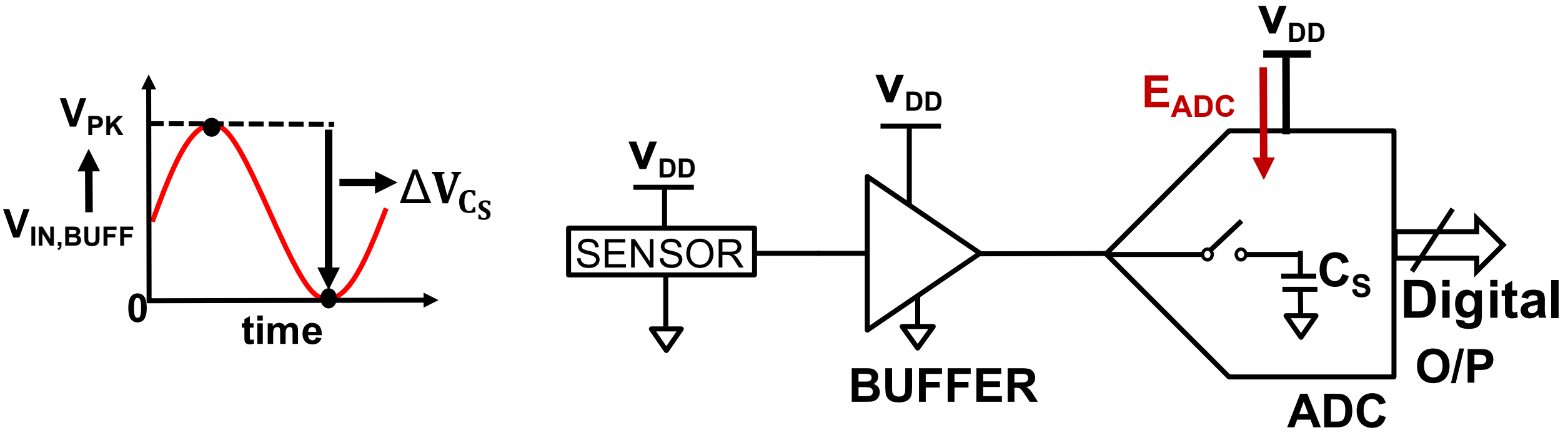
SAR ADC – Block diagram





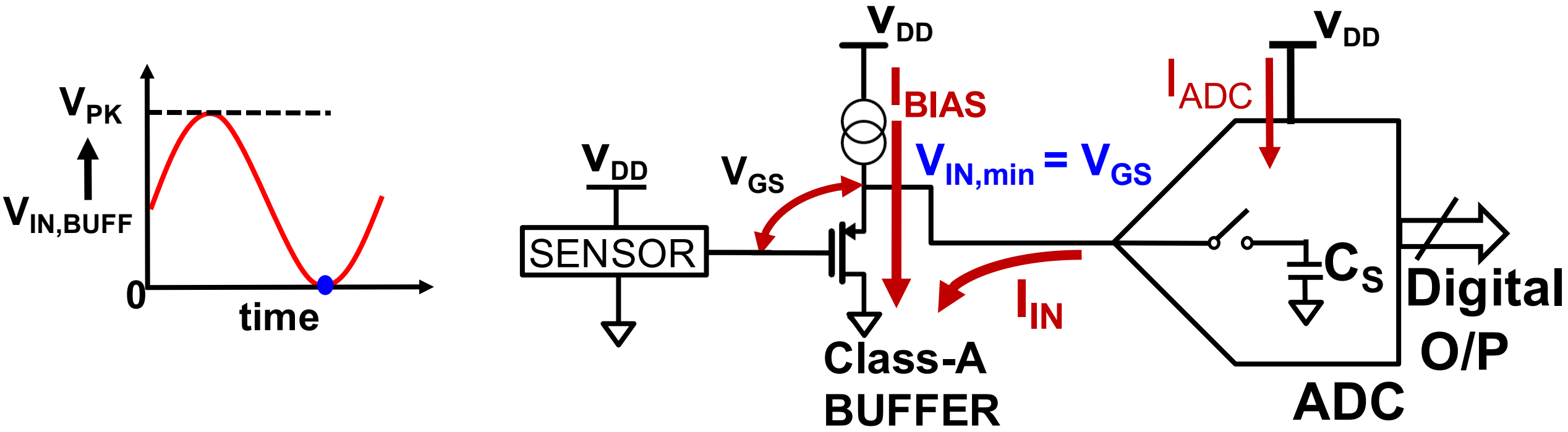
An input signal buffer or driver circuit is required to put V_{IN} on C_{DAC} – Not part of E_{ADC} !!

Motivation – Input Buffer



Buffer the input signal !!

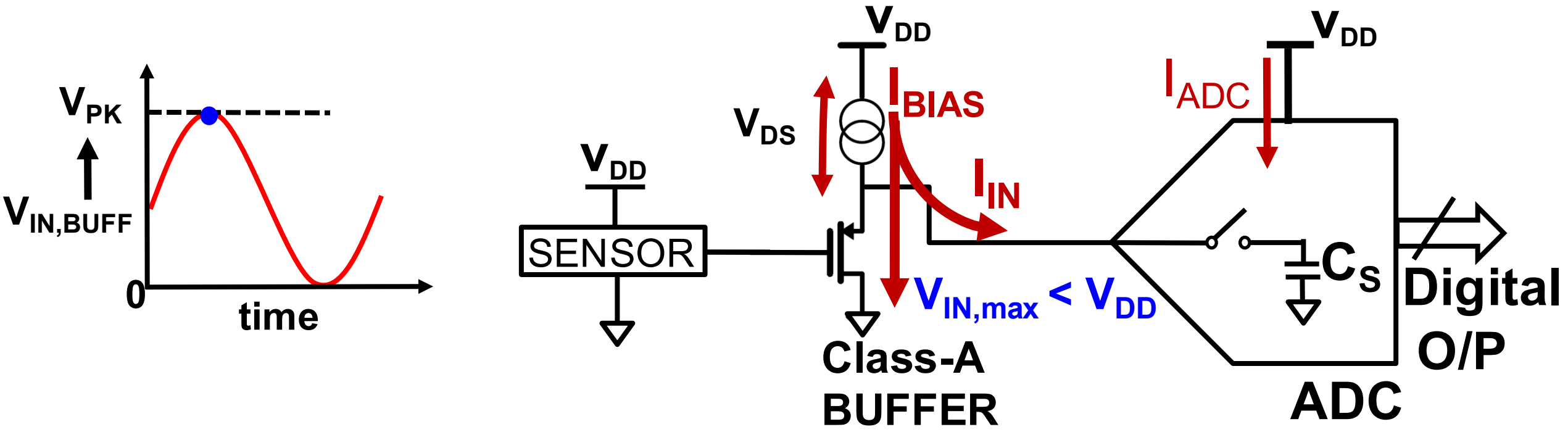
Motivation – Input Buffer Requirement



☺ Class-A buffer : low distortion, always “ON” output stage.

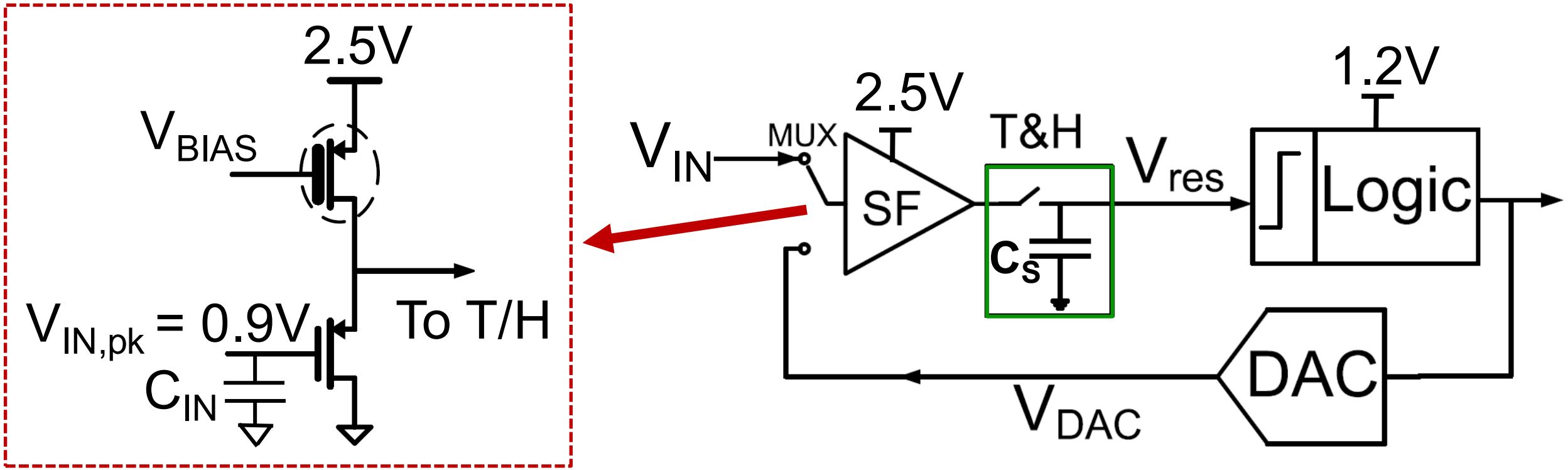
☹ $V_{IN,min} > V_{GS}$. Rail-to-rail operation not feasible with single supply.

Motivation – Input Buffer Requirement



- ☹ $I_{BIAS} \geq I_{IN,MAX}$ and $V_{IN,max} < V_{DD} - V_{DS}$
- ☹ Power hungry and poor efficiency. $P_{BUFF} \geq V_{PK} I_{BIAS}$

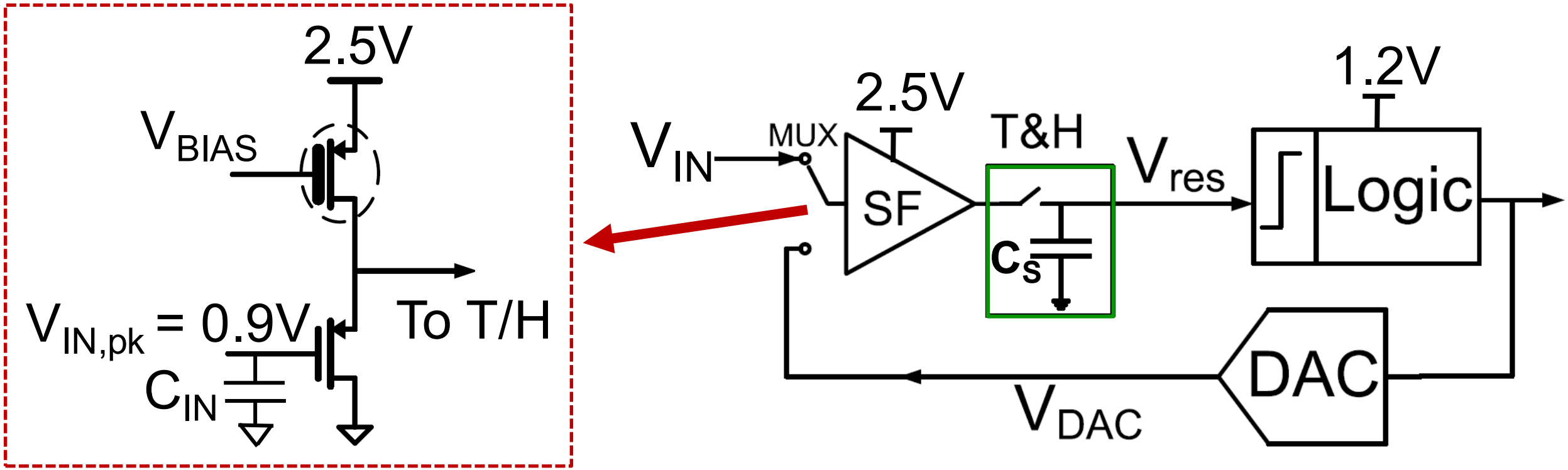
Source Follower embedded in SAR



- ☺ Input capacitance (C_{IN}) is 0.2pF (18x lower than C_S)
- ☺ Non-linearity of SF is cancelled inside SAR loop

[Krämer, *JSSC 2015*]

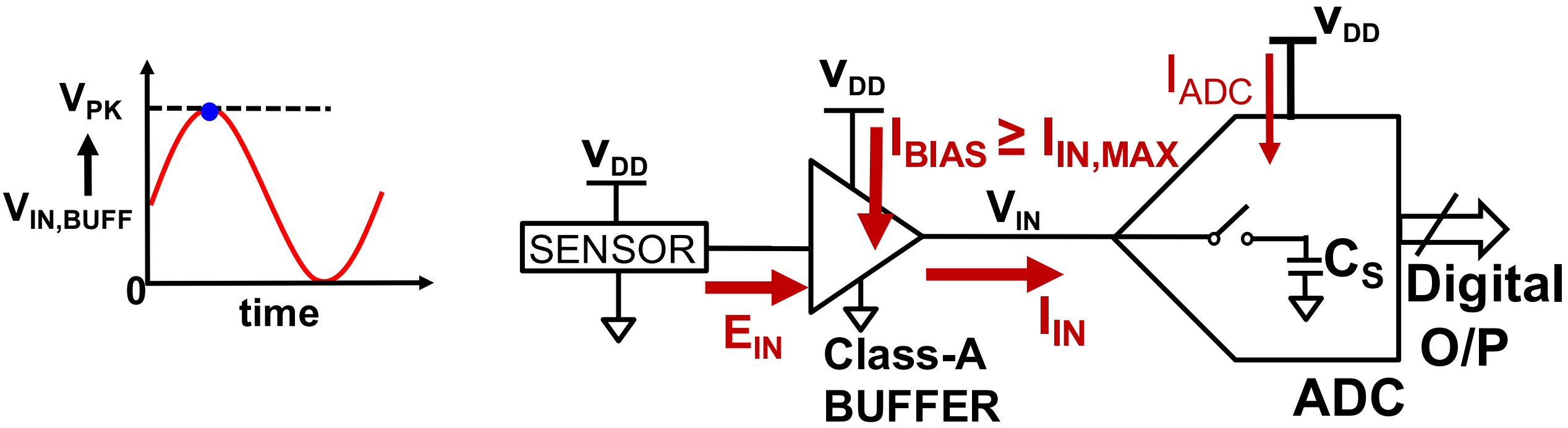
Source Follower embedded in SAR



☹ Additional high supply voltages (≈ 3xV_{PK}) required

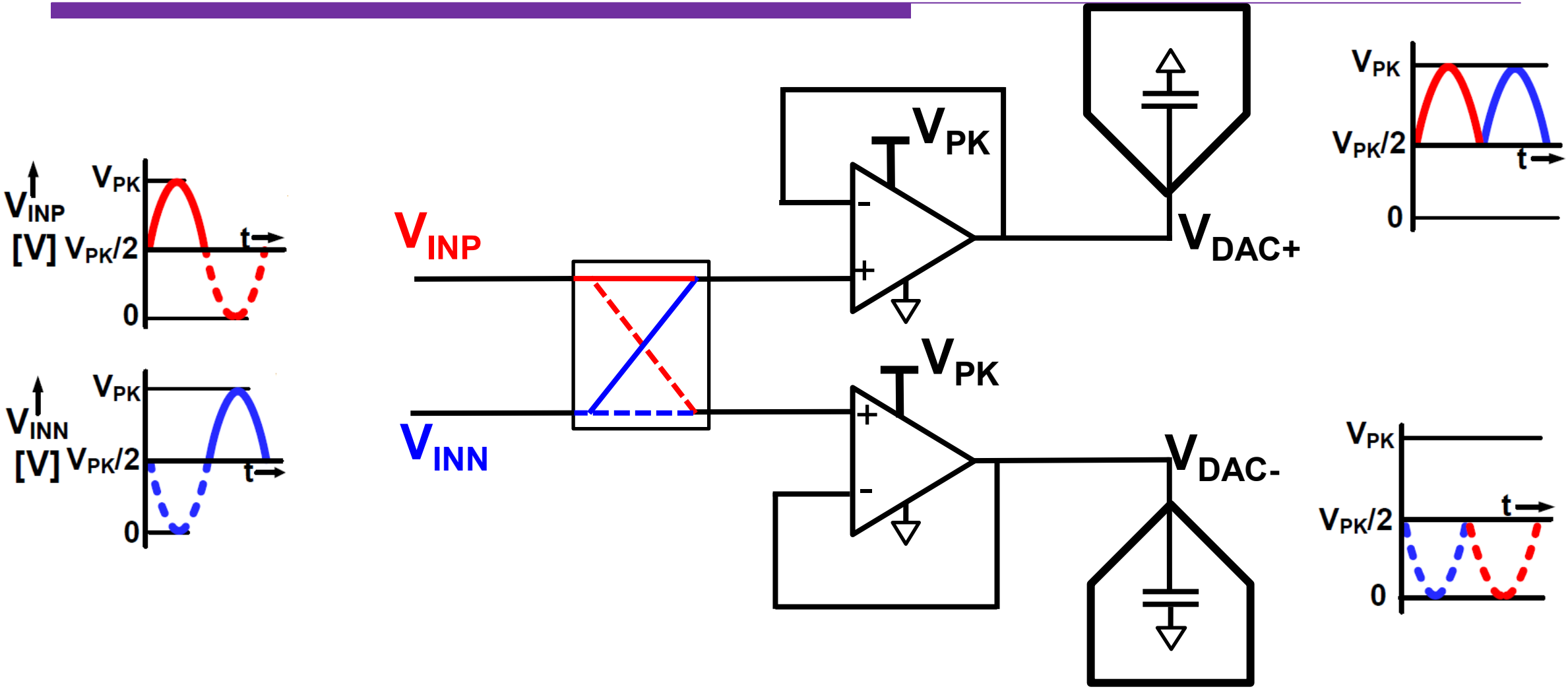
[Krämer, *JSSC 2015*]

Motivation – Input Buffer Requirement

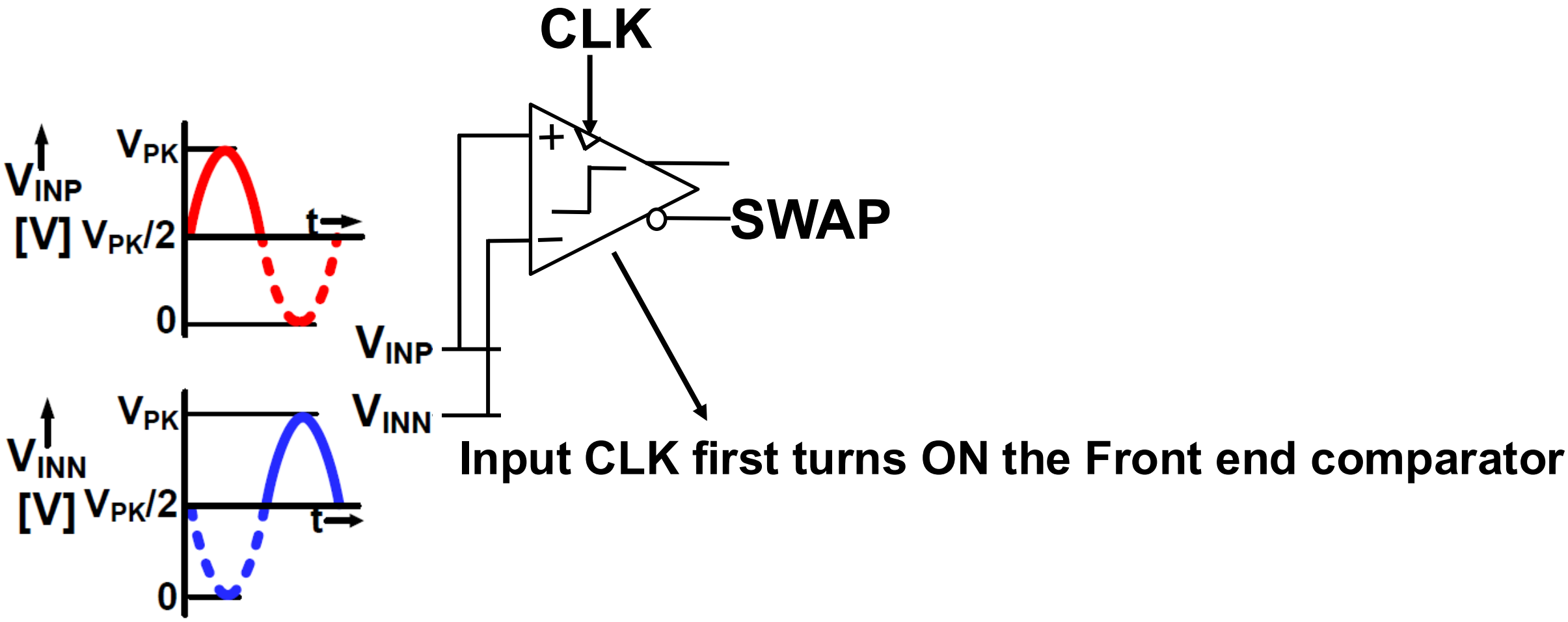


- Single V_{DD} with near rail-to-rail signal swing.
- Minimize I_{BIAS} (or $I_{IN,MAX}$) ⟶ low $FoM_{W,buff}$.
- Minimize E_{IN}.

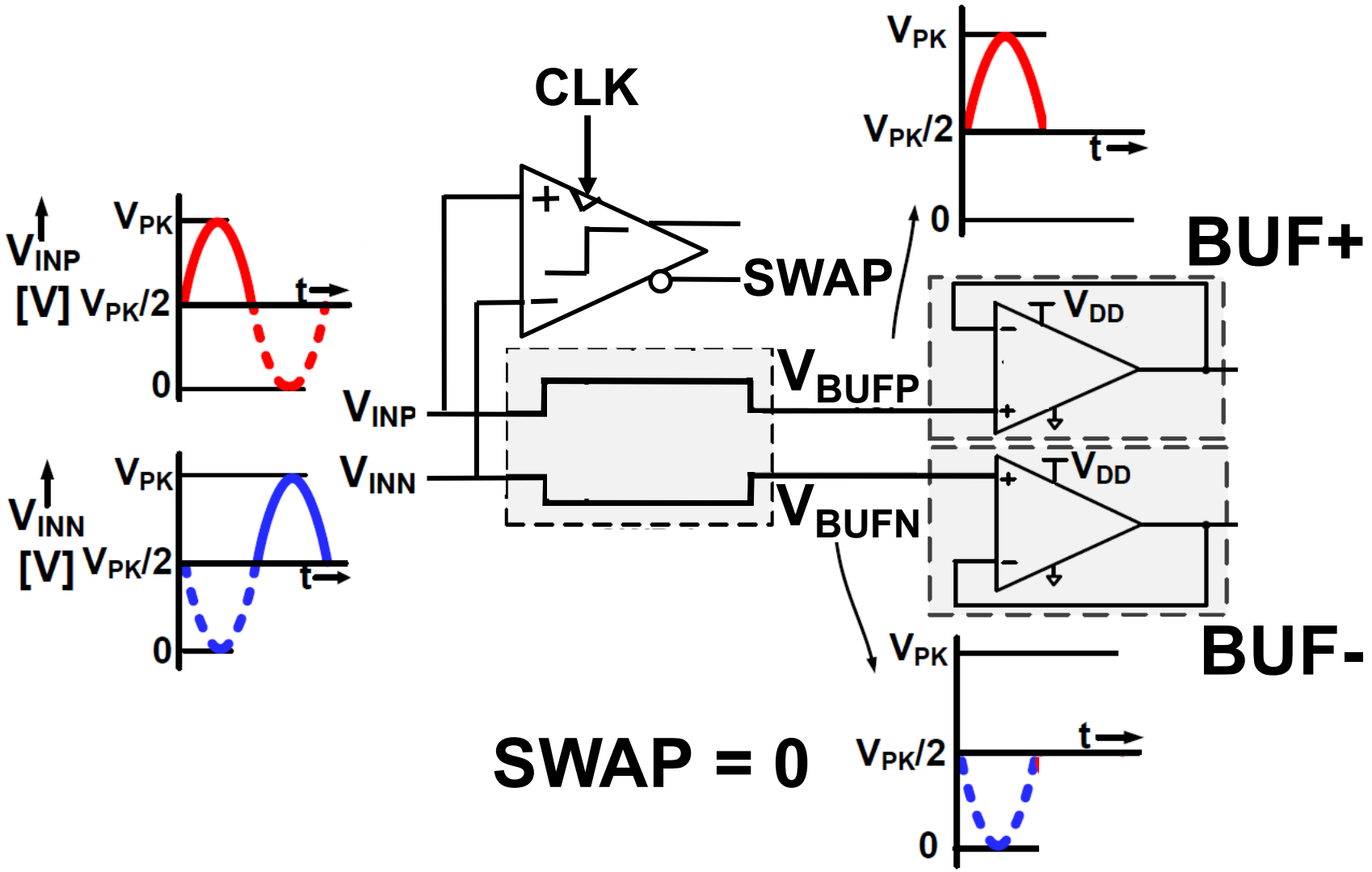
ADC Integrated with Buffer



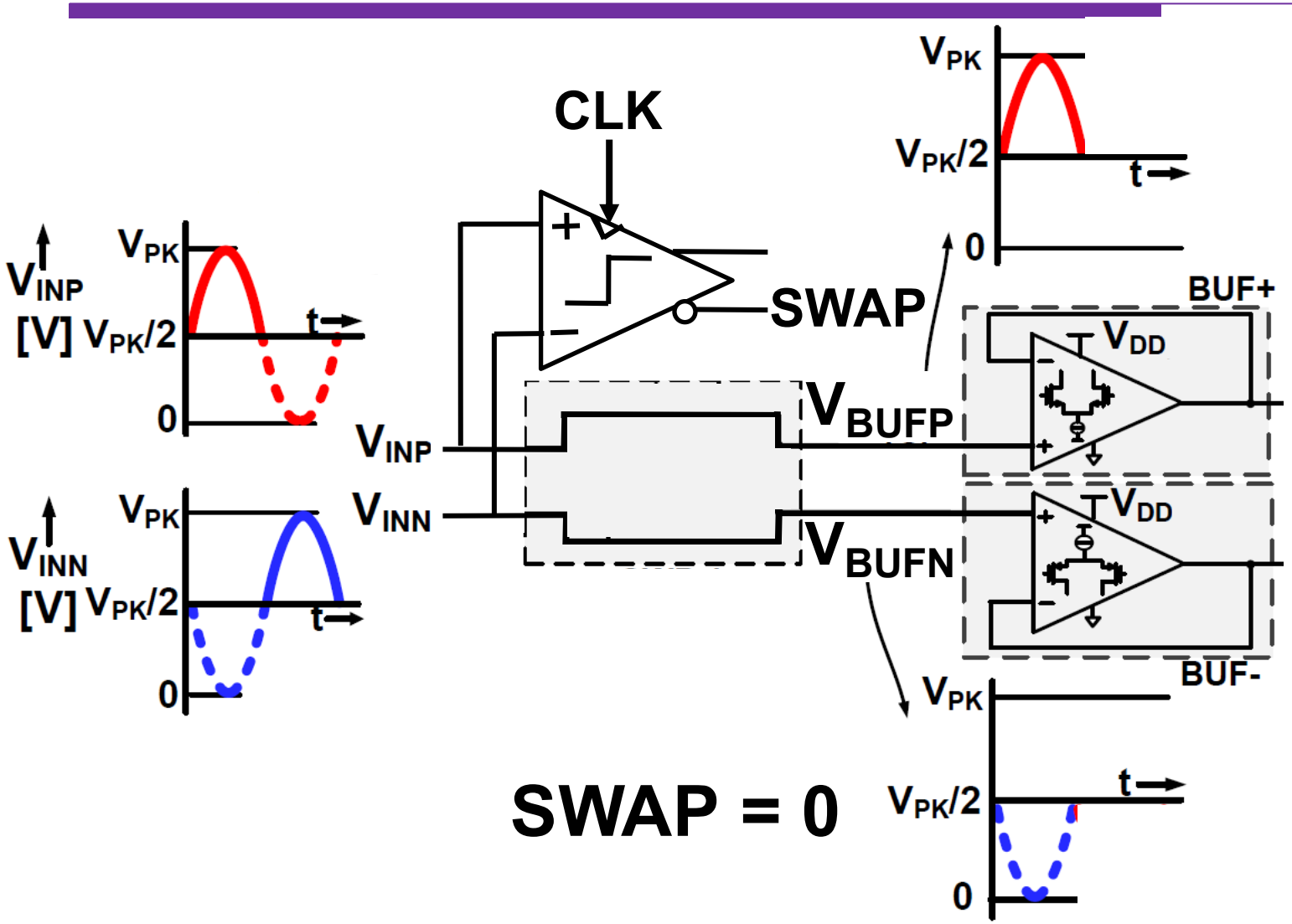
Proposed Architecture - Front End

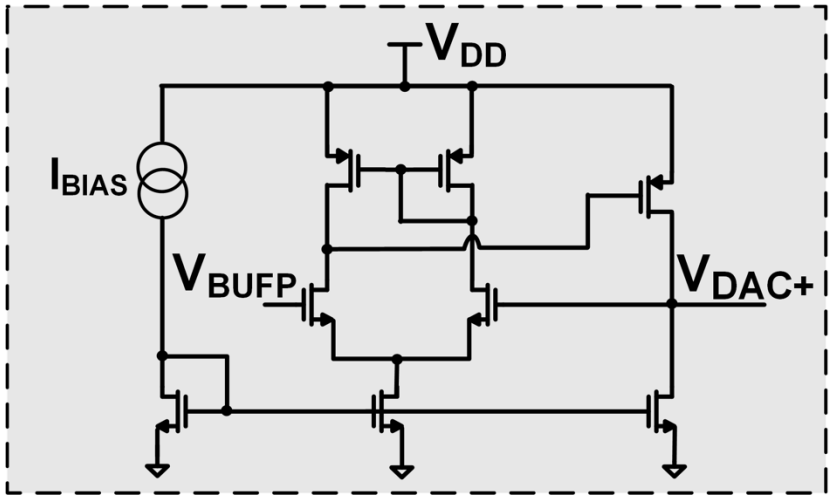


Architecture - Front End

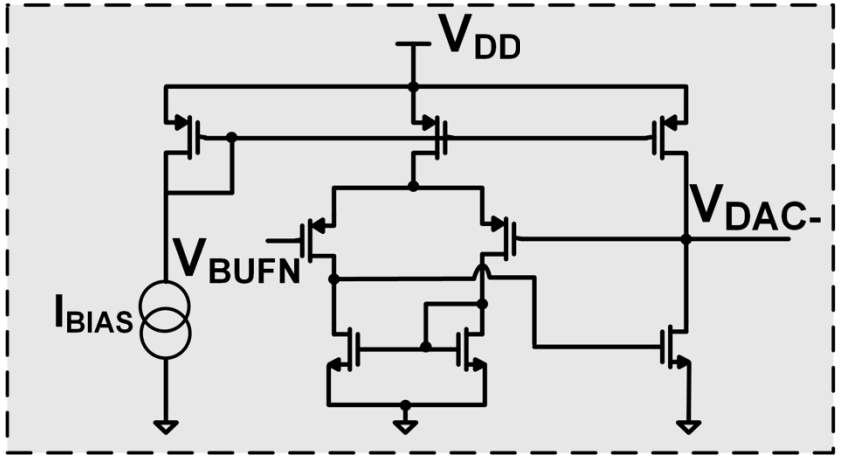


Architecture - Front End

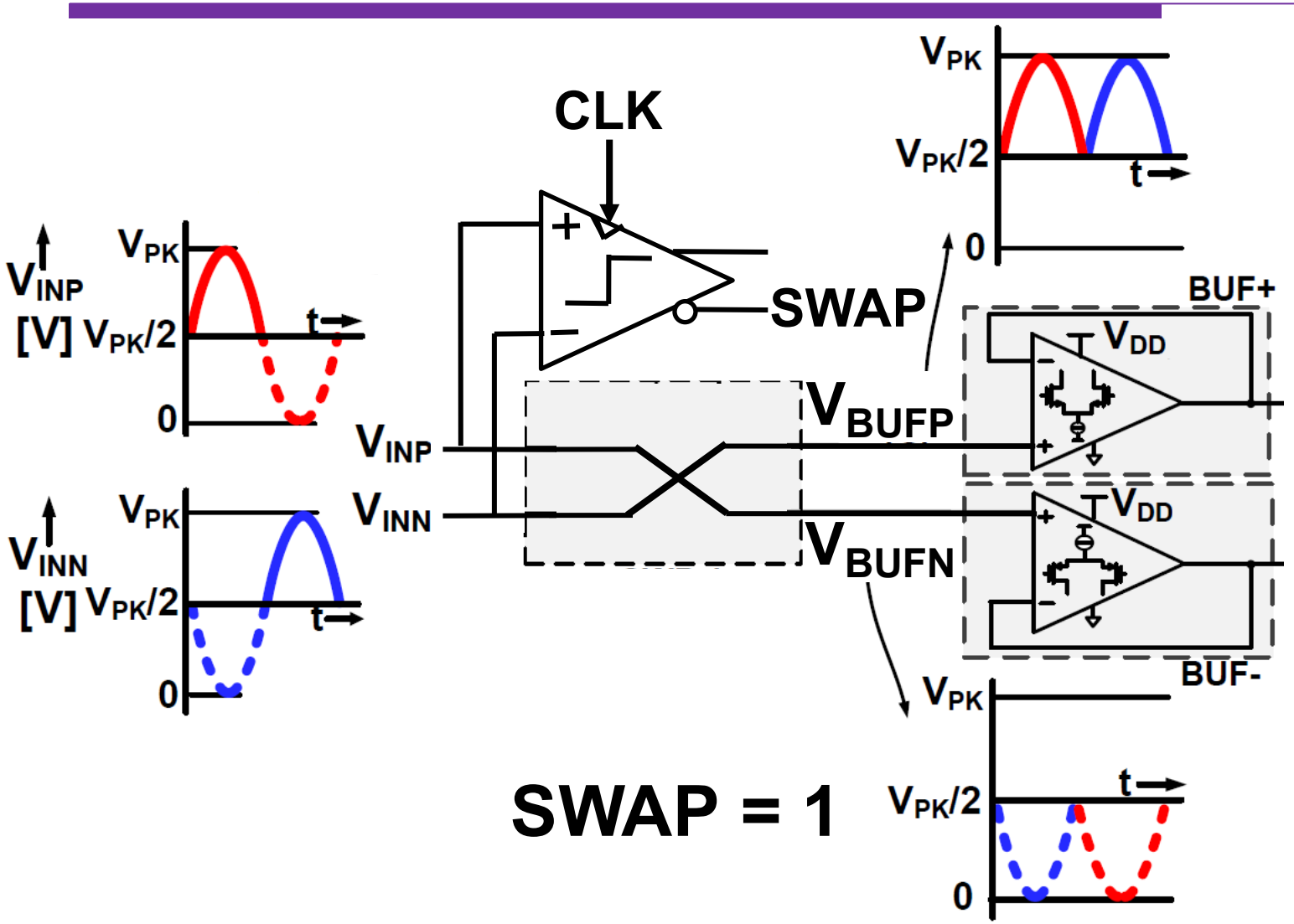




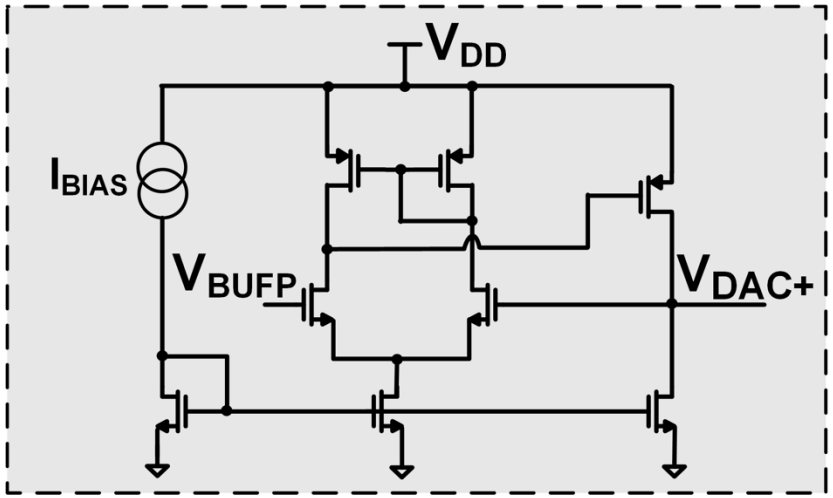
$V_{DD} \approx V_{PK}$



Architecture - Front End

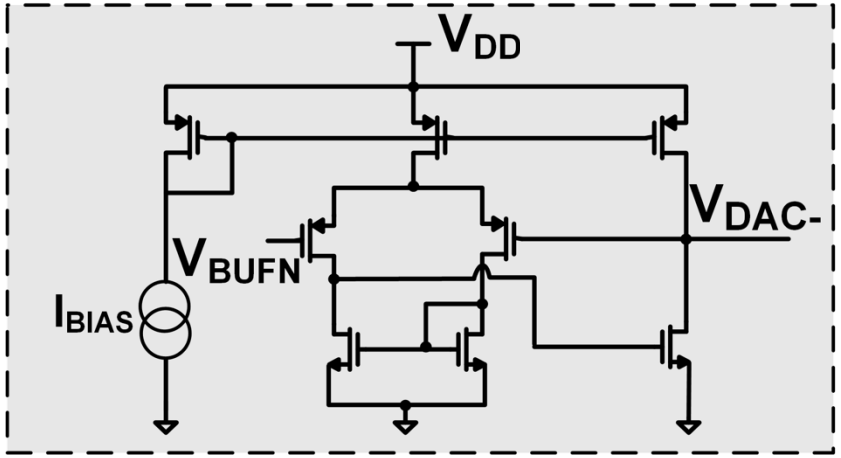


BUF+

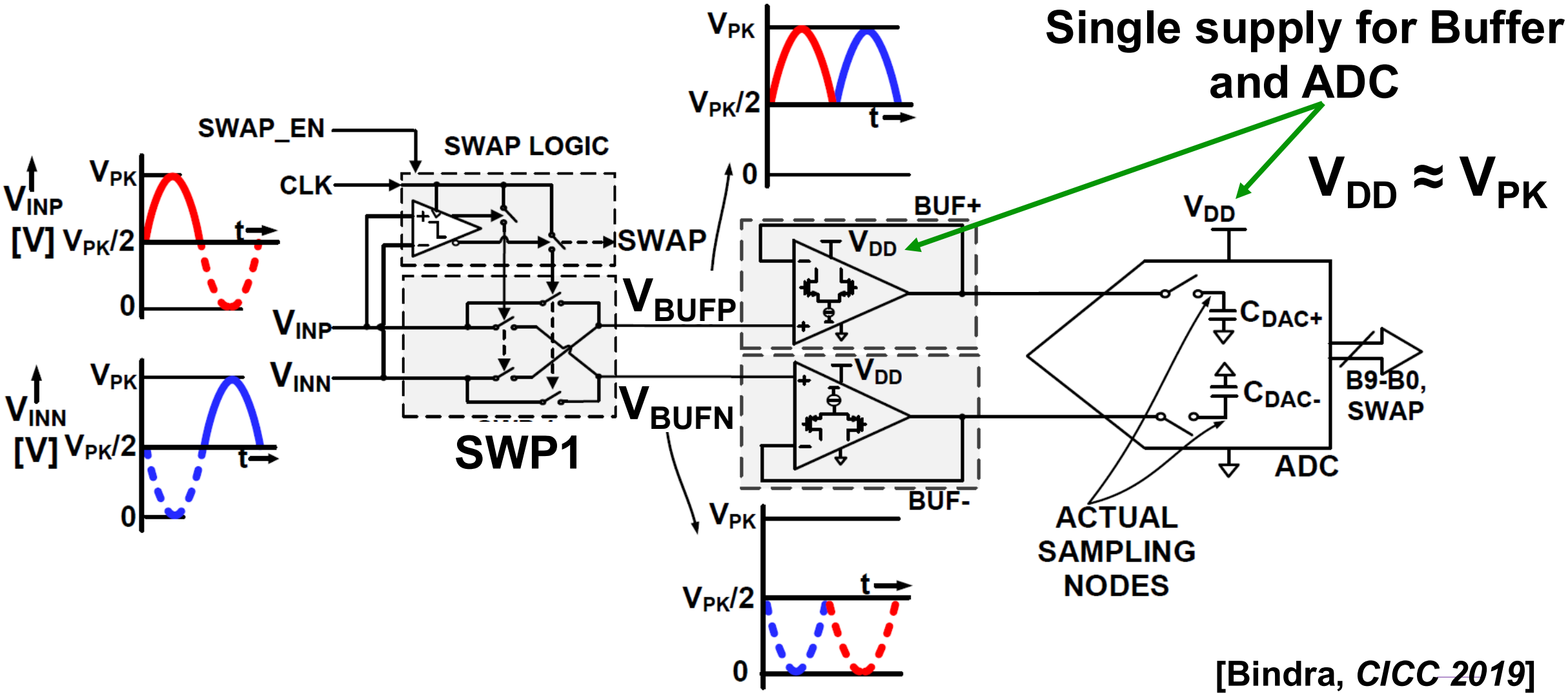


$V_{DD} \approx V_{PK}$

BUF-

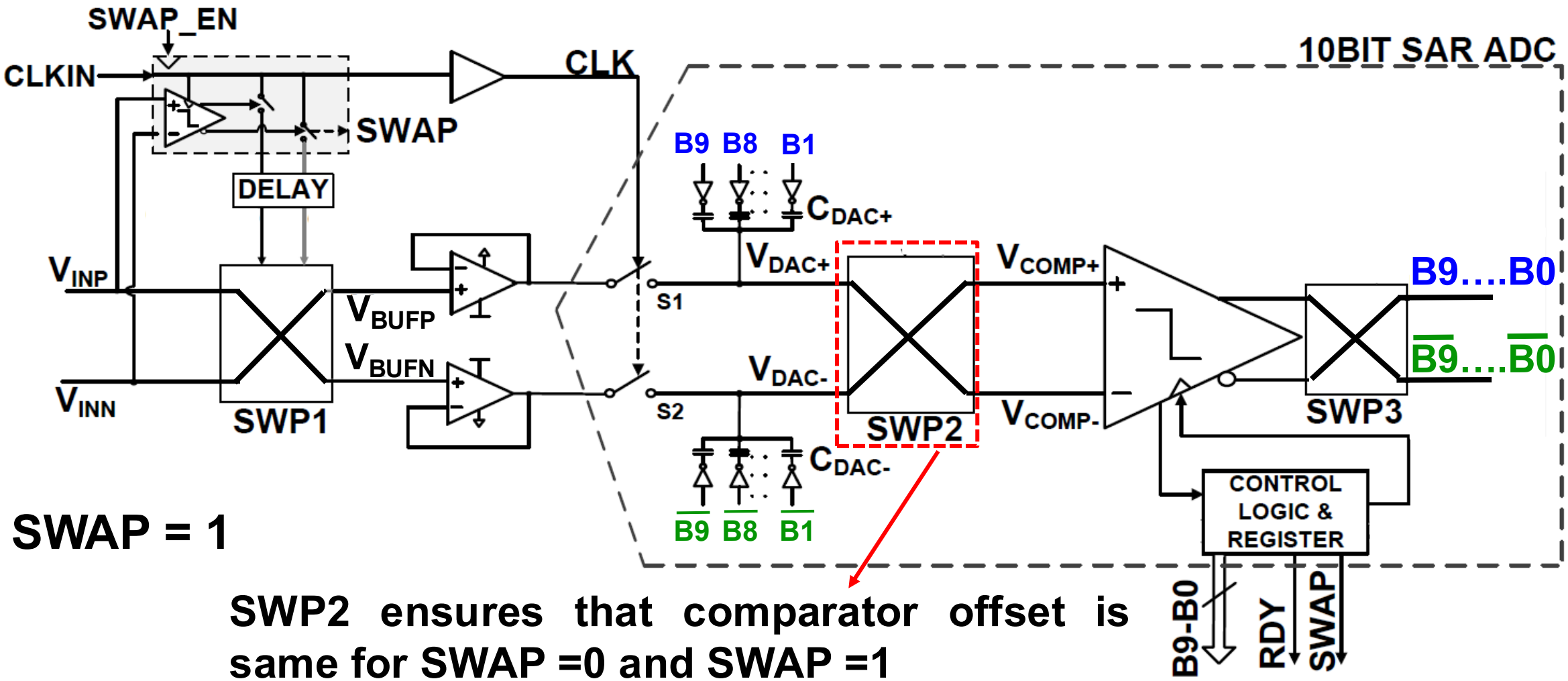


Architecture - Front End



[Bindra, *CICC 2019*]

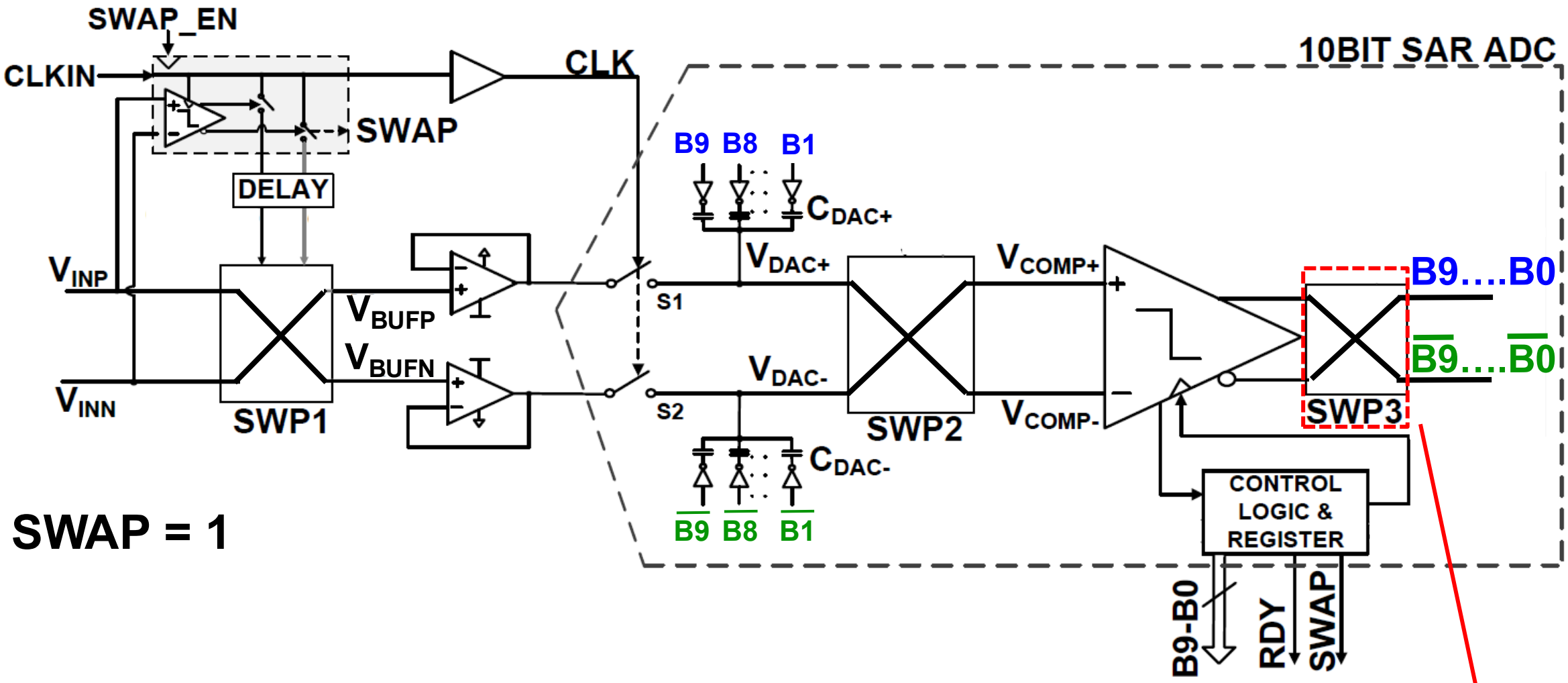
Architecture - Buffers integrated with ADC



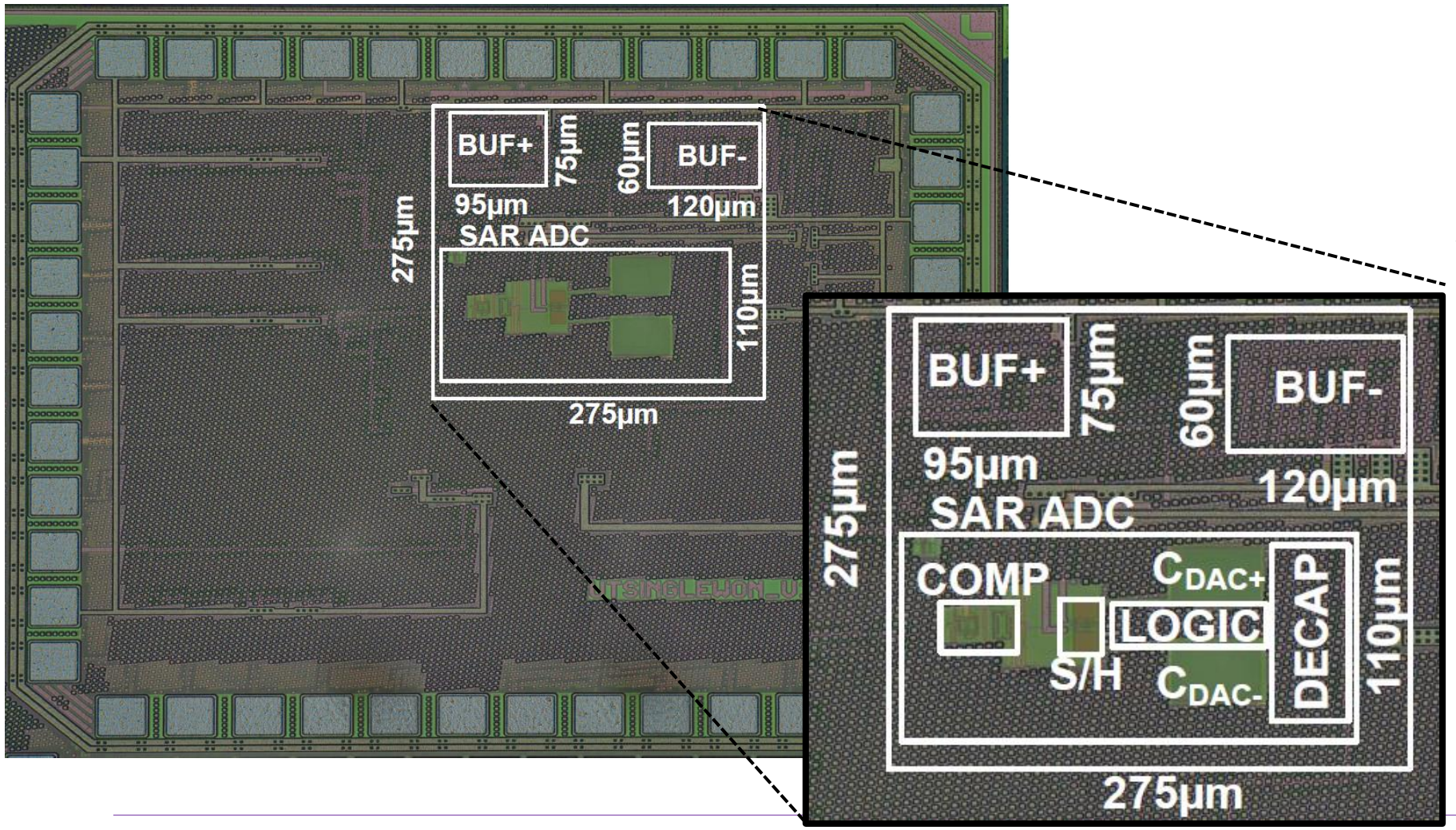
SWAP = 1

SWP2 ensures that comparator offset is same for SWAP =0 and SWAP =1

Architecture - Buffers integrated with ADC



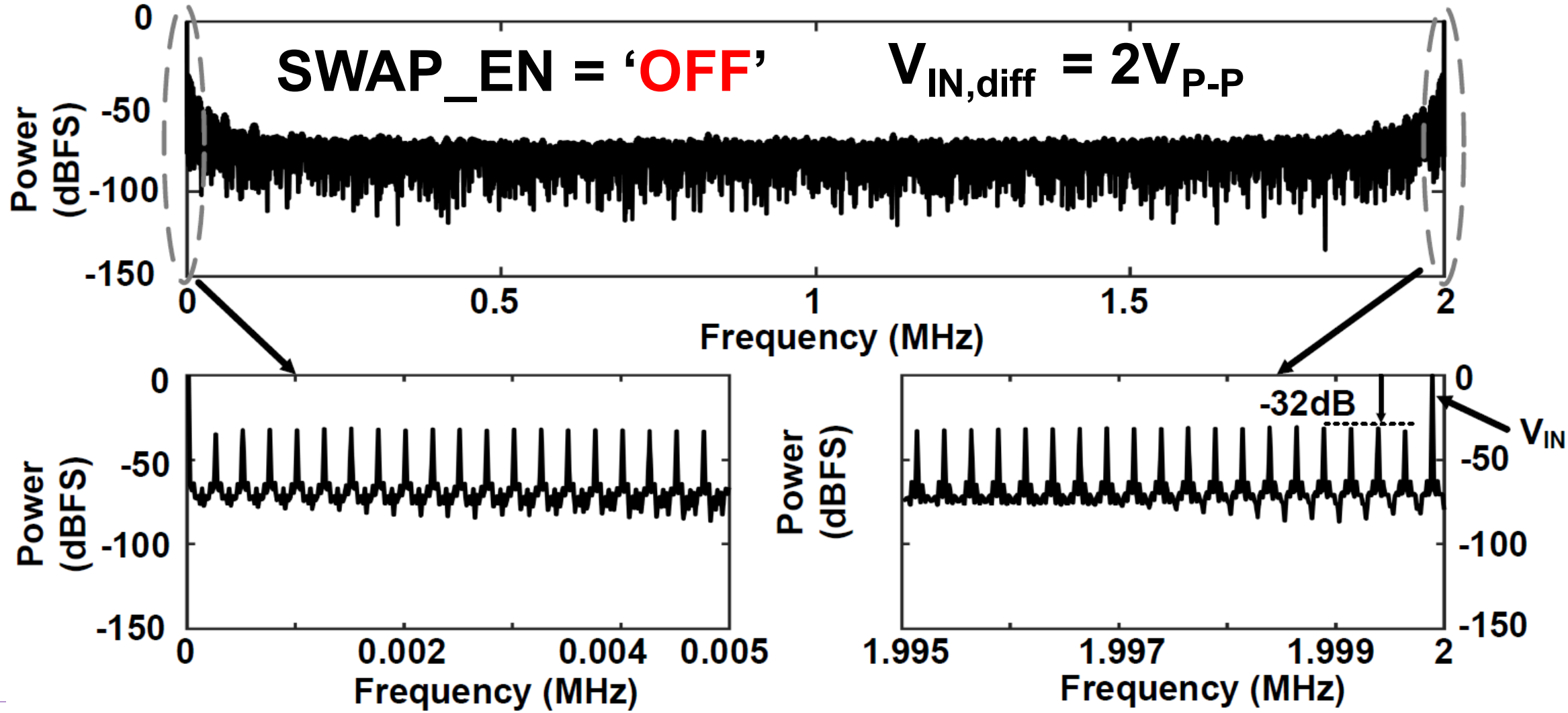
Die micrograph (65nm CMOS)



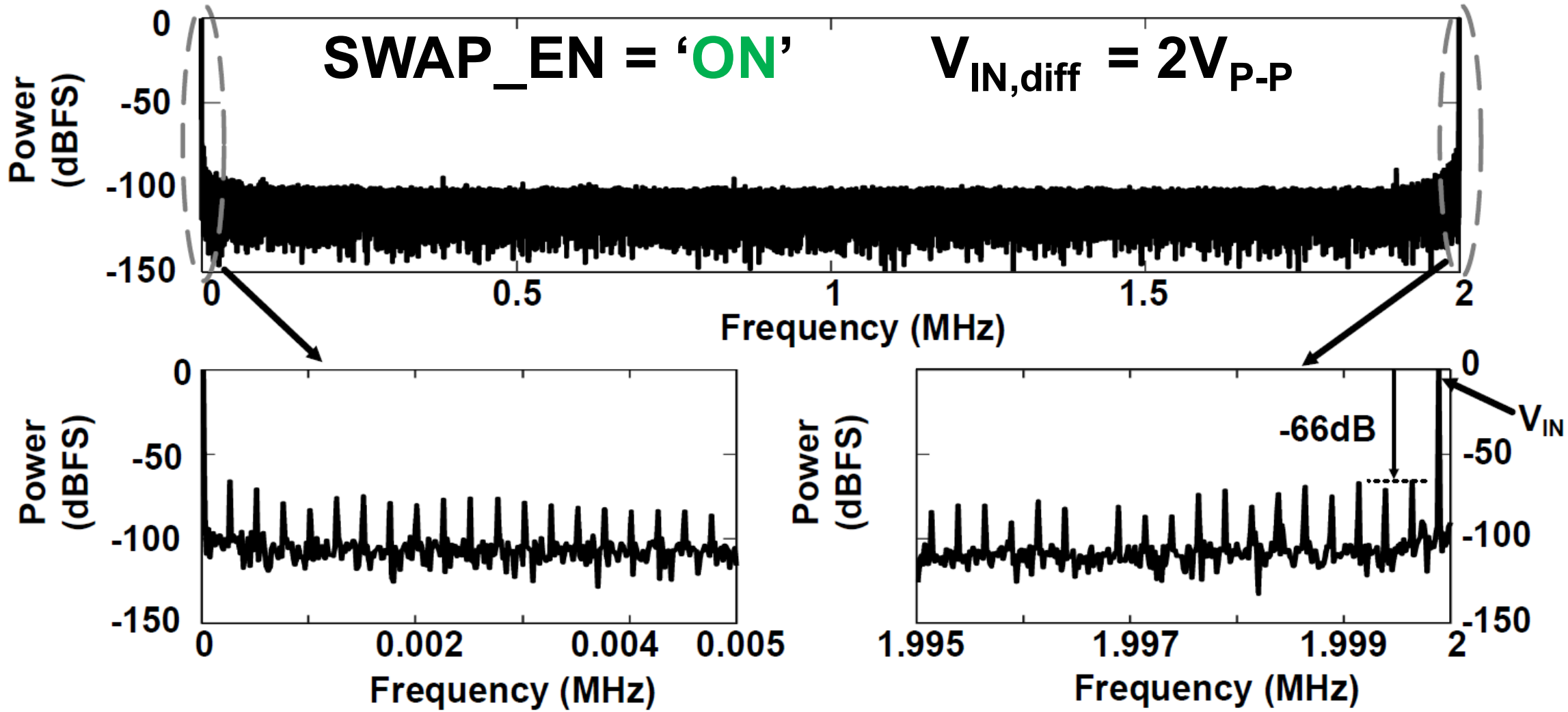
Area : 0.075mm^2 (incl. decaps)

Power Consumption
ADC : 9µW
Buffers : 70µW each

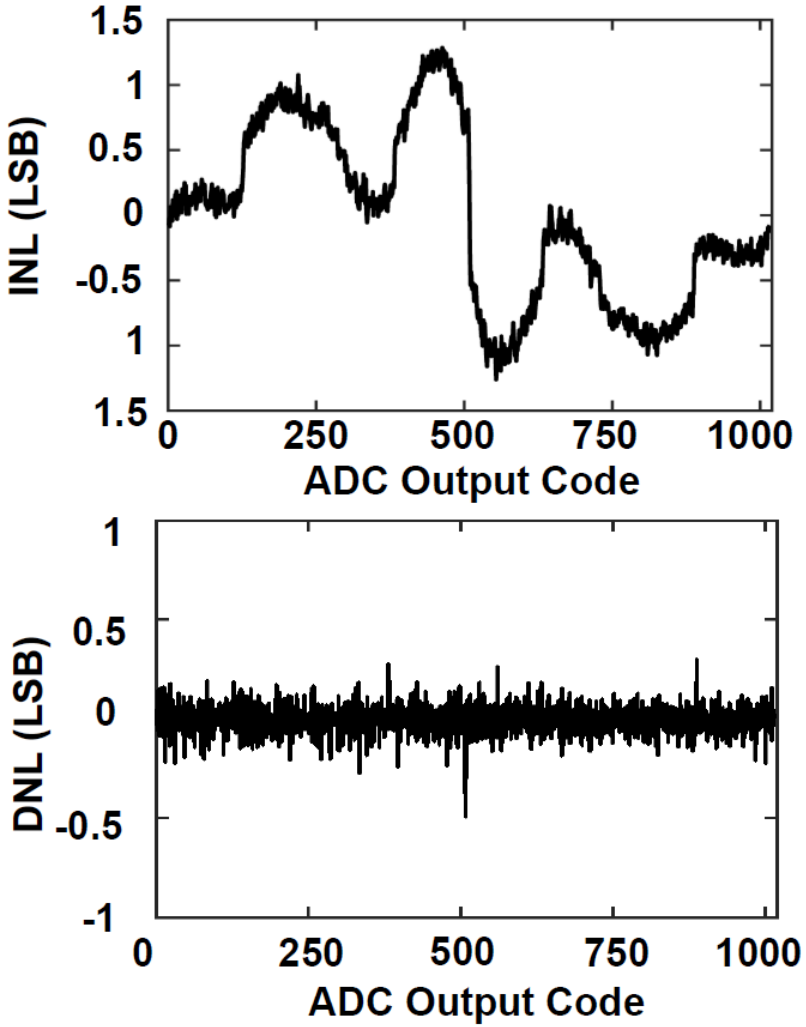
Measurement Results - FFT



Measurement Results - FFT

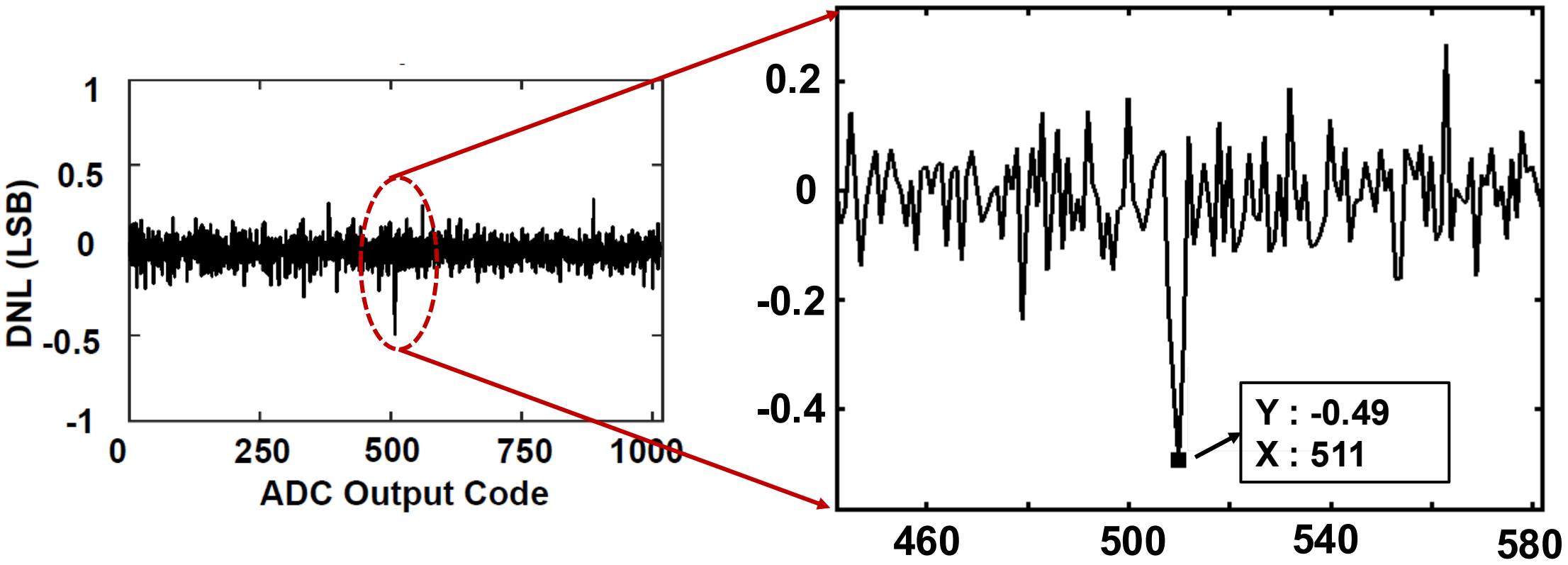


Measurement Results – INL/DNL



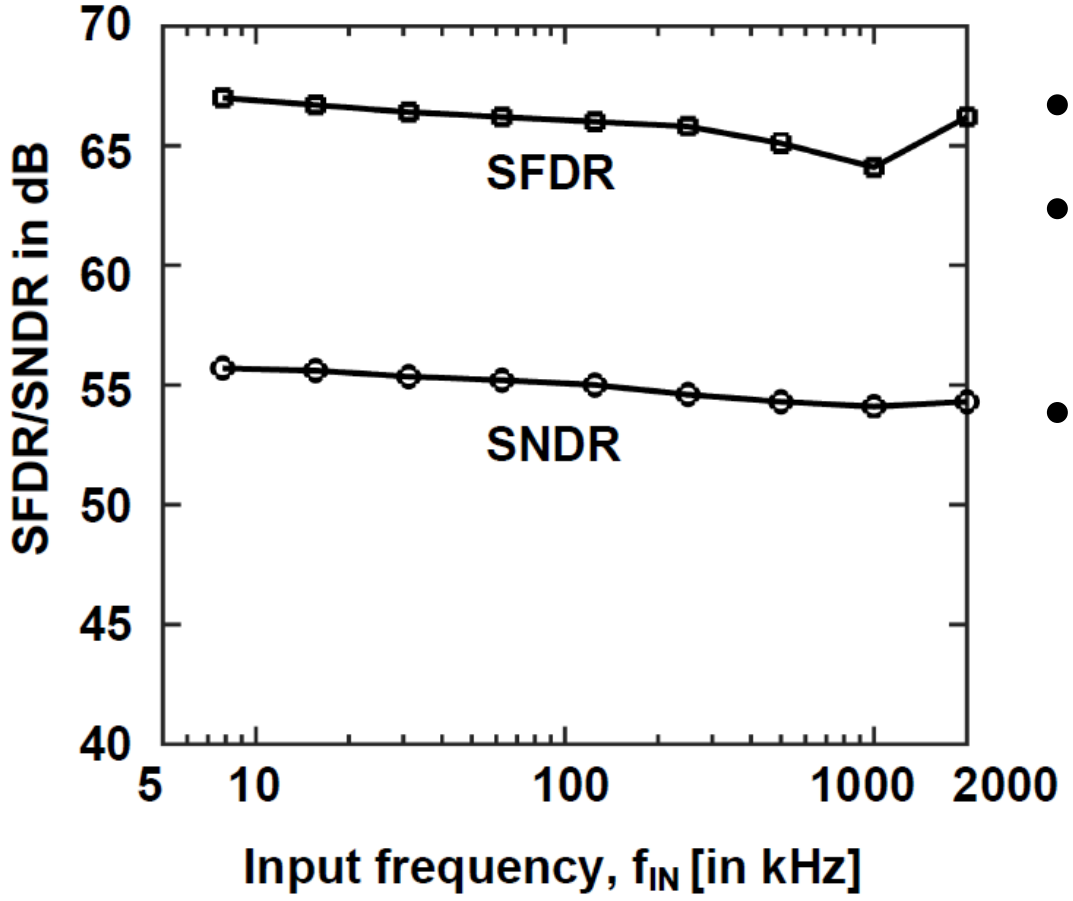
- f_S = 4MS/s
- INL = 1.2LSB
- DNL = 0.5LSB

Measurement Results – INL/DNL



- **Maximum jump is at mid-code due to non-segmented DAC array**

Measurement Results – SFDR/SNDR



- **f_S = 4MS/s**
- **SFDR > 64dB over Nyquist range**
- **SNDR > 54dB over Nyquist range**

Design Highlights

- Single Supply Low Power Buffered SAR ADC design
  - Buffers and ADC operate at single supply voltage (1.2V), process near rail-to-rail inputs.
  - Swapping the inputs allow each of the input buffer to handle one-half the full-scale range.
  - State-of-the-art Walden FoM (ADC + Buffer) of 87fJ/conv-step.

Outline

- Introduction
- Ultra-low-power SAR ADCs overview: Building blocks perspective
- Wide-bandwidth (GS/s) SAR ADCs
- Auxiliary blocks and its requirements
- Conclusion

Conclusion

- Walden FoM describes part of ADC power consumption
  - does not include auxilliary drive power
- Design techniques presented for low-energy
  - Limit the maximum change in voltage at the sample capacitors
  - Reduce the input drive current, energy and charge by at least a factor 2
  - Allow buffer integration at same supply voltage of ADC, near rail-to-rail inputs
  - Achieve lowest Walden FoM (also incl. buffers)

Conclusion

- Hierarchical sampling offers the best trade-off for achieving 100Gbps+ conversion rates
  - Cost of calibration and SNR degradation through the hierarchy

- High-speed quantizers (comparators) are increasingly becoming the bottleneck for energy-speed-noise trade-off